Chapter 10

Elliptic curves over local fields: reduction, Kodaira types, Tate curves

The paper works one bad prime at a time. At each bad prime q of the optimal curve E it reads off a handful of local invariants: the Kodaira symbol, the geometric component group Φ_q^{geom} and the Tamagawa number c_q, the valuations $v_q(\Delta)$ and $v_q(j)$, the conductor exponent, the Tate parameter q_E and its unit part u at a potentially multiplicative prime, the order of the image of inertia at a potentially good prime, and the effect on all of these of the quadratic twist by $\mathbb{Q}(\sqrt{q^*})$. This chapter defines each of them, proves what can be proved with the tools of the book, states the rest precisely with references, and computes everything on the paper's curves. The local field theory (inertia, the tame character, Kummer theory, Artin conductors) is in Chapter 7; Weierstrass equations, the group law, torsion, Tate modules and quadratic twists of models are in Chapter 9; the Galois representations built from the present results are in Chapter 11.

Core results at a glance

Throughout, K is complete for a discrete valuation v with $v(\varpi) = 1$, valuation ring $\mathcal{O}$, perfect residue field k of characteristic p, and $\ell \neq p$ unless stated otherwise.

10.1 Minimal models. A model with all $a_i \in \mathcal{O}$ is integral; it is minimal if $v(\Delta)$ is least among integral models. Minimal models exist; two of them differ by (u, r, s, t) with $u \in \mathcal{O}^\times$, $r, s, t \in \mathcal{O}$; the Néron differential is defined up to $\mathcal{O}^\times$, and $v(\Delta_{\min})$, $v(c_4)$, $v(c_6)$ of a minimal model and $v(j) = 3v(c_4) - v(\Delta)$ are invariants of E/K. *(proved here)* Definition 10.3, Proposition 10.4.

10.2 Minimality criteria. An integral model with $v(\Delta) < 12$, or $v(c_4) < 4$, or $v(c_6) < 6$ is minimal. For $p \geq 5$ it is minimal if and only if $v(\Delta) < 12$ or $v(c_4) < 4$, and otherwise $(\varpi, 0, 0, 0)$ applied to $y^2 = x^3 - 27c_4x - 54c_6$ lowers $v(\Delta)$ by 12. *(proved here)* Over $\mathbb{Z}$, c_4, c_6 with $\Delta = (c_4^3 - c_6^2)/1728 \in \mathbb{Z} \setminus \{0\}$ come from an integral model if and only if $v_3(c_6) \neq 2$ and either $c_6 \equiv -1 \pmod 4$ or $v_2(c_4) \geq 4$, $c_6 \equiv 0, 8 \pmod{32}$ (Kraus). *(stated; reference)* Proposition 10.5.

10.3 Global minimal models. Every $E/\mathbb{Q}$ has a model over $\mathbb{Z}$ minimal at every p, unique up to $(\pm 1, r, s, t)$, $r, s, t \in \mathbb{Z}$, and unique with $a_1, a_3 \in \{0, 1\}$, $a_2 \in \{-1, 0, 1\}$; its discriminant is Δ_E. *(proof sketched here)* Theorem 10.6.

10.4 Reduction types. For a minimal model: good iff $v(\Delta) = 0$; multiplicative iff $v(\Delta) > 0 = v(c_4)$ (node; split iff the tangent slopes are in k); additive iff $v(\Delta) > 0$, $v(c_4) > 0$ (cusp). *(proved here)* Definition 10.9.

10.5 Nonsingular points. $\tilde{E}_{\mathrm{ns}}(k) \simeq k^+$ (cusp), $k^\times$ (split node), $\ker(\mathrm{N}\colon l^\times \to k^\times)$ (nonsplit node, l/k quadratic); over $\mathbb{F}_p$ of orders p, $p-1$, $p+1$. Hence for $E/\mathbb{Q}$ with bad reduction at p, $a_p = p+1-\#\tilde{E}(\mathbb{F}_p)$ is 1, -1, 0 for split, nonsplit, additive. *(proved here)* Proposition 10.10, Corollary 10.11.

10.6 Split versus nonsplit. After moving the node to $(0,0)$, split iff $T^2 + a_1T - a_2$ has a root in k; for p odd, split iff $-c_6$ (equivalently $-c_4/c_6$) is a square in $k^\times$; for p odd and k finite the unramified quadratic twist exchanges split and nonsplit, and over the unramified quadratic extension the reduction is split. *(proved here)* Proposition 10.12.

10.7 The reduction filtration. With $E_0(K) = \{P : \tilde{P} \text{ nonsingular}\}$ and $E_1(K) = \{P : \tilde{P} = \tilde{O}\}$, there is an exact sequence $0 \to E_1(K) \to E_0(K) \to \tilde{E}_{\mathrm{ns}}(k) \to 0$. *(proved here)* Definition 10.14, Proposition 10.15.

10.8 Formal group. $E_1(K) \simeq \hat{E}(\mathfrak{m})$ *(stated; reference)*; for m prime to p, $[m]$ is bijective on $E_1(K)$, which has no prime-to-p torsion *(proved here, given the formal group)*. Theorem 10.16.

10.9 Torsion and reduction. For m prime to p, $E_0(K)[m] \hookrightarrow \tilde{E}_{\mathrm{ns}}(k)$. If E has good reduction, I_K acts trivially on $E[m]$ and on $T_\ell E$, and reduction $E[m] \to \tilde{E}[m](\bar{k})$ is a Galois-equivariant isomorphism. *(proved here)* Corollary 10.17.

10.10 Index of E_0. For k finite, $E(K)/E_0(K)$ is finite, trivial for good reduction. *(proved here)* Proposition 10.18.

10.11 Potential reduction. E has potentially good reduction iff $v(j) \geq 0$, and potentially multiplicative reduction iff $v(j) < 0$; then it is multiplicative over every extension where it is semistable. Multiplicative reduction has $v(j) = -v(\Delta_{\min})$. *(proof sketched here)* Definition 10.20, Theorem 10.21.

10.12 Semistable reduction. E becomes semistable over a finite extension, for instance $K(E[m])$ with $m \geq 3$ prime to p. *(proof sketched here)* Theorem 10.22.

10.13 The four classes. The Kodaira symbols fall into: good reduction, I_0; multiplicative, I_n ($n \geq 1$, $n = v(\Delta_{\min}) = -v(j)$); additive potentially multiplicative, I_n^* ($n \geq 1$); additive potentially good, II, III, IV, I_0^*, IV*, III*, II*, and at $p = 2$ only also some I_m^* ($m \geq 1$). *(proved here, given the classification)* Remark 10.23.

10.14 Néron models. The Néron model $\mathcal{E}$ (smooth separated group scheme of finite type over $\mathcal{O}$, generic fibre E, Néron mapping property) exists and is unique; it is the smooth locus of the minimal regular model. *(stated; reference)* Definition 10.26, Theorem 10.27.

10.15 Component group. $\mathcal{E}(\mathcal{O}) = E(K)$, $\mathcal{E}_k^0 \simeq \tilde{E}_{\mathrm{ns}}$, $\mathcal{E}^0(\mathcal{O}) = E_0(K)$, and the quotient $\mathcal{E}_k/\mathcal{E}_k^0$ is a finite étale group scheme over k, the component group scheme, whose groups of points are written $\Phi_E(k)$ and $\Phi_E(\bar{k}) = \Phi^{\mathrm{geom}}$ *(stated; reference)*; for k finite $E(K)/E_0(K) \simeq \Phi_E(k)$ *(proof sketched here)*. Proposition 10.28.

10.16 Φ^{geom} **and** c**.** For $E/\mathbb{Q}$: $\Phi_p^{\mathrm{geom}} = \Phi_p(\bar{\mathbb{F}}_p)$ and $c_p = \#\Phi_p(\mathbb{F}_p) = [E(\mathbb{Q}_p) : E_0(\mathbb{Q}_p)]$, the Frobenius invariants; $c_p \mid \#\Phi_p^{\mathrm{geom}}$, with equality iff Frobenius acts trivially. The paper's statements are about $\#\Phi_q^{\mathrm{geom}}$. *(proved here)* Definition 10.29, Convention 10.30.

10.17 Unramified base change. Néron models commute with étale base change; type, Kodaira symbol, Φ^{geom}, $v(\Delta_{\min})$ and f are invariant under unramified extension; c is not. *(stated; reference)* Proposition 10.31.

10.18 Tamagawa numbers of I_n**.** $\Phi^{\mathrm{geom}} \simeq \mathbb{Z}/n$; $c = n$ (split), $c = \gcd(2, n)$ (nonsplit); for ℓ odd the ℓ-part of c is $\ell^{v_\ell(n)}$ (split) or 1 (nonsplit). *(proved here, given Tate's theorem)* Remark 10.32, Proposition 10.65.

10.19 Inertia invariants and component groups. Let $K/\mathbb{Q}_q$ be finite, $\ell \neq q$. If the reduction is additive, $E[\ell^t]^{I_K} = E(K^{\mathrm{ur}})[\ell^t] \simeq \Phi^{\mathrm{geom}}[\ell^t]$, Galois-equivariantly, and $E(K)[\ell^\infty] \simeq \Phi_E(k)[\ell^\infty]$. If it is multiplicative, $0 \to \mu_{\ell^t} \otimes \varepsilon \to E[\ell^t]^{I_K} \to \Phi^{\mathrm{geom}}[\ell^t] \to 0$ with $\Phi^{\mathrm{geom}} \simeq \mathbb{Z}/n$, ε trivial iff split. For ℓ odd, $\dim E[\ell]^{I_K}$ is $1 + [\ell \mid n]$ for I_n, 1 for $\mathrm{IV}, \mathrm{IV}^*$ if $\ell = 3$, and 0 for every other additive type. *(proved here)* Proposition 10.33.

10.20 Kodaira-Néron classification. The special fibres and $(m, \Phi^{\mathrm{geom}})$ are: I_0, I_1, II: $m = 1$, 0; I_n: n, $\mathbb{Z}/n$; III: 2, $\mathbb{Z}/2$; IV: 3, $\mathbb{Z}/3$; I_0^*: 5, $(\mathbb{Z}/2)^2$; I_n^*: $n+5$, $\mathbb{Z}/4$ (n odd) or $(\mathbb{Z}/2)^2$ (n even); IV^*: 7, $\mathbb{Z}/3$; III^*: 8, $\mathbb{Z}/2$; II^*: 9, 0. Possible c: $\mathrm{IV}, \mathrm{IV}^*$: $1, 3$; I_0^*: $1, 2, 4$; I_n^*: $2, 4$; $\mathrm{III}, \mathrm{III}^*$: 2; $\mathrm{II}, \mathrm{II}^*$: 1. *(stated; reference)* Theorem 10.36.

10.21 Tate's algorithm. The eleven steps of Theorem 10.37 compute the type, m, $f = v(\Delta) + 1 - m$, c and a minimal model; explicit translations for Step 2 at $p \geq 5$, $p = 3$, $p = 2$, and completing the square for Step 6 at odd p. *(proof sketched here)* Theorem 10.37.

10.22 The tame table. For $p \geq 5$ and a minimal model, $(v(c_4), v(c_6), v(\Delta))$ is $(0, 0, n)$ for I_n; $(\geq 1, 1, 2)$ II; $(1, \geq 2, 3)$ III; $(\geq 2, 2, 4)$ IV; $(\geq 2, \geq 3, 6)$ I_0^*; $(2, 3, n+6)$ I_n^*; $(\geq 3, 4, 8)$ IV^*; $(3, \geq 5, 9)$ III^*; $(\geq 4, 5, 10)$ II^*. So for potentially good reduction the type is determined by $v(\Delta) \in \{0, 2, 3, 4, 6, 8, 9, 10\}$, and $f = 2$ for every additive type. *(proved here, given Tate's algorithm, and Ogg's formula for the index of* I_n^**)* Corollary 10.38.

10.23 The cubic criterion for I_0^***.** For p odd and $y^2 = x^3 + a_2x^2 + a_4x + a_6$ with $\varpi \mid a_2$, $\varpi^2 \mid a_4$, $\varpi^3 \mid a_6$: type I_0^* iff $T^3 + a_{2,1}T^2 + a_{4,2}T + a_{6,3}$ has distinct roots mod ϖ, and then $c = 1 + \#\text{roots in } k$; every I_0^* at odd p has such a model. The twist by q^* of a model with good reduction at odd q has cubic $T^3 + \epsilon a_2 T^2 + a_4 T + \epsilon a_6$, $\epsilon = q^*/q$, hence type I_0^*. *(proved here)* Proposition 10.39.

10.24 PARI's Kodaira codes. 1: I_0; $n+4$: I_n and $-(n+4)$: I_n^* *for* $n \geq 1$ *only*; -1: I_0^*; $\pm 2, \pm 3, \pm 4$: II/II*, III/III*, IV/IV*. `elllocalred` returns $[f, k, [u, r, s, t], c]$. A session on `726a1`, `20a1`, `539c1`, `11a1` shows the output. *(proved here, as a dictionary)* Convention 10.41, Example 10.84.

10.25 Conductor exponent. $f = a(V_\ell E) = \dim V - \dim V^I + \mathrm{sw}(V)$, independent of ℓ *(stated; reference)*; it is 0 (good), 1 (multiplicative, V^I a line), $2 + \mathrm{sw} \geq 2$ (additive, $V^I = 0$), and $\mathrm{sw} = 0$ for $p \geq 5$, and for p odd at an additive potentially multiplicative prime *(proved here, given the criterion of Néron, Ogg and Shafarevich)*. Definition 10.45, Proposition 10.46.

10.26 Ogg's formula. $f = v(\Delta_{\min}) + 1 - m$. *(stated; reference)* Theorem 10.47.

10.27 Conductor bounds. $f_p \le 2$ for $p \ge 5$ *(proved here)*, $f_3 \le 5$, $f_2 \le 8$ *(stated; reference)*. Theorem 10.48.

10.28 The conductor. $N = \prod_p p^{f_p}$; $p \,\|\, N$ iff multiplicative, $p^2 \mid N$ iff additive, $p^3 \nmid N$ for $p \ge 5$; isogenous curves have the same conductor. *(proved here)* Definition 10.49, Proposition 10.50.

10.29 Serre-Tate. For potentially good reduction there is a smallest $L \supseteq K^{\mathrm{ur}}$ with good reduction, $L = K^{\mathrm{ur}}(E[m])$ $(m \ge 3,\ p \nmid m)$; $G_{\mathrm{ss}} = \mathrm{Gal}(L/K^{\mathrm{ur}})$ is the image of I_K in $\mathrm{Aut}(T_\ell E)$ for every $\ell \ne p$; it embeds in $\mathrm{SL}_2(\mathbb{Z}/m)$, its elements have orders $1, 2, 3, 4, 6$, and for $p \ge 5$ it is cyclic. *(proof sketched here)* Theorem 10.51.

10.30 Kraus. For $p \ge 5$ and potentially good reduction, G_{ss} is cyclic of order $12/\gcd(v(\Delta_{\min}), 12)$: $1, 6, 4, 3, 2, 3, 4, 6$ for $\mathrm{I}_0, \mathrm{II}, \mathrm{III}, \mathrm{IV}, \mathrm{I}_0^*, \mathrm{IV}^*, \mathrm{III}^*, \mathrm{II}^*$ *(proved here, given Serre-Tate)*. At $p = 3$: cyclic of order $1, 2, 3, 4, 6$ or $\mathbb{Z}/3 \rtimes \mathbb{Z}/4$; at $p = 2$: cyclic of order $1, 2, 3, 4, 6$, Q_8 or $\mathrm{SL}_2(\mathbb{F}_3)$ *(stated; reference)*. Theorem 10.52.

10.31 Tame additive reduction. Additive potentially good reduction over a finite $K/\mathbb{Q}_p$ is tame $(f = 2)$ iff $p \ge 5$, or $p = 3$ and type $\mathrm{III}, \mathrm{III}^*, \mathrm{I}_0^*$, or $p = 2$ and type $\mathrm{IV}, \mathrm{IV}^*$. *(stated; reference)* Theorem 10.53.

10.32 Types IV, IV* at 2. Such a curve over $\mathbb{Q}_2$ has potentially good reduction, $G_{\mathrm{ss}} \simeq \mathbb{Z}/3$, the image of I_2 on $T_\ell E$ (ℓ odd) is generated by σ of order 3 with eigenvalues $\zeta_3^{\pm 1}$, unipotent on $E[3]$, $f_2 = 2$, and Frob σ $\mathrm{Frob}^{-1} = \sigma^{-1}$ *(proved here)* Corollary 10.54. Consequently (the paper link after it, with Lemma lem:euler and the normalisation of P_q in Chapter 19) $P_2(X) = 1 + 2X$ and $e_\ell(E, 2) = v_\ell(3/2)$ *(consequence; see Chapter 19)*.

10.33 The Tate curve. For $v(\mathfrak{q}) > 0$, $E_{\mathfrak{q}}\colon y^2 + xy = x^3 - 5s_3x - (5s_3 + 7s_5)/12$ has $\Delta = \mathfrak{q}\prod(1 - \mathfrak{q}^n)^{24}$, $j = \mathfrak{q}^{-1} + 744 + 196884\mathfrak{q} + \cdots$, split multiplicative reduction of type $\mathrm{I}_{v(\mathfrak{q})}$ *(proved here)*, and $E_{\mathfrak{q}}(\bar{K}) \simeq \bar{K}^\times/\mathfrak{q}^{\mathbb{Z}}$ Galois-equivariantly *(stated; reference)*. Theorem 10.57.

10.34 Inverting j. $\mathfrak{q}(J) = J + 744J^2 + 750420J^3 + 872769632J^4 + \cdots \in J\mathbb{Z}[[J]]$ inverts $1/j(\mathfrak{q})$; for $v(j_0) < 0$, $\mathfrak{q}(1/j_0)$ is the unique $\mathfrak{q}_0$ with $v(\mathfrak{q}_0) > 0$, $j(E_{\mathfrak{q}_0}) = j_0$, and $v(\mathfrak{q}_0) = -v(j_0)$. *(proved here)* Proposition 10.58.

10.35 Curves with $v(j) < 0$. E is the quadratic twist of E_{q_E} by $\gamma = -c_4/c_6$: split iff γ is a square, nonsplit iff $K(\sqrt{\gamma})/K$ is unramified quadratic, additive iff it is ramified; for multiplicative reduction $n = v(\Delta_{\min}) = -v(j) = v(q_E)$. *(proof sketched here)* Theorem 10.59.

10.36 Tate parameter and unit part. For $E/\mathbb{Q}_q$ with $v_q(j) < 0$: $q_E = \mathfrak{q}(1/j_E)$, $\nu = v_q(q_E) = -v_q(j_E)$ $(= n_q$ at a multiplicative prime$)$, $u = q_E q^{-\nu} \in \mathbb{Z}_q^\times$, $c(u)$ the class of $\bar{u}$ in $\mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{\ell^{v_\ell(q-1)}}$. These depend only on j_E, so $t_\ell(E, q) = v_\ell(\nu)$ is twist invariant *(proved here)*; under an isogeny of prime degree d, $\nu \mapsto d\nu$ or ν/d, so t_ℓ is invariant under isogenies of degree prime to ℓ *(proof sketched here)*. Definition 10.60, Proposition 10.61.

10.37 Torsion of the Tate curve. $E_{\mathfrak{q}}[\ell^t] = \{\zeta^a(\mathfrak{q}^{1/\ell^t})^b\}$, an extension $0 \to \mu_{\ell^t} \to E_{\mathfrak{q}}[\ell^t] \to \mathbb{Z}/\ell^t \to 0$; on $T_\ell E_{\mathfrak{q}}$, $\sigma \mapsto \left(\begin{smallmatrix} \chi_\ell(\sigma) & \kappa(\sigma) \\ 0 & 1 \end{smallmatrix}\right)$ with κ the Kummer cocycle of $\mathfrak{q}$; for the twist by ε, $\left(\begin{smallmatrix} \chi_\ell\varepsilon & \varepsilon\kappa \\ 0 & \varepsilon \end{smallmatrix}\right)$, with corner $\varepsilon\kappa$. *(proved here, given Tate's theorem)* Proposition 10.62.

10.38 Types at additive potentially multiplicative primes. For p odd the type is I_ν^*, $\nu = -v(j)$, with $f = 2$, $v(\Delta_{\min}) = \nu + 6$, and conversely I_n^* $(n \ge 1)$ at odd p forces

$v(j) = -n$ *(proof sketched here, given Ogg's formula).* For $K = \mathbb{Q}_2$ the type is I_m^* with $m = \nu + 2f_2 - 4 \in \{\nu + 4, \nu + 8\}$ *(stated; reference).* Theorem 10.66.

10.39 Unramified twists (the paper's Lemma lem:twist (i)). A twist by a quadratic character unramified at p preserves at p the type, Φ_p^{geom}, $v_p(\Delta_{\min})$, f_p and q_E; c_p may change. *(proved here)* Lemma 10.69.

10.40 Ramified twists (the paper's Lemma lem:twist (ii)). For odd q the twist by q^* exchanges $\mathrm{I}_n \leftrightarrow \mathrm{I}_n^*$ (same $n \geq 1$) and $\mathrm{I}_0 \leftrightarrow \mathrm{I}_0^*$; for $q \geq 5$ also $\mathrm{II} \leftrightarrow \mathrm{IV}^*$, $\mathrm{III} \leftrightarrow \mathrm{III}^*$, $\mathrm{IV} \leftrightarrow \mathrm{II}^*$ with $v_q(\Delta_{\min})$ shifted by 6 mod 12; the conductor exponent of the twist at q is 0 (good), 1 (I_n) or 2 (additive). *(proved here; the equality of the indices in $\mathrm{I}_n \leftrightarrow \mathrm{I}_n^*$ uses Ogg's formula)* Lemma 10.70.

10.41 The exchanges at 3. The twist by -3 exchanges $\mathrm{II} \leftrightarrow \mathrm{IV}^*$, $\mathrm{III} \leftrightarrow \mathrm{III}^*$, $\mathrm{IV} \leftrightarrow \mathrm{II}^*$ at 3 and preserves f_3. *(proved here, given Ogg's formula)* Proposition 10.72.

10.42 Twist-symmetrised weight. For odd q and odd ℓ,

$$w_\ell(E, q) = \max\bigl(v_\ell\#\Phi_q^{\mathrm{geom}}(E),\, v_\ell\#\Phi_q^{\mathrm{geom}}(E^{(q)})\bigr), \qquad w_\ell(E, 2) = v_\ell\#\Phi_2^{\mathrm{geom}}(E).$$

(proved here) Proposition 10.74.

10.43 What changes at 2 and 3. For $p \in \{2, 3\}$: the type determines neither $v(\Delta_{\min})$ nor f (only $f = v(\Delta_{\min}) + 1 - m$); $f \leq 8$ at 2, $f \leq 5$ at 3; G_{ss} can be non-abelian, of order divisible by p; the type determines G_{ss} only for $\mathrm{IV}, \mathrm{IV}^*$ at 2 ($\mathbb{Z}/3$) and $\mathrm{III}, \mathrm{III}^*, \mathrm{I}_0^*$ at 3 ($\mathbb{Z}/4, \mathbb{Z}/4, \mathbb{Z}/2$); at 2 a symbol I_m^* may be potentially good, and if not, $m \neq -v_2(j)$. So the adjoint Euler factor is not a function of the type at 2, 3. *(summary of results proved or stated elsewhere in the chapter)* Remark 10.76.

10.44 Inertia at 2. For additive potentially good reduction over $\mathbb{Q}_2$, the ramification pattern of the roots of ψ_3 is $(1,1,1,1)$, $(1,3,3,3)$, $(2,2,2,2)$, $(4,4,4,4)$ according as G_{ss} is $\mathbb{Z}/2$ (a ramified twist of good reduction), cyclic of order 3 or 6, $\mathbb{Z}/4$, or Q_8 or $\mathrm{SL}_2(\mathbb{F}_3)$; order 3 occurs exactly for IV, IV*. *(proved here, given Kraus's list of groups and Theorem 10.53)* Proposition 10.77.

10.45 The prime 3. Types $\mathrm{III}, \mathrm{III}^*, \mathrm{I}_0^*$ at 3 are tame with G_{ss} cyclic of order $4, 4, 2$; types $\mathrm{II}, \mathrm{II}^*, \mathrm{IV}, \mathrm{IV}^*$ are wild with $3 \mid e$ and $f_3 \in \{3, 4, 5\}$; I_n^* is potentially multiplicative with $f_3 = 2$. *(proved here, given Theorem 10.53)* Remark 10.79.

10.46 Symbols I_m^* at 2. They are potentially good in 61% of the pairs below conductor 10,000; when potentially multiplicative, $t_\ell(E, 2) = v_\ell(\nu)$ with $\nu = m - 4$ or $m - 8$, not $v_\ell(m)$. *(proved here, with data)* Remark 10.81.

10.47 Worked data. Tate's algorithm by hand on `121a1`, `121c2`, `539c1`, `539d1`, `726a1`, `20a1`, `99a1`, `980i1`; minimal models of twists; semistability defects; Tate parameters; twist pairs; local data of the running examples; what Cremona's tables contain. Example 10.7, Example 10.42, Example 10.56, Example 10.67, Example 10.75, Example 10.85, Remark 10.83.

Where the paper uses this chapter

- **Definition def:terms** ($t_\ell(E,q) = v_\ell(-v_q(j_E))$, potentially multiplicative primes, the symbols I_n^* at 2): Proposition 10.4, Theorem 10.21, Proposition 10.58, Definition 10.60, Proposition 10.61, Theorem 10.66, Remark 10.81.
- **The table of Section 1 and Lemma lem:euler** (types at $q \geq 5$, inertia images): Corollary 10.38, Theorem 10.51, Theorem 10.52, Lemma 10.70.
- **Section 1.4** (Φ_q^{geom} versus c_q, the exchange of types, the weight w_ℓ, the inequalities (eq:main), (eq:twist)): Proposition 10.28, Definition 10.29, Theorem 10.36, Lemma 10.70, Proposition 10.72, Proposition 10.74.
- **Lemma lem:twist (i), (ii)**: proved as Lemma 10.69 and Lemma 10.70, using Proposition 10.5, Proposition 10.39, Theorem 10.59 and Theorem 10.66.
- **Lemmas lem:tamagawa-exp and lem:tamfactor** (the Tate curve over the unramified quadratic extension, q_E, u, $c(u)$, the matrices with corner $\varepsilon\kappa$, the check with c_q): Proposition 10.12, Proposition 10.33, Theorem 10.59 to Proposition 10.65; the lemmas themselves are proved in Chapter 11 and Chapter 18.
- **Lemma lem:IV and Remark rem:types** ($E[3]^{I_q}$ a line or 0, Kraus's defect, the case $q = 2$): Proposition 10.33, Theorem 10.51, Theorem 10.52, Corollary 10.54.
- **Proposition prop:shapes and Lemma lem:tamadd** of the revised paper (the image of inertia at a potentially good prime is finite, its elements have orders $1, 2, 3, 4, 6$, it is cyclic for $q \geq 5$; a potentially multiplicative curve is a quadratic twist of a Tate curve): Theorem 10.51, Theorem 10.52, Theorem 10.59, Proposition 10.77.
- **Remark rem:wild, Section sec:wild, Proposition prop:forms**: Theorem 10.53, Corollary 10.54, Remark 10.76, Proposition 10.77, Remark 10.79, Remark 10.81.
- **Theorems thm:qual, thm:quant and the proof of thm:main** (twists at I_0^*, I_n^*, II, II* primes; conductor exponents; the prime 2 of type I_n^*): Proposition 10.50, Lemma 10.70, Theorem 10.59.
- **The appendix** (`ellglobalred`, `R[4]`, `R[5]`, `K[i][2]`, the Kodaira codes, `valuation(j, q)`): Convention 10.41, Example 10.84.
- **The Example of Section 3, the Remark after it, Section sec:anomaly**: Example 10.42, Example 10.67, Example 10.75, Example 10.85.

10.1 Integral and minimal Weierstrass models

This section fixes the local setting and the one choice that every later statement depends on: a Weierstrass model of E over a complete discretely valued field that is integral and has the smallest possible discriminant. Every local invariant the paper reads off at a bad prime q (the Kodaira type, the valuations $v_q(\Delta)$ and $v_q(c_4)$, the Tamagawa number c_q) is defined through such a model. The paper uses minimality explicitly in the proof of Lemma lem:twist (ii), where the twisted model "is minimal exactly when the result is less than 12, otherwise 12 is subtracted": that sentence is Proposition 10.5(ii) below.

Convention 10.1 (the local setting)**.** Throughout the chapter K is a field of characteristic 0, complete for a discrete valuation v, with valuation ring $\mathcal{O}$, maximal ideal $\mathfrak{m} = \varpi\mathcal{O}$, perfect residue field $k = \mathcal{O}/\mathfrak{m}$ of characteristic p, and $v(\varpi) = 1$. In every application K is $\mathbb{Q}_p$, a finite extension of $\mathbb{Q}_p$, or a finite unramified extension of one of these; K^{ur} is the maximal unramified extension and I_K the inertia group (Chapter 7). When the paper's bad prime is meant we write $K = \mathbb{Q}_q$ and $p = q$. The letter q is also used for the Tate parameter in the literature; in this chapter the variable of the Tate curve is written $\mathfrak{q}$ and the Tate curve $E_{\mathfrak{q}}$, and the Tate parameter of a given curve E is always q_E (Section 10.7). The arithmetic Frobenius is Frob; the geometric Frobenius at a named prime is written Frob_q^{-1}, because $\Phi_q(\cdot)$ denotes a component group in this chapter (Appendix B, Appendix B, conventions B-C2 and B-C11).

Convention 10.2 (models)**.** A *model* of an elliptic curve E/K is a Weierstrass equation

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6, \qquad a_i \in K,$$

with the general coefficients $a_1, \dots, a_6$ and the quantities $b_2, b_4, b_6, b_8, c_4, c_6, \Delta, j$ of Chapter 9. Two models of E are related by a change of coordinates $x = u^2x' + r$, $y = u^3y' + su^2x' + t$ with $u \in K^\times$, $r, s, t \in K$, written (u, r, s, t), and then

$$u^4c_4' = c_4, \qquad u^6c_6' = c_6, \qquad u^{12}\Delta' = \Delta, \qquad j' = j,$$

together with the formulas for the a_i' and b_i' recalled in Chapter 9. At $p = 2$ and $p = 3$ a short model $y^2 = x^3 + Ax + B$ is in general not minimal, so short models are used only when $p \geq 5$.

Definition 10.3 (integral and minimal models)**.** A model of E/K is *integral* if all $a_i \in \mathcal{O}$; then $b_2, b_4, b_6, b_8, c_4, c_6, \Delta \in \mathcal{O}$, as they are polynomials with integer coefficients in the a_i. An integral model is *minimal* if $v(\Delta)$ is smallest among all integral models of E; its discriminant is the *minimal discriminant* $\Delta_{\min}$, well defined up to $\mathcal{O}^\times$. For a minimal model the invariant differential $\omega = dx/(2y + a_1x + a_3)$ is the *Néron differential*.

Proposition 10.4 (existence and uniqueness of minimal models)**.** *Let E/K be an elliptic curve.*

(i) E has a minimal model.

(ii) Two minimal models of E are related by a change (u, r, s, t) with $u \in \mathcal{O}^\times$ and $r, s, t \in \mathcal{O}$.

(iii) The Néron differential is well defined up to multiplication by $\mathcal{O}^\times$. The integers $v(\Delta_{\min})$, $v(c_4)$ and $v(c_6)$ of a minimal model depend only on E/K, and $v(j) = 3v(c_4) - v(\Delta)$ is the same for every model.

Proof. (i) Start from any model. The change $(u, 0, 0, 0)$ multiplies a_i by u^{-i}; with $u = \varpi^{-n}$ the new coefficients are $\varpi^{ni}a_i$, which lie in $\mathcal{O}$ for n large. So integral models exist, the values $v(\Delta)$ of integral models are nonnegative integers, and a smallest one is attained.

(ii) Any two models of E are related by some (u, r, s, t) (Chapter 9). If both are minimal, then $v(\Delta) = v(\Delta')$ and $u^{12}\Delta' = \Delta$ give $v(u) = 0$. The transformation formulas for b_6 and b_8 read

$$4r^3 + b_2r^2 + 2b_4r + b_6 = u^6b_6',$$
$$3r^4 + b_2r^3 + 3b_4r^2 + 3b_6r + b_8 = u^8b_8',$$

with right-hand sides in $\mathcal{O}$. Suppose $v(r) < 0$. If $p \neq 3$, in the second equation the term $3r^4$ has valuation $4v(r)$, strictly less than the valuation of every other term ($\geq 3v(r)$, $2v(r)$, $v(r)$ or 0),

which is impossible. If $p = 3$ then 4 is a unit and the same argument applies to $4r^3$ in the first equation. So $r \in \mathcal{O}$. Next, $ua_1' = a_1 + 2s$ and $u^2a_2' = a_2 - sa_1 + 3r - s^2$ show that s is a root of the monic polynomial $X^2 + a_1X + (u^2a_2' - a_2 - 3r)$ with coefficients in $\mathcal{O}$; as $\mathcal{O}$ is integrally closed, $s \in \mathcal{O}$. Likewise $u^6a_6' = a_6 + ra_4 + r^2a_2 + r^3 - ta_3 - t^2 - rta_1$ shows that t is a root of a monic quadratic over $\mathcal{O}$, so $t \in \mathcal{O}$.

(iii) Under (u, r, s, t) the invariant differential is multiplied by u (Chapter 9), and u is a unit by (ii). The relations $u^4c_4' = c_4$, $u^6c_6' = c_6$, $u^{12}\Delta' = \Delta$ with u a unit give the invariance of the three valuations. Finally $j = c_4^3/\Delta$ does not depend on the model at all. □

Proposition 10.5 (criteria for minimality)**.** *Let a model of E/K be integral.*

(i) (any p) If $v(\Delta) < 12$, or $v(c_4) < 4$, or $v(c_6) < 6$, the model is minimal.

(ii) ($p \geq 5$) The model is minimal if and only if $v(\Delta) < 12$ or $v(c_4) < 4$. If it is not minimal, the model $y^2 = x^3 - 27\varpi^{-4}c_4x - 54\varpi^{-6}c_6$ is integral with discriminant of valuation $v(\Delta) - 12$; repeating this step finitely often gives a minimal model.

(iii) (Kraus's conditions; $K = \mathbb{Q}$) Let $c_4, c_6 \in \mathbb{Z}$ with $\Delta = (c_4^3 - c_6^2)/1728 \in \mathbb{Z} \setminus \{0\}$. There is a Weierstrass model with coefficients in $\mathbb{Z}$ whose invariants are c_4, c_6 if and only if $v_3(c_6) \neq 2$, and either $c_6 \equiv -1 \pmod 4$, or $v_2(c_4) \geq 4$ and $c_6 \equiv 0$ or $8 \pmod{32}$. Consequently, for $p \in \{2, 3\}$, an integral model over $\mathbb{Z}$ with $p^4 \mid c_4$, $p^6 \mid c_6$ and $p^{12} \mid \Delta$ fails to be minimal at p exactly when $(c_4/p^4, c_6/p^6)$ satisfies the condition at p ($v_3(c_6/3^6) \neq 2$ for $p = 3$; the congruence condition for $p = 2$).

Proof. (i) Suppose an integral model with invariants c_4', c_6', Δ' and $v(\Delta') < v(\Delta)$ exists, related by (u, r, s, t). Then $u^{12}\Delta' = \Delta$ forces $v(u) \geq 1$, hence $v(\Delta) = v(\Delta') + 12v(u) \geq 12$, $v(c_4) = v(c_4') + 4v(u) \geq 4$ and $v(c_6) = v(c_6') + 6v(u) \geq 6$, as c_4', c_6', Δ' are integral. Each of the three inequalities in the hypothesis contradicts one of these.

(ii) By (i) the condition is sufficient. Conversely let $v(c_4) \geq 4$ and $v(\Delta) \geq 12$. From $c_6^2 = c_4^3 - 1728\Delta$ we get $v(c_6^2) \geq 12$, so $v(c_6) \geq 6$. The curve has the model $y^2 = x^3 - 27c_4x - 54c_6$ (Chapter 9), whose discriminant is $6^{12}\Delta$; applying $(\varpi, 0, 0, 0)$ to it gives $y^2 = x^3 - 27\varpi^{-4}c_4x - 54\varpi^{-6}c_6$, which is integral, with discriminant $6^{12}\varpi^{-12}\Delta$. As $p \geq 5$, 6 is a unit and the valuation is $v(\Delta) - 12$. So the model was not minimal. Each step lowers $v(\Delta)$ by 12, so the process stops.

(iii) We do not prove this; see Kraus [9], and [4, Section 3.2] for its use in algorithms. The consequence follows by applying the criterion to the invariants c_4/p^4, c_6/p^6 of the model obtained by $(p, 0, 0, 0)$: the conditions at the other prime are unchanged, since p is a unit there (for $p = 3$: multiplication by $3^{-6} \equiv 9 \pmod{32}$ preserves the classes 0 and 8 modulo 32 and the class -1 modulo 4). □

We checked (iii) on the data: 20,000 random integral models with coefficients in $[-50, 50]$ all satisfy Kraus's conditions, and in all 8,248 pairs (curve, $p \in \{2, 3\}$) of Cremona's tables below conductor 10,000 whose minimal model has $p^4 \mid c_4$, $p^6 \mid c_6$, $p^{12} \mid \Delta$, the scaled invariants fail the condition at p, as they must.

Theorem 10.6 (global minimal models)**.** *Every elliptic curve $E/\mathbb{Q}$ has a model with coefficients in $\mathbb{Z}$ that is minimal at every prime p. It is unique up to a change (u, r, s, t) with $u = \pm 1$ and $r, s, t \in \mathbb{Z}$, and its discriminant Δ_E is the* minimal discriminant *of E. Imposing $a_1, a_3 \in \{0, 1\}$ and $a_2 \in \{-1, 0, 1\}$ makes it unique.*

Sketch of proof. Start from a model over $\mathbb{Z}$. At each of the finitely many primes p dividing its discriminant choose a change (u_p, r_p, s_p, t_p) over $\mathbb{Q}_p$ to a minimal model. Since $\mathbb{Z}$ is a principal ideal domain the ideal $\prod_p p^{v_p(u_p)}$ is generated by $u = \prod_p p^{v_p(u_p)}$, and the Chinese remainder theorem lets one choose $r, s, t \in \mathbb{Z}$ approximating the r_p, s_p, t_p closely enough that the change (u, r, s, t) is integral and minimal at every p simultaneously. Uniqueness follows from Proposition 10.4(ii) at every prime: u is a unit of $\mathbb{Z}$ and $r, s, t \in \mathbb{Z}_p$ for all p. The normalisation of a_1, a_3, a_2 uses s, t and r modulo 2, 2 and 3. See [23, VIII.8] and [4, Section 3.2]. □

Cremona's tables [4, 5] and PARI's `ellminimalmodel` give exactly this normalised global minimal model. All models printed in this book are of this form.

Example 10.7 (minimal models)**.** (a) The curve `121a1` is $[1, 1, 1, -30, -76]$ with $\Delta = -11^2$. Its twist by -11, written with the b-invariants as in Chapter 9, is $y^2 = x^3 + db_2x^2 + 8d^2b_4x + 16d^3b_6$ with $d = -11$; its discriminant is $2^{12}d^6\Delta$, so $v_{11}(\Delta) = 2 + 6 = 8 < 12$ and the model is minimal at 11 by Proposition 10.5(i). It is the curve `121c2`, with $\Delta = -11^8$. Twisting `121c2` back by -11 gives $v_{11}(\Delta) = 8 + 6 = 14$ and $v_{11}(c_4) = 3 + 2 = 5 \geq 4$, so that model is not minimal at 11; one step of Proposition 10.5(ii) gives $v_{11}(\Delta) = 2$, and the curve is `121a1` again.

(b) The curve `140b1` is $[0, 0, 0, 32, 212]$, with $v_2(\Delta) = 8$, $v_2(c_4) = 9$ and $v_2(j) = 27 - 8 = 19$. It is minimal at 2 by Proposition 10.5(i), although c_4 is very divisible by 2.

(c) The short model $y^2 = x^3 - 27c_4x - 54c_6$ of `11a1` has discriminant $6^{12}\Delta$, so $v_2(\Delta) = v_3(\Delta) = 12$; it is not minimal at 2 or 3, since the model $[0, -1, 1, -10, -20]$ has $\Delta = -11^5$.

In the paper 10.8. Lemma lem:twist (ii), proof: "the twisted model has $v_q(\Delta)$ increased by 6, and is minimal exactly when the result is less than 12, otherwise 12 is subtracted". For $q \geq 5$ and potentially good reduction this is Proposition 10.5: if the result is at least 12, then $v_q(j) \geq 0$ gives $3v_q(c_4) \geq v_q(\Delta) \geq 12$, so $v_q(c_4) \geq 4$ and part (ii) applies once. The full argument is Lemma 10.70.

10.2 Reduction modulo the uniformiser

This section defines good, multiplicative and additive reduction, splits multiplicative reduction into split and nonsplit, and studies the reduction map on points. The paper uses the trichotomy everywhere (the numbers $\delta_p = \dim V^{I_p} = 2, 1, 0$ of its Section 1.1; the PARI test `k >= 5` of its appendix), the values $a_p \in \{1, -1, 0\}$ at bad primes (Lemma lem:old, Definition def:terms), and the split/nonsplit distinction in Lemmas lem:tamagawa-exp and lem:tamfactor. The injectivity of reduction on prime-to-p torsion (Corollary 10.17) is the easy half of the criterion of Néron, Ogg and Shafarevich (Chapter 11) and the input of Proposition 10.33.

Definition 10.9 (reduction types)**.** Let $y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$ be a minimal model of E/K and let $\tilde{E}$ be the plane cubic over k defined by the reduced coefficients $\tilde{a}_i$. The curve E has

- *good reduction* if $\tilde{E}$ is smooth, that is, $v(\Delta) = 0$;
- *multiplicative reduction* if $\tilde{E}$ has a node, that is, $v(\Delta) > 0$ and $v(c_4) = 0$;
- *additive reduction* if $\tilde{E}$ has a cusp, that is, $v(\Delta) > 0$ and $v(c_4) > 0$.

Multiplicative reduction is *split* if the two tangent directions at the node are defined over k, and *nonsplit* otherwise. *Semistable* means good or multiplicative. For $E/\mathbb{Q}$ and a prime p we say that E has good (multiplicative, additive) reduction at p if $E/\mathbb{Q}_p$ has.

The definition does not depend on the minimal model: two minimal models differ by (u, r, s, t) with $u \in \mathcal{O}^\times$ and $r, s, t \in \mathcal{O}$ (Proposition 10.4), which reduces to a change of coordinates over k. A Weierstrass cubic is smooth if and only if its discriminant is nonzero, and if $\tilde{\Delta} = 0$ it has exactly one singular point, a node if $\tilde{c}_4 \neq 0$ and a cusp if $\tilde{c}_4 = 0$ (Chapter 9). The singular point is k-rational: it is unique, hence fixed by $\mathrm{Gal}(\bar{k}/k)$, and k is perfect.

Proposition 10.10 (the group of nonsingular points)**.** *Let $\tilde{E}/k$ be a Weierstrass cubic with a singular point S, and let $\tilde{E}_{\mathrm{ns}}(k)$ be the set of k-points other than S. The chord-tangent construction makes $\tilde{E}_{\mathrm{ns}}(k)$ an abelian group with neutral element $\tilde{O}$, and*

(i) if S is a cusp, $\tilde{E}_{\mathrm{ns}}(k) \simeq k^+$;

(ii) if S is a split node, $\tilde{E}_{\mathrm{ns}}(k) \simeq k^\times$;

(iii) if S is a nonsplit node and l/k is the quadratic extension containing the tangent slopes, $\tilde{E}_{\mathrm{ns}}(k) \simeq \ker(\mathrm{N}_{l/k} \colon l^\times \to k^\times)$.

For $k = \mathbb{F}_p$ the orders are p, $p-1$ and $p+1$.

Proof. A line through two nonsingular points does not pass through S: a line through S meets the cubic there with multiplicity at least 2, hence in at most one further point. So the chord-tangent law, defined exactly as for a smooth cubic (Chapter 9), stays inside $\tilde{E}_{\mathrm{ns}}$; associativity can be checked as in the smooth case, and also follows from the isomorphisms below, which we now construct.

(i) Translate S to $(0,0)$; the cusp has a single tangent direction, which a further change of coordinates makes the line $y = 0$, and the curve becomes $Y^2Z = X^3$ in projective coordinates. Its points other than $S = [0:0:1]$ have $Y \neq 0$ and are $[t:1:t^3]$, $t \in k$, with $\tilde{O} = [0:1:0]$ at $t = 0$. A line $\alpha X + \beta Y + \gamma Z = 0$ not through S has $\gamma \neq 0$ and meets the curve at the points with $\gamma t^3 + \alpha t + \beta = 0$, counted with multiplicity. The coefficient of t^2 is 0, so the three values of t add up to 0. Three points add up to $\tilde{O}$ in the group law exactly when they are the intersection of a line with the curve, so $P \mapsto t(P)$ is a homomorphism $\tilde{E}_{\mathrm{ns}}(k) \to k^+$, and it is bijective.

(ii) Translate S to $(0,0)$. The equation becomes $y^2 + a_1xy - a_2x^2 = x^3$ (with $\tilde{a}_3 = \tilde{a}_4 = \tilde{a}_6 = 0$), and the quadratic part factors as $(y - \alpha x)(y - \beta x)$ with $\alpha \neq \beta$ the roots of $T^2 + a_1T - a_2$; split means $\alpha, \beta \in k$. A nonsingular point $P \neq \tilde{O}$ lies on the line $y = \lambda x$ through S with $\lambda = y/x$, and then $x = (\lambda - \alpha)(\lambda - \beta)$, $y = \lambda x$; so $P \mapsto \lambda$ is a bijection from $\tilde{E}_{\mathrm{ns}}(k) \setminus \{\tilde{O}\}$ to $k \setminus \{\alpha, \beta\}$, and $\tilde{O}$ corresponds to $\lambda = \infty$. Put $\phi(P) = (\lambda - \alpha)/(\lambda - \beta)$ and $\phi(\tilde{O}) = 1$; this is a bijection onto $k^\times$. A line $y = mx + c_0$ with $c_0 \neq 0$ meets the curve at the points with $\lambda x = mx + c_0$, that is, at the roots of $(\lambda - \alpha)(\lambda - \beta)(\lambda - m) - c_0 = \prod_{i=1}^{3}(\lambda - \lambda_i)$. Evaluating at $\lambda = \alpha$ and at $\lambda = \beta$ gives $\prod(\alpha - \lambda_i) = -c_0 = \prod(\beta - \lambda_i)$, so $\prod_i \phi(P_i) = 1$. A vertical line $x = x_0$ meets the curve at $\tilde{O}$ and at the two roots of $(\lambda - \alpha)(\lambda - \beta) - x_0$, and the same evaluation gives $\phi(P_1)\phi(P_2) = 1 = \phi(\tilde{O})^{-1}$. So ϕ is a homomorphism to $k^\times$.

(iii) Apply (ii) over l, where α, β are conjugate under the generator σ of $\mathrm{Gal}(l/k)$. For a point P with slope $\lambda \in l$, the conjugate point has slope $\sigma(\lambda)$, and $\phi(\sigma P) = (\sigma\lambda - \alpha)/(\sigma\lambda - \beta) = \sigma((\lambda - \beta)/(\lambda - \alpha)) = \sigma(\phi(P))^{-1}$. So P is k-rational if and only if $\phi(P)\sigma(\phi(P)) = 1$, that is, $\mathrm{N}_{l/k}\,\phi(P) = 1$.

For $k=\mathbb{F}_p$: $\#\mathbb{F}_p=p$, $\#\mathbb{F}_p^\times=p-1$, and the norm $\mathbb{F}_{p^2}^\times\to\mathbb{F}_p^\times$ is surjective, so its kernel has $(p^2-1)/(p-1)=p+1$ elements. □

Corollary 10.11 (a_p at a bad prime)**.** *Let $E/\mathbb{Q}$ have bad reduction at p and put $a_p=p+1-\#\tilde{E}(\mathbb{F}_p)$, the singular point included in the count. Then $a_p=1,\ -1,\ 0$ according as the reduction is split multiplicative, nonsplit multiplicative or additive.*

Proof. $\#\tilde{E}(\mathbb{F}_p)=\#\tilde{E}_{\mathrm{ns}}(\mathbb{F}_p)+1$, and Proposition 10.10 gives p, $p-1$, $p+1$ for the three cases. □

Proposition 10.12 (split and nonsplit)**.** *Let E/K have multiplicative reduction.*

(i) After a change $(1,r,0,t)$ with $r,t\in\mathcal{O}$ moving the node to $(0,0)$, the reduction is split if and only if $T^2+a_1T-a_2$ has a root in k.

(ii) If p is odd, the reduction is split if and only if $-c_6$ is a square in $k^\times$ for one (equivalently every) minimal model; equivalently, if and only if $-c_4/c_6$ is a square in $k^\times$.

(iii) If p is odd and k is finite, the twist of E by the unramified quadratic extension of K exchanges split and nonsplit reduction, and over the unramified quadratic extension of K the reduction is split.

Proof. (i) is the definition, by the factorisation of the quadratic part in the proof of Proposition 10.10(ii). (ii) After moving the node to $(0,0)$, $a_3,a_4,a_6\in\mathfrak{m}$, so $b_4\equiv0$ and $b_6\equiv0$, and $c_4=b_2^2-24b_4\equiv b_2^2$, $c_6=-b_2^3+36b_2b_4-216b_6\equiv-b_2^3$. The discriminant of $T^2+a_1T-a_2$ is $a_1^2+4a_2=b_2$, which is nonzero mod $\mathfrak{m}$ because c_4 is a unit. So split means $\tilde{b}_2$ is a square, and $-\tilde{c}_6=\tilde{b}_2\cdot\tilde{b}_2^2$ is a square if and only if $\tilde{b}_2$ is, as is $-\tilde{c}_4/\tilde{c}_6=\tilde{b}_2^{-1}$. Another minimal model changes c_6 by u^6 and c_4/c_6 by u^{-2}, with u a unit, which does not change the square classes. (iii) Since k is finite, $k^\times/(k^\times)^2$ has order 2, so K has a unique unramified quadratic extension $K'=K(\sqrt{d})$, with d a unit whose reduction is not a square. For p odd we may take a model $y^2=x^3+a_2x^2+a_4x+a_6$; its twist by d is $y^2=x^3+da_2x^2+d^2a_4x+d^3a_6$ (Chapter 9), which multiplies c_6 by d^3. As $\tilde{d}$ is not a square, this changes the square class of $-\tilde{c}_6$, and the twisted model is still minimal ($v(\Delta)$ is unchanged). Over the unramified quadratic extension K' every element of $k^\times$ becomes a square in the residue field k', so the reduction is split there. □

We checked (ii) on all 127,592 pairs (curve, odd prime of multiplicative reduction) of Cremona's tables below conductor 10,000: in every case $a_p=\left(\frac{-c_6}{p}\right)$.

In the paper 10.13. Lemma lem:tamagawa-exp ends with "This holds for split and for nonsplit reduction", and its proof passes to "the unramified quadratic extension of $\mathbb{Q}_q$ (which does not change the inertia group)", over which E is split by Proposition 10.12(iii) and a Tate curve by Theorem 10.59. In Lemma lem:tamfactor the character ε is unramified of order at most 2 with $\varepsilon(\mathrm{Frob}_q)=a_q$, and in Lemma lem:old $\varepsilon(\mathrm{Frob}_p)=a_p(f)=\pm1$ by Corollary 10.11.

Definition 10.14 (the reduction filtration)**.** For a minimal model of E/K and $P\in E(K)$, write $P=[X:Y:Z]$ with $X,Y,Z\in\mathcal{O}$ not all in $\mathfrak{m}$, and let $\tilde{P}=[\tilde{X}:\tilde{Y}:\tilde{Z}]\in\tilde{E}(k)$ be its reduction. Put

$$E_0(K)=\{P\in E(K):\tilde{P}\in\tilde{E}_{\mathrm{ns}}(k)\},\qquad E_1(K)=\{P\in E(K):\tilde{P}=\tilde{O}\}.$$

Proposition 10.15 (the reduction map)**.** *$E_0(K)$ is a subgroup of $E(K)$, $E_1(K)$ is a subgroup of $E_0(K)$, and reduction is a surjective homomorphism $E_0(K)\to\tilde{E}_{\mathrm{ns}}(k)$ with kernel $E_1(K)$:*

$$0\longrightarrow E_1(K)\longrightarrow E_0(K)\longrightarrow\tilde{E}_{\mathrm{ns}}(k)\longrightarrow0.$$

If E has good reduction, $E_0(K)=E(K)$.

Proof. Reduction is well defined, since the primitive coordinates are unique up to $\mathcal{O}^\times$. Clearly $\tilde{O} = [0:1:0]$ is nonsingular, and $-P = (x, -y - a_1x - a_3)$ reduces to $-\tilde{P}$. Let $P_1, P_2 \in E_0(K)$ and let P_3 be the third intersection of the line L through them (the tangent if $P_1 = P_2$) with E. Write L as $\alpha X + \beta Y + \gamma Z = 0$ with $\alpha, \beta, \gamma \in \mathcal{O}$ not all in $\mathfrak{m}$; its reduction $\tilde{L}$ is a line. Parametrising L linearly and substituting into the cubic gives a cubic form in two variables with coefficients in $\mathcal{O}$ whose three roots are P_1, P_2, P_3; reducing it shows that $\tilde{L}$ meets $\tilde{E}$ exactly in $\tilde{P}_1, \tilde{P}_2, \tilde{P}_3$, counted with multiplicity (the reduced form is not identically zero, as $\tilde{E}$ is irreducible and does not contain the line $\tilde{L}$). If $\tilde{P}_3$ were the singular point, $\tilde{L}$ would meet $\tilde{E}$ with multiplicity at least 2 there and also at the nonsingular points $\tilde{P}_1, \tilde{P}_2$, four intersections in all, which is impossible. So $\tilde{P}_3$ is nonsingular and $\tilde{P}_1 + \tilde{P}_2 + \tilde{P}_3 = \tilde{O}$ in $\tilde{E}_{\mathrm{ns}}(k)$. Hence $E_0(K)$ is closed under addition and negation, and reduction is a homomorphism on it; its kernel is $E_1(K)$ by definition.

Surjectivity: let $(\tilde{x}_0, \tilde{y}_0)$ be a nonsingular affine point of $\tilde{E}(k)$ (the point $\tilde{O}$ is the image of O). With $F(x,y) = y^2 + a_1xy + a_3y - x^3 - a_2x^2 - a_4x - a_6$, one of $\partial F/\partial x$, $\partial F/\partial y$ is nonzero at the point. If $\partial F/\partial y(\tilde{x}_0, \tilde{y}_0) \neq 0$, choose any lift $x_0 \in \mathcal{O}$ of $\tilde{x}_0$; the polynomial $F(x_0, y)$ has the simple root $\tilde{y}_0$ modulo $\mathfrak{m}$, and Hensel's lemma (Chapter 7) lifts it to $y_0 \in \mathcal{O}$ with $F(x_0, y_0) = 0$. The other case is the same with the roles of x and y exchanged. The point $(x_0, y_0) \in E(K)$ reduces to $(\tilde{x}_0, \tilde{y}_0)$. □

Theorem 10.16 (the formal group)**.** *Let a minimal model of E/K be given.*

(i) In the parameter $z = -x/y$ near O, the curve is described by a power series $w(z) = -1/y = z^3 + a_1zw + \cdots$ with coefficients in $\mathbb{Z}[a_1, \ldots, a_6]$, and the group law by a power series $F(z_1, z_2) \in \mathbb{Z}[a_1, \ldots, a_6][[z_1, z_2]]$; F is a one-parameter commutative formal group $\hat{E}$, and $P \mapsto z(P)$ is an isomorphism $E_1(K) \simeq \hat{E}(\mathfrak{m})$, the set $\mathfrak{m}$ with the group law F.

(ii) For $r > v(p)/(p-1)$ the formal logarithm gives an isomorphism $\hat{E}(\mathfrak{m}^r) \simeq \mathfrak{m}^r$ with the additive group.

(iii) For every integer m prime to p, multiplication by m is a bijection of $E_1(K)$. In particular $E_1(K)$ has no nonzero torsion of order prime to p, and it is uniquely m-divisible.

Sketch of proof. (i) and (ii) are [23, IV.1 to IV.6 and VII.2]: one solves the equation of the curve for w as a power series in z by iteration, computes the third intersection of a line with the curve in these coordinates, and checks that the resulting series converge on $\mathfrak{m}$ and match the reduction filtration. (iii) is a complete argument given (i): the series $[m](z)$ of multiplication by m on $\hat{E}$ is $mz +$ (terms of degree ≥ 2), with $m \in \mathcal{O}^\times$. A power series $f(z) = mz + \cdots$ with coefficients in $\mathcal{O}$ and m a unit has a compositional inverse $g(z) = m^{-1}z + \cdots$ with coefficients in $\mathcal{O}$ (solve $g(f(z)) = z$ coefficient by coefficient, dividing only by powers of m). So $[m]$ is a bijection of $\hat{E}(\mathfrak{m})$, hence of $E_1(K)$. □

Corollary 10.17 (torsion and reduction)**.** *Let m be prime to p.*

(i) Reduction $E_0(K)[m] \to \tilde{E}_{\mathrm{ns}}(k)$ is injective.

(ii) If E has good reduction, $E(K)[m] \hookrightarrow \tilde{E}(k)$; the inertia group I_K acts trivially on $E[m]$, so $E[m] \subset E(K^{\mathrm{ur}})$; and reduction is an isomorphism $E[m] \to \tilde{E}[m](\bar{k})$ compatible with the actions of G_K and $\mathrm{Gal}(\bar{k}/k)$. Consequently I_K acts trivially on $T_\ell E$ for every prime $\ell \neq p$.

Proof. (i) The kernel is $E_1(K)[m]$, which is 0 by Theorem 10.16(iii). (ii) The first statement is (i) with $E_0(K) = E(K)$. Let $L = K(E[m])$, a finite extension; the given model has unit discriminant

over L too, so E/L has good reduction and (i) over L shows that $E(L)[m] = E[m]$ injects into $\tilde{E}(k_L)$. Let $\sigma \in I_K$ and $P \in E[m]$. Since σ acts trivially on k_L, the point $\sigma P - P \in E[m]$ reduces to $\widetilde{\sigma P} - \tilde{P} = \tilde{\sigma}\tilde{P} - \tilde{P} = \tilde{O}$, so $\sigma P = P$. Hence $E[m] \subset E(K^{\mathrm{ur}})$. Reduction from $E[m]$ to $\tilde{E}[m](\bar{k})$ is injective, and both groups have m^2 elements (m is prime to the characteristics; Chapter 9), so it is bijective. It commutes with the Galois actions because reduction commutes with automorphisms of K^{ur}. Passing to the limit over $m = \ell^n$ gives the statement for $T_\ell E$. □

Proposition 10.18 (the index of E_0)**.** *$E(K)/E_0(K)$ is finite when k is finite; it is trivial when the reduction is good, and its order (the Tamagawa number) is computed in Proposition 10.28, Theorem 10.36 and Proposition 10.65.*

Proof. For good reduction $E_0(K) = E(K)$ by definition. In general $E_0(K)$ is open in the compact group $E(K)$ (reduction is locally constant on $E(K)$: two points whose primitive coordinates are congruent modulo $\mathfrak{m}$ have the same reduction), so its cosets form an open cover of the compact group $E(K)$ (k is finite, so K is locally compact), and the index is finite. The finer statement that $E(K)/E_0(K)$ is the group of k-points of the component group of the Néron model is Proposition 10.28; see [23, VII.6] and [22, IV.9]. □

In the paper 10.19. Section 1.1 of the paper: $\delta_p = \dim V^{I_p}$ is 2 at a good prime because I_p acts trivially on $T_\ell E$ (Corollary 10.17(ii)); the values 1 and 0 at multiplicative and additive primes come from Proposition 10.62 and Proposition 10.46. Section 1.4: the Tamagawa number is $c_q = [E(\mathbb{Q}_q) : E_0(\mathbb{Q}_q)]$.

10.3 Potentially good and potentially multiplicative reduction

The reduction type can change under finite extensions of K, and the paper's local terms depend on which reduction a curve acquires: the term $t_\ell(E, q)$ of Definition def:terms lives at the potentially multiplicative primes, and the term $e_\ell(E, q)$ at a tame additive prime is read off the potentially good types. This section proves the criterion by $v(j)$ and states the semistable reduction theorem.

Definition 10.20 (potential reduction)**.** E/K has *potentially good* (resp. *potentially multiplicative*) reduction if there is a finite extension L/K such that E/L has good (resp. multiplicative) reduction.

Theorem 10.21 (the j-invariant criterion)**.** *Let E/K be an elliptic curve.*

(i) E has potentially good reduction if and only if $v(j) \geq 0$.

(ii) E has potentially multiplicative reduction if and only if $v(j) < 0$; in that case E has multiplicative reduction over every finite extension over which it is semistable, and never good reduction.

(iii) If E has multiplicative reduction, $v(j) = -v(\Delta_{\min}) < 0$.

Sketch of proof. (iii) For a minimal model c_4 is a unit, so $v(j) = 3v(c_4) - v(\Delta) = -v(\Delta) < 0$.

(i), the direction "⇒": if E/L has good reduction, a minimal model over L has Δ a unit and c_4 integral, so $j = c_4^3/\Delta$ is integral in L, hence $v(j) \geq 0$.

(i), the direction "⇐", for p odd: over the finite extension $L = K(E[2])$ the curve has a model $y^2 = (x-e_1)(x-e_2)(x-e_3)$, and after a further finite extension and a change of coordinates a Legendre model $y^2 = x(x-1)(x-\lambda)$, for which

$$j = 2^8 \frac{(\lambda^2-\lambda+1)^3}{\lambda^2(\lambda-1)^2}, \qquad \Delta = 16\lambda^2(\lambda-1)^2.$$

If $v(\lambda) < 0$ the numerator has valuation $6v(\lambda)$ and the denominator $4v(\lambda)$, so $v(j) = 2v(\lambda) < 0$. If $v(\lambda) > 0$ then $v(j) = -2v(\lambda) < 0$, and likewise if $v(\lambda-1) > 0$. So $v(j) \geq 0$ forces λ and $\lambda - 1$ to be units, and then the Legendre model has unit discriminant (p odd): good reduction over the extension. For $p = 2$ one uses a different cubic normal form; see [23, VII.5.5].

(ii) If $v(j) < 0$, E never has good reduction, by (i). By the semistable reduction theorem (Theorem 10.22) it is semistable over some finite extension, hence multiplicative there, and over every larger extension. Conversely multiplicative reduction over L gives $v_L(j) < 0$ by (iii), hence $v(j) < 0$. □

Theorem 10.22 (semistable reduction)**.** *For every elliptic curve E/K there is a finite extension L/K such that E/L is semistable. One may take $L = K(E[m])$ for any integer $m \geq 3$ prime to p.*

Sketch of proof. If $v(j) \geq 0$, Theorem 10.21(i) gives good reduction over a finite extension. If $v(j) < 0$, E becomes isomorphic to a Tate curve, which has split multiplicative reduction, over an extension of degree at most 2 (Theorem 10.59). The statement about $K(E[m])$ is the theorem of Serre and Tate [21, Section 2]; see also [23, VII.5] and Theorem 10.51 below. □

Remark 10.23 (the four classes). Combining Definition 10.9, Theorem 10.21 and the classification of Section 10.5, the Kodaira symbols fall into four classes:

- good reduction: I_0;
- multiplicative reduction: I_n, $n \geq 1$, with $n = v(\Delta_{\min}) = -v(j)$;
- additive, potentially multiplicative reduction: I_n^* with $n \geq 1$ (Theorem 10.66);
- additive, potentially good reduction: II, III, IV, I_0^*, IV*, III*, II*, and, at $p = 2$ only, also some symbols I_m^* with $m \geq 1$.

Until Section 10.5 the symbols are used only as names for these classes.

Warning 10.24. "Potentially multiplicative" does not mean "multiplicative": a curve of type I_n^* ($n \geq 1$) at an odd prime has additive reduction ($f_p \geq 2$, $a_p = 0$, $\delta_p = 0$), but $v(j) < 0$; at $p = 2$ a symbol I_m^* may also be potentially good (Remark 10.81). The paper's $n_q = v_q(\Delta_E)$ is defined at multiplicative primes; at an additive potentially multiplicative prime the relevant integer is $\nu = -v_q(j_E)$, which for odd q is the index n of the symbol I_n^* (Theorem 10.66) but for $q = 2$ is not.

In the paper 10.25. Definition def:terms: $t_\ell(E,q) = v_\ell(-v_q(j_E))$ if $v_q(j_E) < 0$ and 0 otherwise, "the ℓ-part of the valuation of the Tate parameter at a potentially multiplicative prime". Lemma lem:euler: "the inertia group acts on V through a finite cyclic quotient when the reduction is potentially good, and through a quadratic character times a unipotent when it is potentially multiplicative" (Theorem 10.51, Theorem 10.59). Remark rem:wild: a symbol I_n^* at 2 can be potentially good (Remark 10.81).

10.4 Néron models and component groups

The paper's inequalities (eq:main) and (eq:twist) are stated in terms of the geometric component group Φ_q^{geom} of the Néron model of E at q, not the Tamagawa number c_q. This section defines both, explains why they differ, and proves the fact that links component groups to Galois representations: the inertia invariants of $E[\ell^t]$ at an additive prime are the ℓ^t-torsion of Φ_q^{geom} (Proposition 10.33). That proposition is the decisive input of Lemma lem:IV and Remark rem:types (via Chapter 11), and of the treatment of the types IV, IV* at 2 in Corollary 10.54. We use scheme-theoretic language only as far as the statements need it.

A minimal Weierstrass model gives a scheme over $\mathcal{O}$ whose generic fibre is E and whose special fibre is the cubic $\tilde{E}$. When the reduction is bad this model has two defects: its special fibre is singular, and its points over $\mathcal{O}$ do not see the group structure of $E(K)$ well (only $E_0(K)$ reduces into the group $\tilde{E}_{\mathrm{ns}}(k)$). Néron's idea is to look for the best smooth model: one that is a group scheme and whose $\mathcal{O}$-points are all of $E(K)$. Its special fibre is a smooth commutative algebraic group over k, usually disconnected; the identity component is $\tilde{E}_{\mathrm{ns}}$, and the group of components is a finite abelian group with an action of $\mathrm{Gal}(\bar{k}/k)$.

Definition 10.26 (Néron model)**.** Let $R = \mathcal{O}$. A *Néron model* of E/K is a smooth separated group scheme $\mathcal{E}$ of finite type over R with generic fibre $\mathcal{E}_K = E$, such that for every smooth R-scheme S every K-morphism $S_K \to E$ extends uniquely to an R-morphism $S \to \mathcal{E}$ (the *Néron mapping property*).

Theorem 10.27 (Néron)**.** *Every elliptic curve E/K has a Néron model $\mathcal{E}$, unique up to unique isomorphism. It is the smooth locus of the minimal regular proper model of E over R; if E has good reduction it is the minimal Weierstrass model.*

We do not prove this; see Néron [13], [2, Sections 1.2, 1.4] and [22, IV.5, IV.6].

Proposition 10.28 (the component group)**.** *Let $\mathcal{E}$ be the Néron model of E/K, $\mathcal{E}_k$ its special fibre and $\mathcal{E}_k^0$ the connected component of the identity.*

(i) $\mathcal{E}(R) = E(K)$. For a minimal Weierstrass model, $\mathcal{E}_k^0 \simeq \tilde{E}_{\mathrm{ns}}$ and $\mathcal{E}^0(R) = E_0(K)$, where $\mathcal{E}^0$ is the open subgroup scheme with special fibre $\mathcal{E}_k^0$. The quotient $\mathcal{E}_k/\mathcal{E}_k^0$ is a finite étale group scheme over k, the component group scheme *of E; we write its groups of points with an argument, $\Phi_E(k)$ and $\Phi_E(\bar{k})$, never as a bare letter. Concretely it is a finite abelian group $\Phi_E(\bar{k})$ with a continuous action of $\mathrm{Gal}(\bar{k}/k)$, and $\Phi_E(k) = \Phi_E(\bar{k})^{\mathrm{Gal}(\bar{k}/k)}$.*

(ii) If k is finite, $E(K)/E_0(K) \simeq \Phi_E(k)$.

Sketch of proof. (i) is [22, IV.5, IV.9] and [2, Section 7.1]; the equality $\mathcal{E}(R) = E(K)$ is the mapping property applied to $S = \operatorname{Spec} R$. (ii) Since $\mathcal{E}$ is smooth, Hensel's lemma shows that reduction $\mathcal{E}(R) \to \mathcal{E}_k(k)$ is surjective. Since k is finite and $\mathcal{E}_k^0$ is connected, Lang's theorem applies: a connected algebraic group G over a finite field k has a k-point on every principal homogeneous space (a variety over k on which G acts simply transitively over $\bar{k}$); in the language of Galois cohomology (Chapter 16), $H^1(k, G) = 0$. A component defined over k, that is a point of $\Phi_E(k)$, is such a space for $G = \mathcal{E}_k^0$, so it has a k-point, and $\mathcal{E}_k(k) \to \Phi_E(k)$ is surjective. The kernel of the composite $E(K) = \mathcal{E}(R) \to \Phi_E(k)$ is the set of points reducing into the identity component, $\mathcal{E}^0(R) = E_0(K)$. See [22, IV.9]. □

Definition 10.29 (geometric component group and Tamagawa number)**.** Let $E/\mathbb{Q}$ and let p be a prime. Write $\Phi_p(\mathbb{F}_p)$ and $\Phi_p(\bar{\mathbb{F}}_p)$ for the groups of points of the component group scheme over $\mathbb{F}_p$ of the Néron model of E over $\mathbb{Z}_p$ (Proposition 10.28). The *geometric component group* is $\Phi_p^{\mathrm{geom}} = \Phi_p(\bar{\mathbb{F}}_p)$, and the *Tamagawa number* is

$$c_p = \#\Phi_p(\mathbb{F}_p) = [E(\mathbb{Q}_p) : E_0(\mathbb{Q}_p)].$$

At a good prime $\Phi_p^{\mathrm{geom}} = 0$ and $c_p = 1$. The product $\prod_p c_p$ is the Tamagawa factor in the conjecture of Birch and Swinnerton-Dyer; Cremona's file `allbsd` lists it (Chapter 22).

Convention 10.30 (Φ^{geom} versus c)**.** The component group is always written with an argument: $\Phi_p(\mathbb{F}_p)$ or $\Phi_p^{\mathrm{geom}} = \Phi_p(\bar{\mathbb{F}}_p)$ (over a general local field, $\Phi_E(k)$ or $\Phi^{\mathrm{geom}} = \Phi_E(\bar{k})$); a bare Φ is the geometric Frobenius (Appendix B). Since $\Phi_p(\mathbb{F}_p)$ is the subgroup of $\mathrm{Gal}(\bar{\mathbb{F}}_p/\mathbb{F}_p)$-invariants of Φ_p^{geom}, c_p divides $\#\Phi_p^{\mathrm{geom}}$, with equality if and only if Frobenius acts trivially on Φ_p^{geom}. The paper's statements are about $\#\Phi_q^{\mathrm{geom}}$, never about c_q: for instance `121c1` and `121c2` at 11 and `980i1` at 2 have $\#\Phi^{\mathrm{geom}} = 3$ but $c = 1$ (Example 10.85). We never write c_p for $\#\Phi_p^{\mathrm{geom}}$, and we never use a bare c for the class of the unit part of the Tate parameter, which is $c(u)$ (Definition 10.60).

Proposition 10.31 (unramified base change)**.** *The formation of the Néron model commutes with étale base change; in particular with base change from K to a finite unramified extension K', or to the completion of K^{ur}. Consequently the reduction type, the Kodaira symbol, the group Φ^{geom} with its structure, $v(\Delta_{\min})$ and the conductor exponent are unchanged under unramified extension of K, and a minimal model over K stays minimal over K'. The Tamagawa number can change, since only the invariants of $\mathrm{Gal}(\bar{k}/k')$ on Φ^{geom} are counted.*

We do not prove this; see [2, Section 7.2] and [22, IV.6]. (For the conductor exponent the invariance is also immediate from Definition 10.45, since $I_{K'} = I_K$.)

Remark 10.32 (Tamagawa numbers at multiplicative primes)*.* At a prime of type I_n one has $\Phi^{\mathrm{geom}} \simeq \mathbb{Z}/n$, and $c = n$ for split, $c \in \{1, 2\}$ (according to the parity of n) for nonsplit reduction (Proposition 10.65). So for ℓ odd the ℓ-part of c is $\ell^{v_\ell(n)}$ in the split case and 1 in the nonsplit case, while the ℓ-part of $\#\Phi^{\mathrm{geom}}$ is $\ell^{v_\ell(n)}$ in both. This is why the paper's inequality (eq:main) uses Φ_q^{geom}, and why the history paragraph of its Section 1.5 (Ribet and Takahashi) speaks of the n_q.

The facts $\Phi^{\mathrm{geom}} \simeq \mathbb{Z}/n$ and the values of c at I_n are proved in Section 10.7 (Proposition 10.65), from the Tate curve. That proof uses only the uniformisation of Theorem 10.59 and Proposition 10.31, not the proposition below, so the forward references to it in this remark and in the proof of Proposition 10.33(ii) are not circular.

Proposition 10.33 (inertia invariants and component groups)**.** *Let $K/\mathbb{Q}_q$ be finite, with residue field k, let $\ell \neq q$ be a prime, $t \geq 1$ an integer, and let E/K have bad reduction, with geometric component group $\Phi^{\mathrm{geom}} = \Phi_E(\bar{k})$.*

(i) If the reduction is additive, there is a $\mathrm{Gal}(\bar{k}/k)$-equivariant isomorphism

$$E[\ell^t]^{I_K} = E(K^{\mathrm{ur}})[\ell^t] \;\simeq\; \Phi^{\mathrm{geom}}[\ell^t].$$

In particular $E(K)[\ell^\infty] \simeq \Phi_E(k)[\ell^\infty]$.

(ii) If the reduction is multiplicative, there is an exact sequence of $\mathrm{Gal}(\bar{k}/k)$*-modules*

$$0 \longrightarrow \mu_{\ell^t} \otimes \varepsilon \longrightarrow E[\ell^t]^{I_K} \longrightarrow \Phi^{\mathrm{geom}}[\ell^t] \longrightarrow 0,$$

where $\Phi^{\mathrm{geom}} \simeq \mathbb{Z}/n$ *(*$n = v(\Delta_{\min})$*) and* ε *is the unramified character of order at most* 2 *which is trivial if and only if the reduction is split.*

(iii) Hence, for ℓ *odd,* $\dim_{\mathbb{F}_\ell} E[\ell]^{I_K}$ *is* $1 + [\ell \mid n]$ *for type* I_n *(*$n \geq 1$*),* 1 *for types* IV *and* IV* *when* $\ell = 3$*, and* 0 *for every other additive type.*

Proof. An element of $E[\ell^t] \subset E(\bar{K})$ is fixed by I_K if and only if its coordinates lie in the fixed field $\bar{K}^{I_K} = K^{\mathrm{ur}}$; this is the first equality in (i). Let K' run over the finite unramified extensions of K, with residue fields k'. By Proposition 10.31 a minimal model over K stays minimal over each K', the component group scheme of E/K' is the base change of that of E/K (so $\Phi_E(k')$ is the group of k'-points of the same scheme and Φ^{geom} is unchanged), and by Proposition 10.15 and Proposition 10.28(ii) there are exact sequences

$$0 \to E_1(K') \to E_0(K') \to \tilde{E}_{\mathrm{ns}}(k') \to 0, \qquad 0 \to E_0(K') \to E(K') \to \Phi_E(k') \to 0,$$

compatible with the inclusions $K' \subset K''$ and with the action of $\mathrm{Gal}(K^{\mathrm{ur}}/K) = \mathrm{Gal}(\bar{k}/k)$. Taking the union over K' (a direct limit of exact sequences is exact) gives

$$0 \to E_1^{\mathrm{ur}} \to E_0^{\mathrm{ur}} \to \tilde{E}_{\mathrm{ns}}(\bar{k}) \to 0, \qquad 0 \to E_0^{\mathrm{ur}} \to E(K^{\mathrm{ur}}) \to \Phi^{\mathrm{geom}} \to 0,$$

with $E_i^{\mathrm{ur}} = \bigcup_{K'} E_i(K')$. Now:

- E_1^{ur} is uniquely ℓ-divisible, by Theorem 10.16(iii) applied over each K'.
- In the additive case $\tilde{E}_{\mathrm{ns}}(\bar{k}) \simeq \bar{k}^+$ (Proposition 10.10), which is uniquely ℓ-divisible as $\ell \neq q$. In the multiplicative case $\tilde{E}_{\mathrm{ns}}(\bar{k}) \simeq \bar{k}^\times$, which is ℓ-divisible ($\bar{k}$ is algebraically closed) with ℓ^t-torsion $\mu_{\ell^t}(\bar{k}) \simeq \mathbb{Z}/\ell^t$. The Galois action is the natural one if the reduction is split, and is twisted by the nontrivial unramified quadratic character ε if it is nonsplit, by the formula $\phi(\sigma P) = \sigma(\phi(P))^{-1}$ in the proof of Proposition 10.10(iii); reduction identifies $\mu_{\ell^t}(\bar{k})$ with $\mu_{\ell^t} \subset K^{\mathrm{ur}}$.

Apply the snake lemma (Chapter 3) to multiplication by ℓ^t on the first sequence. Since E_1^{ur} has no ℓ^t-torsion and $E_1^{\mathrm{ur}}/\ell^t = 0$, we get $E_0^{\mathrm{ur}}[\ell^t] \simeq \tilde{E}_{\mathrm{ns}}(\bar{k})[\ell^t]$ and $E_0^{\mathrm{ur}}/\ell^t E_0^{\mathrm{ur}} \simeq \tilde{E}_{\mathrm{ns}}(\bar{k})/\ell^t = 0$. Then the snake lemma for ℓ^t on the second sequence gives

$$0 \to E_0^{\mathrm{ur}}[\ell^t] \to E(K^{\mathrm{ur}})[\ell^t] \to \Phi^{\mathrm{geom}}[\ell^t] \to E_0^{\mathrm{ur}}/\ell^t E_0^{\mathrm{ur}} = 0.$$

In the additive case $E_0^{\mathrm{ur}}[\ell^t] = 0$, which is (i); in the multiplicative case it is $\mu_{\ell^t} \otimes \varepsilon$, which is (ii), with $\Phi^{\mathrm{geom}} \simeq \mathbb{Z}/n$ by Proposition 10.65. Every map is $\mathrm{Gal}(\bar{k}/k)$-equivariant. Taking $\mathrm{Gal}(\bar{k}/k)$-invariants in (i) gives $E(K)[\ell^t] = (E[\ell^t]^{I_K})^{G_K/I_K} \simeq \Phi^{\mathrm{geom}}[\ell^t]^{\mathrm{Gal}(\bar{k}/k)} = \Phi_E(k)[\ell^t]$.

(iii) In case (ii), $\#E[\ell]^{I_K} = \ell \cdot \#(\mathbb{Z}/n)[\ell]$, which is ℓ^2 if $\ell \mid n$ and ℓ otherwise. In case (i), Φ^{geom} is one of 0, $\mathbb{Z}/2$, $\mathbb{Z}/3$, $(\mathbb{Z}/2)^2$, $\mathbb{Z}/4$ (Theorem 10.36), and for ℓ odd its ℓ-torsion is nonzero only when $\ell = 3$ and $\Phi^{\mathrm{geom}} \simeq \mathbb{Z}/3$, that is, for the types IV and IV*, where it is a line. □

In the paper 10.34. The paragraph before Lemma lem:IV in the revised paper states (i) with the proof above ("the snake lemma for multiplication by ℓ^k gives the isomorphism", citing [21, Section 1]). Lemma lem:IV: "$E[3]^{I_q}$ is a line" is (iii) for IV, IV*. Remark rem:types: "$E[3]^{I_q} = 0$" for

the other additive types is (iii). Lemma lem:tamfactor, the final check with $M = T$: the paper's module $(\mathbb{Z}/\ell^{v_\ell(n)})\bar{e}_1$ (the torsion of $H^1(I,T)$, generated by the class of e_1), with Frobenius acting by ε, is isomorphic as a Galois module to $\Phi^{\mathrm{geom}}[\ell^\infty]$, the quotient in (ii) generated by the image of e_2 (Frobenius acts on $\Phi^{\mathrm{geom}} \simeq \mathbb{Z}/n$ by ε, Proposition 10.65); so its Frobenius invariants have the order of $\Phi_E(k)[\ell^\infty]$, the ℓ-part of $c_q(E)$ (Remark 10.32). They are not the invariants of the sub $\mu_{\ell^t} \otimes \varepsilon$, on which Frobenius acts by $q\varepsilon$. Chapter 11 (Chapter 11, its item 2.2) cites this proposition instead of proving it again.

10.5 Kodaira symbols and Tate's algorithm

The paper names the reduction of E at a bad prime by its Kodaira symbol, reads $\#\Phi_q^{\mathrm{geom}}$ off the symbol (Section 1.4), and for a tame prime $q \geq 5$ reads the omitted Euler factor off the symbol (the table of Section 1, Lemma lem:euler). It computes the symbols with Tate's algorithm, through PARI (Section 2, Section sec:data, the appendix). This section states the classification of Kodaira and Néron, states Tate's algorithm in full, derives from it the table for $p \geq 5$ and the criterion for I_0^* that Lemma lem:twist (ii) uses, fixes PARI's codes, and runs the algorithm by hand on the paper's curves.

Regular models, blowing up, multiplicities. The classification below is a statement about a surface over $\mathcal{O}$, and we explain the words it uses for a reader without algebraic geometry. A *model* of E over $\mathcal{O}$ is a two-dimensional scheme, proper and flat over $\mathcal{O}$, with generic fibre E; the minimal Weierstrass model is one, given by one equation $F(x,y) = 0$ in the coordinates x, y over $\mathcal{O}$ (plus the point at infinity). Such a surface is *regular* at a point P of its special fibre if the local ring there has a maximal ideal generated by two elements. For a surface given by one equation, and $P = (x_0, y_0)$ in the special fibre, this holds if and only if F does not lie in $\mathfrak{n}_P^2$, where $\mathfrak{n}_P = (\varpi, x - \tilde{x}_0, y - \tilde{y}_0)$ with lifts $\tilde{x}_0, \tilde{y}_0$; informally, the three partial "derivatives" of F in the directions ϖ, x, y do not all vanish at P. A Weierstrass model is regular at every smooth point of $\tilde{E}$, and may fail to be regular at the singular point. *Blowing up* a point P replaces it by the projective line of directions through it: in coordinates, one substitutes $x - \tilde{x}_0 = \varpi x_1$, $y - \tilde{y}_0 = \varpi y_1$ (and similarly in the other charts, where $x - \tilde{x}_0$ or $y - \tilde{y}_0$ is the divisor) and removes the factor of ϖ that appears. Repeating this at the non-regular points, and then contracting the components that can be contracted without losing regularity, gives the *minimal regular proper model*, unique up to isomorphism (Lichtenbaum, Shafarevich; see [22, IV.7]). Its special fibre is the divisor of ϖ, a union of curves C_i counted with *multiplicities* $m_i \geq 1$: m_i is the order of vanishing of ϖ along C_i. The *dual graph* has a vertex for each C_i and an edge for each intersection point. A point of $E(K)$ extends, by properness, to an $\mathcal{O}$-point of the regular model, which meets the special fibre with intersection number 1; so it reduces to a smooth point of a component of multiplicity 1; this is why only those components contribute to the Néron model and to Φ^{geom}.

Example 10.35 (blowing up a node: type I_2)**.** Let $p \geq 5$ and $E\colon y^2 = x^3 + x^2 + p^2$ over $\mathbb{Q}_p$. Here $b_2 = 4$, $b_4 = 0$, $b_6 = 4p^2$, $b_8 = 4p^2$, so $c_4 = 16$ is a unit and $\Delta = -16p^2(4 + 27p^2)$ has $v_p(\Delta) = 2$: multiplicative reduction, with $\tilde{E}\colon y^2 = x^3 + x^2$ nodal at $(0,0)$, with tangents $y = \pm x$ (split). The equation $F = y^2 - x^2 - x^3 - p^2$ lies in $(p,x,y)^2$, so the model is not regular at the node. Blow it up: with $x = px_1$, $y = py_1$ one gets $F = p^2(y_1^2 - x_1^2 - px_1^3 - 1)$, and the new surface $y_1^2 - x_1^2 - 1 - px_1^3 = 0$ is regular along the special fibre of this chart (the partial derivatives $2y_1$ and $-2x_1$ do not both vanish on $y_1^2 - x_1^2 = 1$; the other charts are checked in the same way). Its special fibre in this chart is the conic $y_1^2 - x_1^2 = 1$, a new component $C_1 \simeq \mathbb{P}^1$ along which

$\varpi = p$ vanishes to order 1. The strict transform C_0 of $\tilde{E}$ is the nodal cubic with its node pulled apart into the two points of C_1 at infinity in the directions $y_1 = \pm x_1$, which are the tangent directions of the node. So the special fibre is two rational curves of multiplicity 1 meeting in two points: the cycle $\tilde{A}_1$ of type I_2. Removing the two intersection points leaves two copies of $\mathbb{G}_m$, and $\Phi^{\mathrm{geom}} \simeq \mathbb{Z}/2$, as in Table 10.1. In the additive types higher multiplicities appear: in type I_0^* the central component has multiplicity 2, so it contributes nothing to the Néron model, and Φ^{geom} comes from the four outer components.

Theorem 10.36 (Kodaira, Néron)**.** *Let E/K be an elliptic curve and let m be the number of irreducible components (over $\bar{k}$) of the special fibre of the minimal regular proper model of E over $\mathcal{O}$. The special fibre over $\bar{k}$ is one of the configurations of Table 10.1. The Néron model's special fibre is the union of the multiplicity-one components minus their intersection points with the other components, and Φ^{geom} is the group of multiplicity-one components, with the structure given in the table. Over a finite residue field k the Tamagawa number takes only the values in the last column.*

symbol	m	special fibre over $\bar{k}$	Φ^{geom}	$\#\Phi^{\mathrm{geom}}$	possible c
I_0	1	smooth curve of genus 1	0	1	1
I_1	1	rational curve with a node	0	1	1
I_n, $n \geq 2$	n	n smooth rational curves of multiplicity 1 forming a cycle ($\tilde{A}_{n-1}$)	$\mathbb{Z}/n$	n	n (split); 1 or 2 for n odd or even (nonsplit)
II	1	rational curve with a cusp	0	1	1
III	2	two rational curves meeting at one point with a common tangent	$\mathbb{Z}/2$	2	2
IV	3	three rational curves through one point, with distinct tangents	$\mathbb{Z}/3$	3	1 or 3
I_0^*	5	one curve of multiplicity 2 meeting four of multiplicity 1 ($\tilde{D}_4$)	$(\mathbb{Z}/2)^2$	4	1, 2 or 4
I_n^*, $n \geq 1$	$n+5$	a chain of $n+1$ curves of multiplicity 2, with two curves of multiplicity 1 at each end ($\tilde{D}_{n+4}$)	$\mathbb{Z}/4$ (n odd), $(\mathbb{Z}/2)^2$ (n even)	4	2 or 4
IV^*	7	$\tilde{E}_6$, multiplicities $1,2,3,2,1,2,1$	$\mathbb{Z}/3$	3	1 or 3
III^*	8	$\tilde{E}_7$, multiplicities $1,2,3,4,3,2,1,2$	$\mathbb{Z}/2$	2	2
II^*	9	$\tilde{E}_8$, multiplicities $1,2,3,4,5,6,4,2,3$	0	1	1

Table 10.1: The Kodaira-Néron classification. The components are smooth rational curves except in the first, second and fourth rows; $\tilde{A}$, $\tilde{D}$, $\tilde{E}$ are the extended Dynkin diagrams formed by the dual graph (a vertex for each component, an edge for each intersection point).

We do not prove Theorem 10.36; see Kodaira [7], Néron [13], Tate [24] and [22, IV.8, IV.9 and Table 4.1]. The orders $\#\Phi^{\mathrm{geom}}$ are the ones quoted in Section 1.4 of the paper (n; 1; 2; 3; 4 for I_n; $\mathrm{II}, \mathrm{II}^*$; $\mathrm{III}, \mathrm{III}^*$; $\mathrm{IV}, \mathrm{IV}^*$; $\mathrm{I}_0^*, \mathrm{I}_n^*$). In the language of Lie theory (a physics reader's aside): Φ^{geom} is the centre of the simply connected compact Lie group of type $A_{n-1}, D_{n+4}, E_6, E_7, E_8$ for $\mathrm{I}_n, \mathrm{I}_n^*, \mathrm{IV}^*, \mathrm{III}^*, \mathrm{II}^*$ (III and IV correspond to A_1 and A_2; I_0, I_1 and II have trivial Φ^{geom}), which is the dual of the root lattice modulo the root lattice; the same diagrams classify the finite subgroups of $\mathrm{SU}(2)$, and the binary tetrahedral group $\mathrm{SL}_2(\mathbb{F}_3)$ met in Theorem 10.52 is one of them.

On Cremona's tables below conductor 10,000 (211,455 pairs of a curve and a bad prime) every value of c in the last column occurs, and no other: for $\mathrm{IV}, \mathrm{IV}^*$ one finds $c \in \{1,3\}$, for I_0^* $c \in \{1,2,4\}$, for I_n^* $c \in \{2,4\}$, and for I_n the values predicted by Proposition 10.65 in all 153,343 multiplicative pairs.

Theorem 10.37 (Tate's algorithm)**.** *Let an integral model of E/K be given, with ϖ, k as in Convention 10.1, and write $a_{i,j} = a_i/\varpi^j$. The following procedure terminates and returns the Kodaira symbol, the number m of components, the conductor exponent $f = v(\Delta) + 1 - m$ (Theorem 10.47), the Tamagawa number c and a minimal model (the model at the step where it stops).*

Step 1. *If $\varpi \nmid \Delta$: type I_0, $m = 1$, $f = 0$, $c = 1$.*

Step 2. *Change coordinates by $(1, r, 0, t)$, $r, t \in \mathcal{O}$, so that the singular point of $\tilde{E}$ is $(0,0)$; then $\varpi \mid a_3, a_4, a_6$. If $\varpi \nmid b_2$: type I_n with $n = v(\Delta)$, $m = n$, $f = 1$; the reduction is split if $T^2 + a_1 T - a_2$ has a root in k, and then $c = n$; otherwise $c = 1$ (n odd) or 2 (n even).*

Step 3. *If $\varpi^2 \nmid a_6$: type II, $m = 1$, $f = v(\Delta)$, $c = 1$.*

Step 4. *If $\varpi^3 \nmid b_8$: type III, $m = 2$, $f = v(\Delta) - 1$, $c = 2$.*

Step 5. *If $\varpi^3 \nmid b_6$: type IV, $m = 3$, $f = v(\Delta) - 2$; $c = 3$ if $Y^2 + a_{3,1}Y - a_{6,2}$ has a root in k, else $c = 1$.*

Step 6. *Change coordinates by $(1, 0, s, t)$ so that $\varpi \mid a_1, a_2$, $\varpi^2 \mid a_3, a_4$, $\varpi^3 \mid a_6$. Let $P(T) = T^3 + a_{2,1}T^2 + a_{4,2}T + a_{6,3}$. If $\tilde{P}$ has distinct roots: type I_0^*, $m = 5$, $f = v(\Delta) - 4$, $c = 1 + \#\{\text{roots of } \tilde{P} \text{ in } k\}$.*

Step 7. *If $\tilde{P}$ has one simple and one double root: type I_n^* for some $n \geq 1$, $m = n + 5$, $f = v(\Delta) - 4 - n$. To find n, translate the double root to 0 (a change $(1, \varpi r_0, 0, 0)$); then $\varpi \parallel a_2$, $\varpi^3 \mid a_4$, $\varpi^4 \mid a_6$. For $s = 1, 2, \ldots$ alternately consider the quadratics*

$$Y^2 + a_{3,(s+3)/2}Y - a_{6,s+3} \ (s \text{ odd}), \qquad a_{2,1}X^2 + a_{4,(s+4)/2}X + a_{6,s+3} \ (s \text{ even});$$

if the current one has distinct roots in $\bar{k}$, stop with $n = s$, and $c = 4$ if its roots lie in k, else $c = 2$; if it has a double root, translate it to 0 (by t for s odd, by r for s even, with the appropriate power of ϖ) and continue with $s + 1$.

Step 8. *If $\tilde{P}$ has a triple root, translate it to 0; then $\varpi^2 \mid a_2$, $\varpi^3 \mid a_4$, $\varpi^4 \mid a_6$. If $Y^2 + a_{3,2}Y - a_{6,4}$ has distinct roots: type IV^*, $m = 7$, $f = v(\Delta) - 6$, $c = 3$ if the roots are in k, else $c = 1$.*

Step 9. *Otherwise translate its double root to 0, so that $\varpi^3 \mid a_3$, $\varpi^5 \mid a_6$. If $\varpi^4 \nmid a_4$: type III^*, $m = 8$, $f = v(\Delta) - 7$, $c = 2$.*

Step 10. *If $\varpi^6 \nmid a_6$: type II^*, $m = 9$, $f = v(\Delta) - 8$, $c = 1$.*

Step 11. *Otherwise the model is not minimal: replace a_i by a_i/ϖ^i (the change $(\varpi, 0, 0, 0)$) and return to Step 1.*

The translations can be made explicit. For $K = \mathbb{Q}_p$, Step 2 is achieved by the following (r, t) modulo p:

- *$p \geq 5$: $r = -b_2/12$ if $p \mid c_4$, $r = -(c_6 + b_2c_4)/(12c_4)$ otherwise, and $t = -(a_1r + a_3)/2$;*
- *$p = 3$: $r = -b_6$ if $3 \mid b_2$, $r = -b_2b_4$ otherwise, and $t = a_1r + a_3$;*
- *$p = 2$: $r = a_4$, $t = r(1 + a_2 + a_4) + a_6$ if $2 \mid b_2$, and $r = a_3$, $t = r + a_4$ otherwise.*

For p odd, Step 6 is completing the square: $s = -a_1/2$, $t = -a_3/2$, after which $a_1 = a_3 = 0$; at $p = 2$ one solves the congruences for s and t directly.

Sketch of proof. Each step either recognises a configuration of the special fibre of the minimal regular model, by blowing up the singular point of the current model and examining the exceptional curves, or performs the change of coordinates that puts the next singularity at the origin. The conditions $\varpi^i \mid a_j$ record which monomials vanish to which order at the singular point. That the procedure stops and returns a minimal model is part of the theorem; the values of f come from Ogg's formula. See Tate [24], [22, IV.9] and [4, Section 3.2]. The formulas for Step 2 are elementary: for $p \geq 5$ the singular point of $\tilde{E}$ has x-coordinate the multiple root of $4x^3 + b_2x^2 + 2b_4x + b_6$, which is $-b_2/12$ for a triple root and $-(c_6 + b_2c_4)/(12c_4)$ for a double root, and its y-coordinate is the zero of $\partial F/\partial y = 2y + a_1x + a_3$; the formulas at 3 and 2 come from the same computation in characteristic 3 and 2. We checked the three formulas on all 211,455 pairs (curve, bad prime) below conductor 10,000: in every case the translated model has $p \mid a_3, a_4, a_6$. □

Corollary 10.38 (the tame table)**.** *Let $p \geq 5$ and let a minimal model of E/K be given. Then*

(i) $v(\Delta) = 0$ if and only if the type is I_0;

(ii) $v(c_4) = 0 < v(\Delta) = n$ if and only if the type is I_n;

(iii) if $v(c_4) > 0$ and $v(j) \geq 0$, the type is determined by $v(\Delta) \in \{2, 3, 4, 6, 8, 9, 10\}$, which gives $\mathrm{II}, \mathrm{III}, \mathrm{IV}, \mathrm{I}_0^*, \mathrm{IV}^*, \mathrm{III}^*, \mathrm{II}^*$ respectively;

(iv) if $v(c_4) > 0$ and $v(j) < 0$, then $v(c_4) = 2$, $v(c_6) = 3$, $v(\Delta) = n + 6$ with $n = -v(j) \geq 1$, and the type is I_n^*.

More precisely, the triples $(v(c_4), v(c_6), v(\Delta))$ are as in Table 10.2. In particular $f = 2$ for every additive type.

Proof. As $p \geq 5$, the change $(1/6, *, *, *)$ to the short model $y^2 = x^3 + Ax + B$ with $A = -27c_4$, $B = -54c_6$ has u a unit, so the short model is integral and minimal, with $v(A) = v(c_4)$, $v(B) = v(c_6)$ and the same $v(\Delta)$, where $\Delta = -16(4A^3 + 27B^2)$ up to a unit. For it $a_1 = a_2 = a_3 = 0$, $a_4 = A$, $a_6 = B$, $b_2 = 0$, $b_6 = 4B$, $b_8 = -A^2$. Run Theorem 10.37.

If $v(\Delta) = 0$ the type is I_0. If $v(\Delta) > 0$ and $v(A) = 0$ the reduction is multiplicative, of type I_n with $n = v(\Delta)$; then $v(B) = 0$ too, since $4A^3 \equiv -27B^2$. Otherwise $v(A) \geq 1$, hence $v(B) \geq 1$, the singular point is already at $(0, 0)$ and $b_2 = 0$, so we pass to Step 3.

- Step 3: $\varpi^2 \nmid a_6$ means $v(B) = 1$: type II; then $v(27B^2) = 2 < v(4A^3)$, so $v(\Delta) = 2$.
- Step 4: $\varpi^3 \nmid b_8 = -A^2$ means $v(A) = 1$ (with $v(B) \geq 2$): type III, $v(\Delta) = \min(3, 2v(B)) = 3$.
- Step 5: $\varpi^3 \nmid b_6 = 4B$ means $v(B) = 2$ (with $v(A) \geq 2$): type IV, $v(\Delta) = 4$.

type	$v(c_4)$	$v(c_6)$	$v(\Delta)$	m	f
I_0	≥ 0	≥ 0	0	1	0
I_n	0	0	n	n	1
II	≥ 1	1	2	1	2
III	1	≥ 2	3	2	2
IV	≥ 2	2	4	3	2
I_0^*	≥ 2	≥ 3	6	5	2
I_n^*	2	3	$n+6$	$n+5$	2
IV^*	≥ 3	4	8	7	2
III^*	3	≥ 5	9	8	2
II^*	≥ 4	5	10	9	2

Table 10.2: Valuations of a minimal model at $p \geq 5$. For I_0^* at least one of $v(c_4) = 2$, $v(c_6) = 3$ holds; $v(c_4) = \infty$ or $v(c_6) = \infty$ (when $c_4 = 0$ or $c_6 = 0$) is allowed wherever "$\geq$" appears.

- Step 6: now $v(A) \geq 2$, $v(B) \geq 3$, and the conditions of Step 6 hold without change. $P(T) = T^3 + aT + b$ with $a = A/\varpi^2$, $b = B/\varpi^3$, whose discriminant is $-(4a^3 + 27b^2) = \Delta/(16\varpi^6)$ up to a unit. So $\tilde{P}$ has distinct roots if and only if $v(\Delta) = 6$: type I_0^*.

- $\tilde{P}$ has a triple root ρ if and only if $3\rho = 0$ (no T^2 term), so $\rho = 0$, that is, $v(A) \geq 3$ and $v(B) \geq 4$. If $v(\Delta) > 6$ and the root is not triple, then $v(A) = 2$ or $v(B) = 3$; but $v(A) = 2$, $v(B) \geq 4$ or $v(A) \geq 3$, $v(B) = 3$ would give $v(\Delta) = 6$. So Step 7 occurs exactly when $v(A) = 2$, $v(B) = 3$, $v(\Delta) > 6$; then $v(j) = 3v(c_4) - v(\Delta) = 6 - v(\Delta) < 0$, and by Theorem 10.66 the type is I_n^* with $n = -v(j) = v(\Delta) - 6$.

- Step 8: $v(A) \geq 3$, $v(B) \geq 4$, root already at 0; $Y^2 - a_{6,4}$ has distinct roots if and only if $v(B) = 4$: type IV^*, $v(\Delta) = 8$.

- Step 9: $v(B) \geq 5$, no translation needed; $\varpi^4 \nmid a_4$ means $v(A) = 3$: type III^*, $v(\Delta) = 9$.

- Step 10: $v(A) \geq 4$; $\varpi^6 \nmid a_6$ means $v(B) = 5$: type II^*, $v(\Delta) = 10$.

- Step 11 would need $v(A) \geq 4$, $v(B) \geq 6$, impossible for a minimal model (Proposition 10.5(ii)).

This gives Table 10.2. In (iii), $v(j) \geq 0$ excludes I_n and I_n^* ($v(j) = -n$ there), and the seven remaining additive types have the distinct values $v(\Delta) = 2, 3, 4, 6, 8, 9, 10$. The values of f follow from $f = v(\Delta) + 1 - m$. □

All 9,699 potentially good additive pairs (curve, $p \geq 5$) of Cremona's tables below conductor 10,000 have $v_p(\Delta)$ as in (iii), and all pairs with $p \geq 5$ fit Table 10.2. At $p = 2, 3$ no such table exists: Papadopoulos [15] gives the corresponding tables, which need congruence conditions on the unit parts of c_4, c_6, Δ besides the valuations.

Proposition 10.39 (the cubic criterion for I_0^*)**.** *Let p be odd and let E/K have a model $y^2 = x^3 + a_2x^2 + a_4x + a_6$ with $\varpi \mid a_2$, $\varpi^2 \mid a_4$, $\varpi^3 \mid a_6$. Then E has type I_0^* if and only if*

$$P(T) = T^3 + (a_2/\varpi)T^2 + (a_4/\varpi^2)T + a_6/\varpi^3$$

has distinct roots modulo ϖ, and then $c = 1 + \#\{\text{roots of } \tilde{P} \text{ in } k\}$. Conversely every curve of type I_0^* at an odd p has such a model. In particular, if $y^2 = x^3 + a_2x^2 + a_4x + a_6$ is a model over $\mathbb{Z}$ with good reduction at an odd prime q, its twist by q^*,

$$y^2 = x^3 + q^*a_2x^2 + (q^*)^2a_4x + (q^*)^3a_6,$$

has cubic $T^3 + \epsilon a_2T^2 + a_4T + \epsilon a_6$ with $\epsilon = q^*/q = \pm 1$, whose roots are ϵ times those of $x^3 + a_2x^2 + a_4x + a_6$; so it has type I_0^* at q.

Proof. For the given model $a_1 = a_3 = 0$, so $b_2 = 4a_2$, $b_6 = 4a_6$, $b_8 = 4a_2a_6 - a_4^2$, and $(0, 0)$ is the singular point of $\tilde{E}$ (the reduced equation is $y^2 = x^3$). Run Theorem 10.37: Step 2 passes as $\varpi \mid b_2$; Step 3 passes as $v(a_6) \geq 3$; Step 4 passes as $v(b_8) \geq \min(v(a_2) + v(a_6), 2v(a_4)) \geq 4$; Step 5 passes as $v(b_6) \geq 3$; and the conditions of Step 6 already hold. So the type is I_0^* exactly when $\tilde{P}$ has distinct roots, with the stated c.

Conversely let E have type I_0^* at odd p. Completing the square (a change $(1, 0, s, t)$ with $s = -a_1/2$, $t = -a_3/2$, integral since p is odd) gives a minimal model $y^2 = f(x) = x^3+a_2x^2+a_4x+a_6$. The singular point of $\tilde{E}$ is $(x_0, 0)$ with x_0 a multiple root of $\tilde{f}$; translating x by a lift of x_0 keeps the shape of the equation and gives $\varpi \mid a_4, a_6$. As the reduction is additive, $\tilde{f}$ has a triple root at 0, so $\varpi \mid a_2$. Since Steps 3, 4 and 5 are passed, $\varpi^2 \mid a_6$, $\varpi^3 \mid b_8 = 4a_2a_6 - a_4^2$ and $\varpi^3 \mid b_6 = 4a_6$. Then $v(4a_2a_6) \geq 3$ gives $v(a_4^2) \geq 3$, so $\varpi^2 \mid a_4$, and $\varpi^3 \mid a_6$: the model has the stated shape.

For the twist: $P(T) = T^3+(q^*/q)a_2T^2+(q^*/q)^2a_4T+(q^*/q)^3a_6 = T^3+\epsilon a_2T^2+a_4T+\epsilon a_6$, as $\epsilon^2 = 1$. If θ is a root of $g(x) = x^3+a_2x^2+a_4x+a_6$ then $(\epsilon\theta)^3+\epsilon a_2(\epsilon\theta)^2+a_4\epsilon\theta+\epsilon a_6 = \epsilon g(\theta) = 0$. The roots of $\tilde{g}$ are distinct because the model has good reduction at q ($\Delta = 16\,\mathrm{disc}(g)$ and q is odd), so those of $\tilde{P}$ are too. □

Warning 10.40. The sign $\epsilon = q^*/q$ matters when $q \equiv 3 \pmod 4$: the cubic of the twist is $T^3 + \epsilon a_2T^2 + a_4T + \epsilon a_6$, not the cubic of E. Its roots are ϵ times the old ones, so the conclusion "distinct roots" is unaffected, and so is the number of roots in k, which gives c. The twist must be by q^*, not by q; see Warning 10.71.

Convention 10.41 (PARI's Kodaira codes)**.** For a curve `E` and a prime `p`, PARI's function `elllocalred(E,p)` returns the vector $[f_p, k_p, [u, r, s, t], c_p]$: the conductor exponent, the Kodaira code, the change of coordinates to a minimal model at p, and the Tamagawa number. The function `ellglobalred(E)` returns $[N, [u, r, s, t], \prod_p c_p, F, L]$ with F the factorisation matrix of N and L the list of local data at the bad primes, in the order of F; in PARI 2.17.2 the entries of L are $[f_p, k_p, 0, c_p]$. So the paper's `R[4]` is the factorised conductor and `R[4][,1]` its column of primes; with `K = R[5]`, the entry `K[i][2]` is the Kodaira code at the i-th bad prime. The codes are given in Table 10.3.

So `k >= 5` detects multiplicative reduction, `k <= -5` the symbols I_n^* with $n \geq 1$, and `abs(k) == 3` the types III, III*. Since `ellglobalred` lists only the bad primes, the code 1 never occurs in its output.

Example 10.42 (Tate's algorithm on the paper's curves)**.** All models are the global minimal models of Cremona's tables; every number below was recomputed with PARI.

(a) `121a1` $= [1, 1, 1, -30, -76]$ at 11: $v(\Delta) = 2$, $v(c_4) = 1$. Step 2 with $(r, t) = (6, 2)$ gives $[1, 19, 11, 88, -22]$, with $11 \mid b_2 = 77$; Step 3: $v(a_6) = 1 < 2$. Type II, $c = 1$, $f = 2$.

code k_p	symbol	m	$\#\Phi^{\text{geom}}$	f at $p \geq 5$	$v(\Delta_{\min})$ at $p \geq 5$
1	I_0	1	1	0	0
$n+4$ $(n \geq 1)$	I_n	n	n	1	n
2	II	1	1	2	2
3	III	2	2	2	3
4	IV	3	3	2	4
-1	I_0^*	5	4	2	6
$-(n+4)$ $(n \geq 1)$	I_n^*	$n+5$	4	2	$n+6$
-4	IV^*	7	3	2	8
-3	III^*	8	2	2	9
-2	II^*	9	1	2	10

Table 10.3: PARI's Kodaira codes. The formulas $n+4$ and $-(n+4)$ hold for $n \geq 1$ only: $-(0+4) = -4$ is IV^*, and I_0^* is -1.

(b) `121c2` $= [1, 1, 0, -3632, 82757]$ at 11: $v(\Delta) = 8$. Step 2 with $(r, t) = (6, 8)$ gives $[1, 19, 22, -3520, 61105]$, with $v(a_6) = 2$, $v(b_8) \geq 3$, $v(b_6) \geq 3$, so Steps 3 to 5 are passed. Step 6 with $(s, t) = (5, 110)$ gives $[11, -11, 242, -4840, 46585]$ and $P(T) = T^3 - T^2 - 40T + 35 \equiv (T-4)^3 \pmod{11}$: a triple root, so Step 8. Type IV^*, $c = 1$, $f = 2$.

(c) `539d1` $= [0, 1, 1, -16, -66]$ at 7: Step 2 with $(r, t) = (2, 3)$ gives $[0, 7, 7, 0, -98]$; Step 6 with $(s, t) = (0, 21)$ gives $[0, 7, 49, 0, -686]$ and $P(T) = T^3 + T^2 - 2 = (T-1)(T^2 + 2T + 2)$, where $T^2 + 2T + 2$ has discriminant $-4 \equiv 3$, a non-square modulo 7. Distinct roots, one in $\mathbb{F}_7$: type I_0^*, $c = 2$, $f = 2$.

(d) `539c1` $= [1, 0, 1, 170, -3237]$ at 7: Step 2 with $(r, t) = (4, 1)$, then Step 6 with $(s, t) = (3, 21)$ give $[7, 0, 49, 49, -3087]$ and $P(T) = T^3 + T - 9 \equiv (T-1)(T-3)^2 \pmod{7}$. Step 7: translating the double root by $r = 21$ gives $[7, 63, 196, 1372, 7203]$; the quadratic $Y^2 + 4Y - 3 \equiv (Y-5)^2$ has a double root, and $t = 245$ gives $[7, 63, 686, -343, -100842]$; the quadratic $2X^2 + 6X + 1 \equiv 2(X-2)^2$ has a double root, and $r = 98$ gives $[7, 357, 1372, 40817, 1411788]$; the quadratic $Y^2 + 4Y + 2 \equiv (Y-1)(Y-2)$ has distinct roots in $\mathbb{F}_7$. So $n = 3$: type I_3^*, $c = 4$, and $f = v(\Delta) - 4 - n = 9 - 7 = 2$.

(e) `726a1` $= [1, 1, 0, -35, -51]$: at 11, Step 2 with $(r, t) = (6, 8)$ gives $[1, 19, 22, 77, -121]$, with $v(a_6) = 2$ and $v(b_8) = 2 < 3$: type III, $c = 2$. At 2: $v(\Delta) = 6$, $v(c_4) = 0$, and the tangent cone gives nonsplit reduction: I_6, $c = 2$. At 3: I_3 nonsplit, $c = 1$.

(f) `20a1` $= [0, 1, 0, 4, 4]$ at 2: $v(\Delta) = 8$, $v(c_4) = 4$. The singular point is already at $(0, 0)$, and Steps 3 to 5 are passed ($v(a_6) = 2$, $b_8 = 0$, $v(b_6) = 4$). Step 6 with $(s, t) = (1, 2)$ gives $[2, 0, 4, 0, 0]$ and $P(T) = T^3$, a triple root; Step 8: $Y^2 + a_{3,2}Y - a_{6,4} = Y^2 + Y$ has the distinct roots $0, 1 \in \mathbb{F}_2$. Type IV^*, $c = 3$, $f = 8 + 1 - 7 = 2$.

(g) `99a1` $= [1, -1, 1, -2, 0]$ at 3: Step 2 with $(r, t) = (2, 0)$ gives $[1, 5, 3, 6, 0]$, $v(b_2) = 1$, $a_6 = 0$, and $v(b_8) = 2 < 3$: type III, $c = 2$, $f = v(\Delta) - 1 = 2$.

(h) `980i1` $= [0, 0, 0, 1568, -72716]$: at 2, type IV^* with $c = 1$; at 5, I_1; at 7, the model already satisfies the conditions of Step 6 ($v(a_4) = 2$, $v(a_6) = 3$), with $P(T) = T^3 + 32T - 212 \equiv (T-2)(T+1)^2 \pmod{7}$; the subprocedure of Step 7 passes four double roots and stops at $s = 5$ with the quadratic $Y^2 + 1$, irreducible over $\mathbb{F}_7$: type I_5^*, $c = 2$, $v_7(\Delta) = 11 = 5 + 6$.

In the paper 10.43. The Example of Section 3 of the paper (types II of `121a1`, I_3^* of `539c1`, I_5^* of `980i1`), the Remark after it (`99a1`, `539d1`, `726a1`), Section sec:anomaly (`726a1`, type III at 11) and the appendix (`rule` on `121a1` and `726a1`) all use the types computed here. In the appendix, `k >= 5` selects the multiplicative primes (where the naive factor is removed), `abs(k) != 3` excludes III, III* unless `iii = 1`, and `k == 4 || k == 2`, `k == -4 || k == -2`, `k == -3` select IV, II, then IV*, II*, then III* for the weights w_3 at $q = 3$.

10.6 The conductor and the semistability defect

This section defines the conductor exponent f_p and the conductor N, and studies how inertia acts on the Tate module at a potentially good prime: through a finite group G_{ss}, whose order is given by Kraus's formula $12/\gcd(v(\Delta), 12)$ when $p \geq 5$. The paper uses the conductor exponents in Lemma lem:twist and Theorem thm:qual, the defect in Lemma lem:IV, Remark rem:types and Lemma lem:euler, and the wild primes in Remark rem:wild and Section sec:wild. The section ends with the theorem that types IV and IV* at 2 are tame with inertia image cyclic of order 3 (Corollary 10.54), which removes the exception that the original version of the paper made in Proposition prop:forms.

Convention 10.44 (wild part and defect)**.** The Swan conductor (wild part of the conductor) is written sw, as in Chapter 7; the letter δ_p is reserved for the paper's $\dim V^{I_p}$. For potentially good reduction the group measuring the defect of semistability is written G_{ss} (it is called Φ_K in some sources; we avoid that letter, which denotes component groups here), and $e = \#G_{\mathrm{ss}}$ is the *semistability defect*.

Definition 10.45 (conductor exponent)**.** Let E/K and $\ell \neq p$ a prime, and let $V = V_\ell E$ as a representation of G_K. The *conductor exponent* of E is the Artin conductor of V (Chapter 7):

$$f(E) = a(V) = \dim V - \dim V^{I_K} + \mathrm{sw}(V),$$

where $\mathrm{sw}(V) \geq 0$ is the Swan conductor, which vanishes if and only if the wild inertia group P_K acts trivially on V. For $E/\mathbb{Q}$ and a prime p we write $f_p = f_p(E)$ for the exponent of $E/\mathbb{Q}_p$.

That $f(E)$ does not depend on ℓ is a theorem of Ogg and of Serre and Tate [14, 21]; we do not prove it. The following values are proved here, given the criterion of Néron, Ogg and Shafarevich (Chapter 11: E has good reduction if and only if I_K acts trivially on $T_\ell E$) and two results of this chapter proved below.

Proposition 10.46 (values of the conductor exponent)**.** *Let $\ell \neq p$.*

(i) Good reduction: $V^{I_K} = V$ and $f = 0$.

(ii) Multiplicative reduction: V^{I_K} is a line, I_K acts tamely through a nontrivial unipotent, and $f = 1$.

(iii) Additive reduction: $V^{I_K} = 0$ and $f = 2 + \mathrm{sw}(V) \geq 2$.

(iv) If $p \geq 5$, $\mathrm{sw}(V) = 0$, so $f = 2$ for additive reduction. If p is odd and the reduction is additive and potentially multiplicative, also $\mathrm{sw}(V) = 0$ and $f = 2$.

Proof. (i) is Corollary 10.17(ii). (ii) By Proposition 10.62(iii), in a suitable basis G_K acts on $T_\ell E$ by $\left(\begin{smallmatrix}\chi_\ell\varepsilon & \varepsilon\kappa\\ 0 & \varepsilon\end{smallmatrix}\right)$ with ε unramified. On I_K the cyclotomic character χ_ℓ is trivial ($\mu_{\ell^\infty}\subset K^{\mathrm{ur}}$), ε is trivial, and κ restricts to ν times the tame character $t_\ell\colon I_K\to\mathbb{Z}_\ell(1)$, with $\nu=v(q_E)\geq 1$ (Chapter 7); as in Appendix B, the tame character always carries its domain or an argument, and $t_\ell(E,q)$ always its two arguments. So $\sigma\in I_K$ acts by $\left(\begin{smallmatrix}1 & \nu t_\ell(\sigma)\\ 0 & 1\end{smallmatrix}\right)$: unipotent, nontrivial, trivial on P_K, with invariants the line spanned by e_1; so $f=2-1+0=1$.

(iii) Suppose first that the reduction is potentially good. By Theorem 10.51 the image $\rho(I_K)$ is finite, and it lies in SL_2 since $\det\rho=\chi_\ell$ is trivial on I_K. If $V^{I_K}\neq 0$, then in a basis (v,w) with v fixed every element of $\rho(I_K)$ has the form $\left(\begin{smallmatrix}1 & b\\ 0 & d\end{smallmatrix}\right)$ with $d=1$; a unipotent matrix of finite order in characteristic 0 is the identity, so $\rho(I_K)=1$ and the reduction would be good by the criterion of Néron, Ogg and Shafarevich. If the reduction is potentially multiplicative, then by Theorem 10.59 E is the twist of a Tate curve by a ramified quadratic character χ, and by Proposition 10.62 $\sigma\in I_K$ acts by $\chi(\sigma)\left(\begin{smallmatrix}1 & \nu t_\ell(\sigma)\\ 0 & 1\end{smallmatrix}\right)$. An element with $\chi(\sigma)=-1$ has the single eigenvalue -1 and fixes no nonzero vector. In both cases $V^{I_K}=0$ and $f=2+\mathrm{sw}(V)$.

(iv) If the reduction is potentially good, $\rho(I_K)\simeq G_{\mathrm{ss}}$ has order dividing 24 (Theorem 10.51(iii)); for $p\geq 5$ this is prime to p, so the pro-p group P_K acts trivially and $\mathrm{sw}(V)=0$. If it is potentially multiplicative and p is odd, the quadratic character χ is trivial on the pro-p group P_K, and so is $t_\ell\colon I_K\to\mathbb{Z}_\ell(1)$; so P_K acts trivially again. □

Theorem 10.47 (Ogg's formula)**.** *Let E/K, with $K/\mathbb{Q}_p$ finite, and let m be the number of irreducible components over $\bar{k}$ of the special fibre of its minimal regular model (Table 10.1). Then*

$$f=v(\Delta_{\min})+1-m.$$

We do not prove this; it is due to Ogg [14], whose argument has a gap in some cases at $p=2$, and in general to Saito [19]; see [22, IV.11]. For $p\geq 5$ it can be read off Table 10.2. PARI computes f through this formula, so a comparison with PARI's output is a consistency check and not an independent test.

Theorem 10.48 (bounds for the conductor exponent)**.** *Let $E/\mathbb{Q}$. Then $f_p\leq 2$ for $p\geq 5$, $f_3\leq 5$ and $f_2\leq 8$.*

The first bound is Proposition 10.46; we do not prove the other two. They are special cases of the bounds of Brumer and Kramer [3] and of Lockhart, Rosen and Silverman [11] for abelian varieties. Below conductor 10,000 every value $f_2\in\{2,\dots,8\}$ and $f_3\in\{2,\dots,5\}$ occurs at an additive prime (4,750, 5,390, 7,067, 2,278, 3,946, 892, 388 pairs at 2; 13,194, 3,014, 914, 356 at 3).

Definition 10.49 (the conductor)**.** The *conductor* of $E/\mathbb{Q}$ is $N=\prod_p p^{f_p}$.

Proposition 10.50 (properties of the conductor)**.** *Let $E/\mathbb{Q}$ have conductor N.*

(i) $p\mid N$ if and only if E has bad reduction at p; $p\,\|\,N$ if and only if the reduction is multiplicative; $p^2\mid N$ if and only if it is additive; and $p^3\nmid N$ for $p\geq 5$.

(ii) Isogenous curves over $\mathbb{Q}$ have the same conductor.

Proof. (i) is Proposition 10.46. (ii) An isogeny $E\to E'$ over $\mathbb{Q}$ induces a $G_{\mathbb{Q}}$-equivariant isomorphism $V_\ell E\simeq V_\ell E'$ (Chapter 9), and f_p depends only on $V_\ell E$ restricted to $G_{\mathbb{Q}_p}$, for any $\ell\neq p$. □

The conductors of the paper's curves, all checked with PARI, are in the column N of Example 10.85.

Theorem 10.51 (Serre and Tate). *Let E/K have potentially good reduction, $K/\mathbb{Q}_p$ finite.*

(i) There is a smallest extension L of K^{ur} over which E has good reduction. It is Galois over K^{ur}, and $L = K^{\mathrm{ur}}(E[m])$ for every integer $m \geq 3$ prime to p.

(ii) Put $G_{\mathrm{ss}} = \mathrm{Gal}(L/K^{\mathrm{ur}})$ and $e = \#G_{\mathrm{ss}}$. For every prime $\ell \neq p$ the kernel of $I_K \to \mathrm{Aut}(T_\ell E)$ is $\mathrm{Gal}(\bar{K}/L)$, so the image of I_K in $\mathrm{Aut}(T_\ell E)$ is isomorphic to G_{ss}, independently of ℓ.

(iii) G_{ss} embeds in $\mathrm{SL}_2(\mathbb{Z}/m)$ for every $m \geq 3$ prime to p; in particular e divides 24. Every element of G_{ss} has order 1, 2, 3, 4 or 6. If $p \geq 5$, G_{ss} is cyclic of order 1, 2, 3, 4 or 6.

(iv) G_{ss} acts faithfully on the reduction $\tilde{E}_L$ of E over L; if $e > 2$ then $j(\tilde{E}_L) \in \{0, 1728\}$.

Sketch of proof. (ii) given (i): for an extension L' of K^{ur} inside $\bar{K}$, E has good reduction over L' if and only if $\mathrm{Gal}(\bar{K}/L')$ (the inertia group of L') acts trivially on $T_\ell E$, by the criterion of Néron, Ogg and Shafarevich over L' (Chapter 11). So the fixed field of the kernel of $I_K \to \mathrm{Aut}(T_\ell E)$ is the smallest such L'; this proves (i) except for the description by $E[m]$, and (ii). For (iii): the image of I_K in $\mathrm{Aut}(E[m])$ is $\mathrm{Gal}(K^{\mathrm{ur}}(E[m])/K^{\mathrm{ur}}) = G_{\mathrm{ss}}$ by (i), and its determinant is the cyclotomic character modulo m, trivial on I_K. So e divides $\#\,\mathrm{SL}_2(\mathbb{Z}/m)$ for every $m \geq 3$ prime to p. If $p \neq 3$, take $m = 3$: $\#\,\mathrm{SL}_2(\mathbb{F}_3) = 24$. If $p = 3$, take $m = 4$ and $m = 5$: e divides $\#\,\mathrm{SL}_2(\mathbb{Z}/4) = 48$ and $\#\,\mathrm{SL}_2(\mathbb{F}_5) = 120$, hence $\gcd(48, 120) = 24$. The characteristic polynomial of an element of $\rho(I_K)$ has integer coefficients independent of ℓ [21, Section 2], and its roots are roots of unity of degree at most 2 over $\mathbb{Q}$, of order $1, 2, 3, 4$ or 6. If $p \geq 5$, $p \nmid 24$, so L/K^{ur} is tamely ramified and G_{ss} is a quotient of the procyclic tame inertia group (Chapter 7), hence cyclic, of an order that is the order of one of its elements. The description $L = K^{\mathrm{ur}}(E[m])$ in (i) and statement (iv) are in [21, Section 2]; see also [20, Section 5.6]. We do not prove them. □

Theorem 10.52 (Kraus). *Let $K/\mathbb{Q}_p$ be finite and let E/K have potentially good reduction.*

(i) If $p \geq 5$, G_{ss} is cyclic of order $e = 12/\gcd(v(\Delta_{\min}), 12)$. By type: I_0, II, III, IV, I_0^*, IV^*, III^*, II^* give $e = 1, 6, 4, 3, 2, 3, 4, 6$.

(ii) If $p = 3$, G_{ss} is cyclic of order $1, 2, 3, 4$ or 6, or the semidirect product $\mathbb{Z}/3 \rtimes \mathbb{Z}/4$ of order 12.

(iii) If $p = 2$, G_{ss} is cyclic of order $1, 2, 3, 4$ or 6, or the quaternion group Q_8, or $\mathrm{SL}_2(\mathbb{F}_3)$ of order 24. For $K = \mathbb{Q}_2$ and $K = \mathbb{Q}_3$, Kraus determines G_{ss} from $v(c_4)$, $v(c_6)$, $v(\Delta)$ and congruences on their unit parts.

Proof. (i), given Theorem 10.51. Put $e = 12/\gcd(v(\Delta), 12)$ for a minimal model, so that $ev(\Delta)/12$ is an integer, and let $K' = K^{\mathrm{ur}}(\varpi^{1/e})$, a cyclic extension of K^{ur} of degree e, totally ramified with uniformiser $\varpi' = \varpi^{1/e}$ (K^{ur} contains μ_e as $p \nmid e$). Let v' be the valuation of K', so $v' = ev$ on K. Apply $(u, 0, 0, 0)$ with $u = \varpi'^{\,ev(\Delta)/12}$. The new invariants satisfy

$$\begin{aligned} v'(\Delta') &= ev(\Delta) - 12 \cdot ev(\Delta)/12 = 0, \\ v'(c_4') &= ev(c_4) - 4ev(\Delta)/12 = e\,v(j)/3 \geq 0, \\ v'(c_6') &= ev(c_6) - 6ev(\Delta)/12 \geq 0, \end{aligned}$$

using $v(j) \geq 0$ (Theorem 10.21) for c_4', and $2v(c_6) \geq \min(3v(c_4), v(\Delta)) = v(\Delta)$ (from $c_6^2 = c_4^3 - 1728\Delta$) for c_6'. As $p \geq 5$, the model $y^2 = x^3 - 27c_4'x - 54c_6'$ is integral with unit discriminant, so E has good reduction over K'. Conversely, if E has good reduction over an extension L' of K^{ur} of ramification index e', a minimal model over L' has unit discriminant $u^{-12}\Delta$, so $e'v(\Delta) \equiv 0 \pmod{12}$ and $e \mid e'$. Apply this to the smallest extension L of Theorem 10.51: e divides $[L : K^{\mathrm{ur}}]$. But $L \subseteq K'$, as E has good reduction over K', so $[L : K^{\mathrm{ur}}] \leq e$. Hence $L = K'$ and $G_{\mathrm{ss}} = \mathrm{Gal}(K'/K^{\mathrm{ur}}) \simeq \mu_e$ is cyclic of order e. The list by type is Corollary 10.38(iii): $v(\Delta) = 0, 2, 3, 4, 6, 8, 9, 10$ gives $e = 1, 6, 4, 3, 2, 3, 4, 6$.

(ii) and (iii) we do not prove; see Kraus [10] and [20, Section 5.6]. □

The theorem numbers inside [10] are quoted inconsistently in the secondary literature, so we cite the paper as a whole.

Theorem 10.53 (tame additive reduction)**.** *Let $K/\mathbb{Q}_p$ be finite and let E/K have additive, potentially good reduction. Then the reduction is tame (equivalently* $\mathrm{sw} = 0$*, equivalently $f = 2$) if and only if*

(a) $p \geq 5$; or

(b) $p = 3$ and the type is III*,* III* *or* I_0^**; or*

(c) $p = 2$ and the type is IV *or* IV**.*

We do not prove this; see Dokchitser and Dokchitser [6, Theorem 3.1], who deduce it from Kramer and Tunnell [8, Proposition 8.20] at $p = 2$ and from Kraus [10] at $p = 3$. The direction "type IV or IV* at 2 implies tame" is proved independently in Corollary 10.54. The data agree: below conductor 10,000, $f_2 = 2$ occurs exactly for the pairs of type IV or IV* at 2 and for no other additive type (Table 10.4), and at 3 the types III, III*, I_0^* always have $f_3 = 2$ while II, II*, IV, IV* always have $f_3 \in \{3, 4, 5\}$.

Corollary 10.54 (types IV and IV* at 2)**.** *Let $E/\mathbb{Q}_2$ have Kodaira type* IV *or* IV**. Then:*

(i) E has potentially good reduction;

(ii) G_{ss} is cyclic of order 3, so for every odd prime ℓ the image of I_2 in $\mathrm{Aut}(T_\ell E)$ is cyclic of order 3, generated by an element σ with $\det\sigma = 1$ and eigenvalues the two primitive cube roots of unity; on $E[3]$ the image is a nontrivial unipotent group and $E[3]^{I_2}$ is a line;

(iii) the reduction is tame and $f_2 = 2$;

(iv) every Frobenius lift $\mathrm{Frob} \in G_{\mathbb{Q}_2}$ satisfies $\rho(\mathrm{Frob})\,\sigma\,\rho(\mathrm{Frob})^{-1} = \sigma^{-1}$.

Proof. By Proposition 10.33(i) and Theorem 10.36, $E[3]^{I_2} \simeq \Phi^{\mathrm{geom}}[3] \simeq \mathbb{Z}/3$ is a line.

(i) Suppose $v_2(j) < 0$. By Theorem 10.59, E is the twist of the Tate curve E_{q_E} by a quadratic character χ, ramified since the reduction is additive, and by Proposition 10.62 an element $\sigma \in I_2$ acts on T_3E by $\chi(\sigma)\left(\begin{smallmatrix} 1 & \kappa(\sigma) \\ 0 & 1 \end{smallmatrix}\right)$, where $\kappa(\sigma) = \nu\, t_3(\sigma)$ is ν times the 3-adic tame character $t_3 \colon I_2 \to \mathbb{Z}_3(1)$ (Chapter 7; not the term $t_3(E, 2)$). The set $H = \{\sigma \in I_2 : \kappa(\sigma) \equiv 0 \pmod 3\}$ is a subgroup of index 1 or 3, and $\ker\chi|_{I_2}$ has index 2; so $H \not\subseteq \ker\chi$, and some $\sigma \in H$ acts on $E[3]$ by -1, which fixes no line. This contradicts $E[3]^{I_2} \neq 0$. So $v_2(j) \geq 0$.

(ii) By Theorem 10.51 the image $\rho_3(I_2)$ is a finite subgroup of $\mathrm{SL}_2(\mathbb{Z}_3)$. Reduction modulo 3 is injective on it: if $g = 1 + 3^k A$ with $k \geq 1$ and $A \not\equiv 0 \pmod 3$ had finite order, a power

of it would have prime order r and the same shape, and $g^r \equiv 1 + r3^k A \pmod{3^{k+1}}$ if $r \neq 3$, $g^3 \equiv 1 + 3^{k+1}A \pmod{3^{k+2}}$ if $r = 3$, neither equal to 1. The image modulo 3 is nontrivial (otherwise $\rho_3(I_2) = 1$ and the reduction would be good) and fixes the line $E[3]^{I_2}$ pointwise. In a basis (v, w) of $E[3]$ with v spanning that line, its elements are $\left(\begin{smallmatrix} 1 & b \\ 0 & d \end{smallmatrix}\right)$ with determinant $d = 1$: the group $\{\left(\begin{smallmatrix} 1 & b \\ 0 & 1 \end{smallmatrix}\right)\}$ of order 3. So $\rho_3(I_2)$ is cyclic of order 3, hence $G_{\mathrm{ss}} \simeq \mathbb{Z}/3$ and $\rho_\ell(I_2) \simeq \mathbb{Z}/3$ for every $\ell \neq 2$ (Theorem 10.51(ii)). A generator σ has determinant 1 and $\sigma^3 = 1$, $\sigma \neq 1$; its eigenvalues λ, λ^{-1} are cube roots of unity, and not both 1 (a unipotent element of finite order is trivial), so they are $\zeta_3^{\pm 1}$.

(iii) The order 3 is prime to 2, so the wild inertia group, a pro-2 group, acts trivially: $\mathrm{sw} = 0$. With $V^{I_2} = 0$ (Proposition 10.46(iii)), $f_2 = 2$.

(iv) As $\rho(P_2) = 1$, $\rho|_{I_2}$ factors through the tame quotient, and $\sigma = \rho(t)$ for a topological generator t of it. The tame relation $\mathrm{Frob}\, t\, \mathrm{Frob}^{-1} = t^2$ (Chapter 7) gives $\rho(\mathrm{Frob})\sigma\rho(\mathrm{Frob})^{-1} = \sigma^2 = \sigma^{-1}$. □

We checked the conclusions on all pairs of type IV or IV* at 2 below conductor 10,000 (the first two rows of Table 10.4): all have $v_2(j) \geq 0$ and $f_2 = 2$, and the four roots of the 3-division polynomial generate over $\mathbb{Q}_2$ fields of ramification indices $1, 3, 3, 3$, as (ii) predicts (Proposition 10.77).

In the paper 10.55. Lemma lem:IV of the revised paper is stated for every $q \neq 3$; its proof is the argument above (component group, then injectivity of reduction on finite subgroups of $\mathrm{GL}_2(\mathbb{Z}_3)$). For $q \geq 5$, Kraus's formula gives the order 3 directly: the revised paper notes after the lemma that "for $q \geq 5$ the order 3 is also the semistability defect $12/\gcd(v_q(\Delta_E), 12)$ of Kraus" (the original version wrote "for $v_q(\Delta_E) \in \{4, 8\}$ this is 3"). The Galois-representation consequences, including the Serre conductor exponent 1 of $\bar{\rho}_{E,3}$ at q, are proved in Chapter 11. With (ii) and (iv), the computation of Lemma lem:euler for $\mathrm{II}, \mathrm{II}^*, \mathrm{IV}, \mathrm{IV}^*$ applies verbatim at $q = 2$: the trace-zero endomorphisms commuting with σ form the line spanned by $2\sigma + 1$, Frobenius inverts σ and so acts on that line by $-1 = \chi_{-3}(2)$, so $L_2(A_f, s)^{-1} = 1 + 2^{-s}$, $P_2(X) = 1 + 2X$ and $e_\ell(E, 2) = v_\ell(3/2)$, which is 1 for $\ell = 3$ and 0 for $\ell \geq 5$ (Chapter 19). This is why Proposition prop:forms holds with no exception (the original version excepted these types and checked $e_3(E, 2) \geq 1$ on 92,812 curves), why Remark rem:wild can say that a type IV or IV* at 2 "always has $P_2(X) = 1 + 2X$", and why case (a) of Theorem thm:qual holds at $q = 2$.

Example 10.56 (semistability defects of the paper's curves)**.** At the additive primes of the running examples: `121a1` at 11 (type II, $v(\Delta) = 2$): $e = 6$; `121c1` (IV, $v = 4$) and `121c2` (IV*, $v = 8$) at 11: $e = 3$; `539d1` at 7 (I_0^*): $e = 2$; `726a1` at 11 (III, $v = 3$): $e = 4$. At the wild primes: `99a1` at 3 (type III, tame by Theorem 10.53) has G_{ss} cyclic of order 4 (Remark 10.79); `20a1`, `140b1` and `980i1` at 2 (type IV*) have $e = 3$ by Corollary 10.54; `80b1`, the twist of `20a1` by -1, has type I_0^* and $f_2 = 4$ at 2, and G_{ss} cyclic of order 6 (the order-3 group of `20a1` times the image of the wildly ramified character of $\mathbb{Q}_2(i)$; Proposition 10.77). The potentially multiplicative primes (`539c1` and `980i1` at 7, `121d1` at 11) have no defect: the image of inertia is infinite (Proposition 10.62).

10.7 The Tate curve

At a prime of multiplicative reduction, and at an additive prime with $v(j) < 0$, the curve is (a quadratic twist of) a Tate curve $\bar{K}^\times / q_E^{\mathbb{Z}}$. Everything the paper says at such a prime comes from this description: the term $t_\ell(E, q) = v_\ell(-v_q(j_E))$ of Definition def:terms is the ℓ-part of the

valuation of the Tate parameter; Lemma lem:tamagawa-exp reads the ramification of $E[\ell^t]$ off an ℓ^t-th root of q_E; Lemma lem:tamfactor uses the unit part u of q_E and the matrices of the Galois action. This section constructs the Tate curve, defines q_E and u, describes $E_{\mathfrak{q}}[\ell^t]$, computes the Tamagawa number, determines the type at an additive potentially multiplicative prime, and tabulates the Tate parameters of the paper's curves.

For the reader with a physics background: over $\mathbb{C}$, the exponential $z \mapsto e^{2\pi i z}$ identifies the torus $\mathbb{C}/(\mathbb{Z}+\mathbb{Z}\tau)$ with $\mathbb{C}^\times/\mathfrak{q}^{\mathbb{Z}}$, $\mathfrak{q} = e^{2\pi i\tau}$, the "multiplicative" picture of the same lattice. Tate's observation is that the second description makes sense over a p-adic field, where $\mathbb{C}/\Lambda$ does not.

Theorem 10.57 (Tate)**.** *Let $\mathfrak{q} \in K^\times$ with $v(\mathfrak{q}) > 0$. Put*

$$s_k(\mathfrak{q}) = \sum_{n\geq 1} \frac{n^k \mathfrak{q}^n}{1-\mathfrak{q}^n}, \qquad a_4(\mathfrak{q}) = -5s_3(\mathfrak{q}), \qquad a_6(\mathfrak{q}) = -\frac{5s_3(\mathfrak{q}) + 7s_5(\mathfrak{q})}{12}.$$

(i) The series a_4, a_6 have coefficients in $\mathbb{Z}$ and converge. The equation

$$E_{\mathfrak{q}}\colon\ y^2 + xy = x^3 + a_4(\mathfrak{q})x + a_6(\mathfrak{q})$$

defines an elliptic curve over K with

$$c_4 = 1 + 240 s_3(\mathfrak{q}), \qquad c_6 = -1 + 504 s_5(\mathfrak{q}), \qquad \Delta = \mathfrak{q}\prod_{n\geq 1}(1-\mathfrak{q}^n)^{24},$$

$$j(E_{\mathfrak{q}}) = \frac{1}{\mathfrak{q}} + 744 + 196884\,\mathfrak{q} + 21493760\,\mathfrak{q}^2 + \cdots.$$

(ii) This model is minimal, and $E_{\mathfrak{q}}$ has split multiplicative reduction of type I_n with $n = v(\mathfrak{q}) = -v(j(E_{\mathfrak{q}}))$.

(iii) With

$$X(w,\mathfrak{q}) = \sum_{n\in\mathbb{Z}} \frac{\mathfrak{q}^n w}{(1-\mathfrak{q}^n w)^2} - 2s_1(\mathfrak{q}), \qquad Y(w,\mathfrak{q}) = \sum_{n\in\mathbb{Z}} \frac{(\mathfrak{q}^n w)^2}{(1-\mathfrak{q}^n w)^3} + s_1(\mathfrak{q}),$$

the map $\phi\colon \bar{K}^\times/\mathfrak{q}^{\mathbb{Z}} \to E_{\mathfrak{q}}(\bar{K})$, $w \mapsto (X(w,\mathfrak{q}), Y(w,\mathfrak{q}))$ for $w \notin \mathfrak{q}^{\mathbb{Z}}$ and $w \mapsto O$ for $w \in \mathfrak{q}^{\mathbb{Z}}$, is an isomorphism of groups commuting with the action of G_K. In particular $E_{\mathfrak{q}}(L) \simeq L^\times/\mathfrak{q}^{\mathbb{Z}}$ for every algebraic extension L/K.

Sketch of proof. (i) $s_k(\mathfrak{q}) = \sum_{n\geq 1}\sigma_k(n)\mathfrak{q}^n$ with $\sigma_k(n) = \sum_{d\mid n} d^k$, so $s_k \in \mathfrak{q}\mathbb{Z}[[\mathfrak{q}]]$. For a_6 it suffices that $5d^3 + 7d^5 \equiv 0 \pmod{12}$ for every integer d: modulo 3, $d^3(5+7d^2) \equiv d^3 \cdot 12 \equiv 0$ if $3 \nmid d$, and trivially $d^3 \equiv 0$ if $3 \mid d$; modulo 4, if d is odd then $d^2 \equiv 1$ and $5+7 = 12$, and if d is even then $8 \mid d^3$. Power series with integer coefficients converge at $\mathfrak{q}$ with $v(\mathfrak{q}) > 0$. For the model, $b_2 = 1$, $b_4 = 2a_4$, $b_6 = 4a_6$, so $c_4 = 1 - 48a_4 = 1 + 240 s_3$ and $c_6 = -1 + 72a_4 - 864a_6 = -1 + 504 s_5$. These are E_4 and $-E_6$, the normalised Eisenstein series of weights 4 and 6 in the variable $\mathfrak{q}$ (Chapter 12), and the classical identity $E_4^3 - E_6^2 = 1728\,\mathfrak{q}\prod(1-\mathfrak{q}^n)^{24}$ of power series with integer coefficients gives Δ; $j = c_4^3/\Delta$ follows, and its coefficients are those of the q-expansion of the modular function j (computed with PARI). As $v(\Delta) = v(\mathfrak{q})$ is finite, $\Delta \neq 0$.

(ii) $c_4 \in 1 + \mathfrak{m}$ is a unit and $v(\Delta) = v(\mathfrak{q})$, so the model is minimal (Proposition 10.5(i)) with multiplicative reduction, of type I_n, $n = v(\Delta)$. The reduced curve is $y^2 + xy = x^3$ with the node

at $(0,0)$, and the tangent cone $y^2+xy=y(y+x)$ has the slopes 0 and -1 in k: split. Finally $v(j)=3v(c_4)-v(\Delta)=-n$.

(iii) Over $\mathbb{C}$, with $\mathfrak{q}=e^{2\pi i\tau}$ and $w=e^{2\pi iz}$, one has $X=(2\pi i)^{-2}\wp(z;\mathbb{Z}+\mathbb{Z}\tau)-\frac{1}{12}$ and $Y=(2\pi i)^{-3}\frac{1}{2}\wp'(z)-\frac{1}{2}X$, so the identities $Y^2+XY=X^3+a_4X+a_6$ and the addition law hold as identities of formal series with rational coefficients (we also checked the first numerically to 40 digits). Tate's theorem is that they hold p-adically and that ϕ is bijective; see [22, V.3]. We do not prove it. □

Proposition 10.58 (inverting j)**.** *There is a unique power series*

$$\mathfrak{q}(J)=J+744J^2+750420J^3+872769632J^4+1102652742882J^5+\cdots\in J\,\mathbb{Z}[[J]]$$

such that $j(\mathfrak{q}(J))=1/J$ as formal series. For $j_0\in K$ with $v(j_0)<0$, the element $\mathfrak{q}_0=\mathfrak{q}(1/j_0)$ is the unique $\mathfrak{q}_0\in K^\times$ with $v(\mathfrak{q}_0)>0$ and $j(E_{\mathfrak{q}_0})=j_0$, and $v(\mathfrak{q}_0)=-v(j_0)$.

Proof. Write $j(\mathfrak{q})=\mathfrak{q}^{-1}U(\mathfrak{q})$ with $U\in 1+\mathfrak{q}\mathbb{Z}[[\mathfrak{q}]]$. Then $J(\mathfrak{q}):=1/j(\mathfrak{q})=\mathfrak{q}\,U(\mathfrak{q})^{-1}\in\mathfrak{q}+\mathfrak{q}^2\mathbb{Z}[[\mathfrak{q}]]$. A power series $J(\mathfrak{q})=\mathfrak{q}+\sum_{n\ge2}b_n\mathfrak{q}^n$ with $b_n\in\mathbb{Z}$ has a compositional inverse $\mathfrak{q}(J)=J+\sum_{n\ge2}c_nJ^n$ with $c_n\in\mathbb{Z}$: the coefficient of J^n in $J(\mathfrak{q}(J))$ is c_n plus a polynomial with integer coefficients in $c_2,\dots,c_{n-1}$ and the b_i, so the equation $J(\mathfrak{q}(J))=J$ determines the c_n recursively in $\mathbb{Z}$. The coefficients printed were computed with PARI's `serreverse`. Since both series have integer coefficients, substituting an element of $\mathfrak{m}$ gives convergent series, and substitution respects composition. Put $\mathfrak{q}_0=\mathfrak{q}(1/j_0)$; then $v(\mathfrak{q}_0)=v(1/j_0)=-v(j_0)>0$ (the leading term dominates), and $J(\mathfrak{q}_0)=1/j_0$, so $j(E_{\mathfrak{q}_0})=j_0$. If also $v(\mathfrak{q}_1)>0$ and $j(E_{\mathfrak{q}_1})=j_0$, then $\mathfrak{q}_1=\mathfrak{q}(J(\mathfrak{q}_1))=\mathfrak{q}(1/j_0)=\mathfrak{q}_0$. □

Theorem 10.59 (curves with $v(j)<0$)**.** *Let E/K have $v(j)<0$, let $q_E=\mathfrak{q}(1/j_E)$ be as in Proposition 10.58, and let $\gamma=-c_4/c_6\in K^\times/(K^\times)^2$, computed from any model of E. Then E is the quadratic twist of E_{q_E} by γ: $E\simeq E_{q_E}$ over $K(\sqrt{\gamma})$, and over K if and only if $\gamma\in(K^\times)^2$. Moreover:*

(i) E has split multiplicative reduction if and only if $\gamma\in(K^\times)^2$;

(ii) E has nonsplit multiplicative reduction if and only if $K(\sqrt{\gamma})/K$ is the unramified quadratic extension;

(iii) E has additive reduction if and only if $K(\sqrt{\gamma})/K$ is ramified.

In the multiplicative case $n=v(\Delta_{\min})=-v(j)=v(q_E)$.

Sketch of proof. As $v(j)<0$, $j\neq0,1728$, so $c_4c_6\neq0$. For $E_{\mathfrak{q}}$, $\gamma(E_{\mathfrak{q}})=(1+240s_3)/(1-504s_5)=1+24(10s_3+21s_5)/(1-504s_5)$ lies in $1+24\mathfrak{m}$, which consists of squares by Hensel's lemma (for p odd, $1+\mathfrak{m}$ consists of squares; for $p=2$, $x^2=1+8m$ with $v(8m)>v(4)$ has a root near 1). Two curves with the same $j\neq0,1728$ are twists of each other by $d=c_6(E)c_4(E_{\mathfrak{q}})/(c_6(E_{\mathfrak{q}})c_4(E))=\gamma(E_{\mathfrak{q}})/\gamma(E)$ (Chapter 9), and $d\equiv\gamma(E)$ modulo squares. This proves the first statement. If γ is a square, $E\simeq E_{q_E}$ has split multiplicative reduction by Theorem 10.57(ii). If $K(\sqrt{\gamma})/K$ is unramified, E acquires multiplicative reduction over it, hence has multiplicative reduction (Proposition 10.31); for p odd, computing γ from a minimal model makes it a unit (c_4 and c_6 are units), which is not a square, so by Hensel's lemma its reduction is not a square, and Proposition 10.12(ii) shows the reduction is nonsplit. If E has multiplicative reduction and p is odd, c_4 and c_6 are units for a minimal model, so $K(\sqrt{\gamma})$ is unramified; hence a ramified γ gives additive reduction. This proves (i) to (iii) for p odd; for $p=2$ see [22, V.5]. The last sentence is Theorem 10.21(iii) and Proposition 10.58. □

Definition 10.60 (Tate parameter and unit part)**.** Let $E/\mathbb{Q}_q$ have $v_q(j_E) < 0$. The *Tate parameter* of E is $q_E = \mathfrak{q}(1/j_E) \in \mathbb{Q}_q^\times$, and

$$\nu = v_q(q_E) = -v_q(j_E) \geq 1;$$

at a multiplicative prime $\nu = n_q = v_q(\Delta_E)$ is the paper's n. The *unit part* is $u = q_E q^{-\nu} \in \mathbb{Z}_q^\times$, with reduction $\bar{u} \in \mathbb{F}_q^\times$; for a prime $\ell \neq q$, $c(u)$ denotes the class of $\bar{u}$ in $\mathbb{F}_q^\times/(\mathbb{F}_q^\times)^{\ell^{v_\ell(q-1)}}$ (the paper's c in Lemma lem:tamfactor; the arithmetic of this class is in Chapter 2).

Proposition 10.61 (invariance of the Tate parameter)**.** *Let $E/\mathbb{Q}_q$ have $v_q(j_E) < 0$.*

(i) q_E, ν, u and $c(u)$ depend only on j_E; in particular they are the same for E and for every quadratic twist of E, and $t_\ell(E, q) = v_\ell(\nu)$ is invariant under quadratic twist.

(ii) If $\varphi\colon E \to E'$ is an isogeny over $\mathbb{Q}_q$ of prime degree d, then $\nu(E') \in \{d\,\nu(E),\ \nu(E)/d\}$. Consequently $v_\ell(\nu)$ and $t_\ell(E, q)$ are unchanged by isogenies of degree prime to ℓ.

Proof. (i) is the definition: q_E is a function of j_E, and twisting does not change j. (ii) *Sketch.* Over a finite extension L of $\mathbb{Q}_q$ over which E and E' are both split, $E(\bar{L}) = \bar{L}^\times/q_E^{\mathbb{Z}}$, and $\ker\varphi$ is a subgroup C of order d. If $C = \mu_d$, then $w \mapsto w^d$ identifies $\bar{L}^\times/\mu_d q_E^{\mathbb{Z}}$ with $\bar{L}^\times/q_E^{d\mathbb{Z}}$, so $E' \simeq E_{q_E^d}$ and $\nu(E') = d\nu(E)$. Otherwise C is generated by the class of $\zeta q_E^{1/d}$ for some $\zeta \in \mu_d$, and $\bar{L}^\times/\langle \zeta q_E^{1/d}\rangle$ is the Tate curve with parameter $\zeta q_E^{1/d}$, so $\nu(E') = \nu(E)/d$. That an isogeny of Tate curves is induced by the corresponding map of multiplicative groups is part of Tate's theory [22, V.3, V.5]. For the consequence: since $d \neq \ell$, $v_\ell(d\nu) = v_\ell(\nu/d) = v_\ell(\nu)$. (A second proof: an isogeny of degree prime to ℓ induces an isomorphism $T_\ell E \simeq T_\ell E'$ of Galois modules, and by Lemma lem:tamagawa-exp (proved in Chapter 11) $v_\ell(\nu)$ is the largest t such that $E[\ell^t]$ is unramified after twisting E by the quadratic character of Theorem 10.59, which commutes with isogenies.) □

Proposition 10.62 (torsion of the Tate curve)**.** *Let $\mathfrak{q} \in K^\times$ with $v(\mathfrak{q}) > 0$, ℓ a prime and $t \geq 1$.*

(i) In $E_\mathfrak{q}(\bar{K}) = \bar{K}^\times/\mathfrak{q}^{\mathbb{Z}}$, fix an ℓ^t-th root $\mathfrak{q}^{1/\ell^t}$ and a primitive ℓ^t-th root of unity ζ. Then $E_\mathfrak{q}[\ell^t] = \{\zeta^a(\mathfrak{q}^{1/\ell^t})^b : a, b \in \mathbb{Z}/\ell^t\}$, and there is an exact sequence of G_K-modules

$$0 \longrightarrow \mu_{\ell^t} \longrightarrow E_\mathfrak{q}[\ell^t] \longrightarrow \mathbb{Z}/\ell^t \longrightarrow 0,$$

the quotient being generated by the class of $\mathfrak{q}^{1/\ell^t}$.

(ii) Let $\ell \neq p$, choose compatible roots ζ_{ℓ^k} and $\mathfrak{q}^{1/\ell^k}$, and let $e_1 = (\zeta_{\ell^k})_k$, $e_2 = (\mathfrak{q}^{1/\ell^k})_k$ be the corresponding basis of $T_\ell E_\mathfrak{q}$. Then $\sigma \in G_K$ acts by

$$\begin{pmatrix} \chi_\ell(\sigma) & \kappa(\sigma) \\ 0 & 1 \end{pmatrix},$$

where κ is the Kummer cocycle of $\mathfrak{q}$: $\sigma(\mathfrak{q}^{1/\ell^k}) = \zeta_{\ell^k}^{\kappa(\sigma)}\mathfrak{q}^{1/\ell^k}$ for all k (Chapter 5).

(iii) If E is the twist of $E_\mathfrak{q}$ by a quadratic character ε of G_K (Theorem 10.59), then G_K acts on $T_\ell E$ by

$$\begin{pmatrix} \chi_\ell\varepsilon & \varepsilon\kappa \\ 0 & \varepsilon \end{pmatrix}.$$

Proof. (i) A class $w \in \bar{K}^\times/\mathfrak{q}^{\mathbb{Z}}$ is killed by ℓ^t if and only if $w^{\ell^t} = \mathfrak{q}^b$ for some $b \in \mathbb{Z}$, that is, $w = \zeta^a \mathfrak{q}^{b/\ell^t}$ with $\mathfrak{q}^{b/\ell^t} = (\mathfrak{q}^{1/\ell^t})^b$. Changing b by ℓ^t multiplies w by $\mathfrak{q}$, so b matters only modulo ℓ^t, and w is trivial if and only if $a \equiv b \equiv 0$. The map $w \mapsto b$ is a homomorphism onto $\mathbb{Z}/\ell^t$ with kernel μ_{ℓ^t}, and it commutes with G_K because $\sigma(\mathfrak{q}^{1/\ell^t})$ is again an ℓ^t-th root of $\mathfrak{q}$. By Theorem 10.57(iii) these are the ℓ^t-torsion points of $E_{\mathfrak{q}}$, with their Galois action. (ii) In additive notation $\sigma(e_1) = \chi_\ell(\sigma)e_1$ by definition of χ_ℓ, and $\sigma(e_2) = e_2 + \kappa(\sigma)e_1$ by definition of κ; the columns of the matrix are the images of e_1 and e_2. (iii) $T_\ell E \simeq T_\ell E_{\mathfrak{q}} \otimes \varepsilon$ as G_K-modules (Chapter 9): every matrix of (ii) is multiplied by $\varepsilon(\sigma)$. In particular the upper right entry is $\varepsilon\kappa$, which satisfies the cocycle relation $c(gh) = \chi_\ell\varepsilon(g)c(h) + c(g)\varepsilon(h)$ for these matrices; κ alone does not when $\varepsilon \neq 1$. □

Warning 10.63. In the nonsplit and the additive potentially multiplicative cases the corner of the matrix is $\varepsilon\kappa$, not κ. The proof of Lemma lem:tamfactor in the original version of the paper wrote κ; the revised version writes $\varepsilon\kappa$. In the nonsplit multiplicative case (the setting of Lemma lem:tamfactor) ε is unramified, so nothing in the conclusion changes: ε is trivial on inertia and only changes the Frobenius entry by a sign. In the additive potentially multiplicative case ε is the ramified character of Theorem 10.59(iii), so it is not trivial on inertia; there the paper does not use the matrix of T but applies the lemma to E_{q_E}, since $\mathrm{ad}^0 T$ is unchanged by a quadratic twist (Chapter 18).

In the paper 10.64. Lemma lem:tamagawa-exp: "The ℓ^t-torsion is generated by μ_{ℓ^t} and an ℓ^t-th root of q_E" is Proposition 10.62(i), applied over the unramified quadratic extension, where a nonsplit curve becomes a Tate curve (Theorem 10.59) and the inertia group is unchanged. The proof of the lemma (inertia acts trivially on $E[\ell^t]$ if and only if $\ell^t \mid n_q$) is given in Chapter 11. Lemma lem:tamfactor uses the matrices of (iii), the decomposition $q_E = q^\nu u$, and the restriction of κ to inertia, which is ν times the tame character (Chapter 7); the cohomology is done in Chapter 16, Chapter 17 and Chapter 18.

Proposition 10.65 (Tamagawa numbers at multiplicative primes)**.** *Let E/K have multiplicative reduction of type I_n, k finite.*

(i) If the reduction is split, $E(K) \simeq K^\times/q_E^{\mathbb{Z}}$, $E_0(K)$ corresponds to $\mathcal{O}^\times q_E^{\mathbb{Z}}/q_E^{\mathbb{Z}}$, and the valuation induces $E(K)/E_0(K) \simeq \mathbb{Z}/n$; so $c = n$.

(ii) In both cases $\Phi^{\mathrm{geom}} \simeq \mathbb{Z}/n$. If the reduction is nonsplit, Frobenius acts on Φ^{geom} by -1, so $c = \#\Phi_E(k)$ is 1 for n odd and 2 for n even.

Proof. (i) By Theorem 10.59, $E \simeq E_{q_E}$, and $E(K) \simeq K^\times/q_E^{\mathbb{Z}}$ by Theorem 10.57(iii). Every class has a representative w with $0 \le v(w) < n$. Write $X(w, q_E) = \sum_{i\in\mathbb{Z}} X_i - 2s_1(q_E)$ and $Y(w, q_E) = \sum_i Y_i + s_1(q_E)$ termwise. For $i = -m \le -1$ one has $X_i = q_E^m w/(q_E^m - w)^2$ and $Y_i = q_E^m w^2/(q_E^m - w)^3$, and $s_1(q_E) \in \mathfrak{m}$. If $v(w) = 0$ and $\bar{w} \neq 1$, every term other than X_0, Y_0 lies in $\mathfrak{m}$, so $\phi(w)$ reduces to $(\bar{w}/(1-\bar{w})^2,\ \bar{w}^2/(1-\bar{w})^3)$, a point of $y^2 + xy = x^3$ other than the node $(0,0)$. If $\bar{w} = 1$, X has a pole and $\phi(w)$ reduces to $\tilde{O}$. If $0 < v(w) < n$, then $v(X_0) = v(w)$, $v(Y_0) = 2v(w)$, $v(X_{-1}) = v(Y_{-1}) = n - v(w)$, and all the other terms have even larger valuation; all are positive, so $\phi(w)$ reduces to the node. Hence $E_0(K)$ is the image of $\mathcal{O}^\times q_E^{\mathbb{Z}}$, and $w \mapsto v(w) \bmod n$ gives $E(K)/E_0(K) \simeq \mathbb{Z}/n$.

(ii) Let K' be the unramified quadratic extension, over which E is split (Proposition 10.12(iii) for p odd, Theorem 10.59 in general). By (i) over every finite unramified extension of K' the component group has $\mathbb{Z}/n$ points, with trivial Galois action, so $\Phi^{\mathrm{geom}} \simeq \mathbb{Z}/n$ (Proposition 10.31).

If E is nonsplit it is the twist of E_{q_E} by the unramified quadratic character, and the nontrivial element σ of $\mathrm{Gal}(K'/K)$ acts on $E(K') \simeq K'^{\times}/q_E^{\mathbb{Z}}$ by $w \mapsto \sigma(w)^{-1}$ (the twisted action, Chapter 9). On $E(K')/E_0(K') \simeq K'^{\times}/\mathcal{O}'^{\times}q_E^{\mathbb{Z}} \simeq \mathbb{Z}/n$ this is $a \mapsto -a$. The invariants of -1 on $\mathbb{Z}/n$ are $\{a : 2a = 0\}$, of order $\gcd(2, n)$. □

All 153,343 pairs (curve, prime of multiplicative reduction) of Cremona's tables below conductor 10,000 obey this rule.

Theorem 10.66 (types at additive potentially multiplicative primes)**.** *Let E/K have additive reduction with $v(j) < 0$, and put $\nu = -v(j) \geq 1$.*

(i) If p is odd, E has type I_ν^*, $f = 2$ and $v(\Delta_{\min}) = \nu + 6$. Conversely, a curve of type I_n^* with $n \geq 1$ at an odd p has $v(j) < 0$, hence $n = -v(j)$.

(ii) If $p = 2$ and $K = \mathbb{Q}_2$, E has type I_m^* with $m = \nu + 2f_2 - 4$, where $f_2 = 2a(\chi) \in \{4, 6\}$ and $a(\chi) \in \{2, 3\}$ is the conductor exponent of the ramified quadratic character χ of Theorem 10.59; so $m = \nu + 4$ or $m = \nu + 8$.

Sketch of proof. (i) By Theorem 10.59, γ is in a ramified class, so $\gamma = \varpi w$ with w a unit up to squares, and $E \simeq E''^{(\varpi)}$ where $E'' = E_{q_E}^{(w)}$ has multiplicative reduction (its γ is the unramified class of w) with $v(\Delta''_{\min}) = \nu$. Take a minimal model of E'' of the form $y^2 = x^3 + a_2x^2 + a_4x + a_6$ (complete the square; p is odd) and translate the node to $(0, 0)$, so that $\varpi \mid a_4, a_6$ and $\varpi \nmid a_2$ (the reduced cubic is $x^2(x + \tilde{a}_2)$ with a double, not triple, root). The twist by ϖ is $y^2 = x^3 + \varpi a_2x^2 + \varpi^2 a_4x + \varpi^3 a_6$. As in the proof of Proposition 10.39, Tate's algorithm passes Steps 2 to 5 and reaches Step 6 with $P(T) = T^3 + a_2T^2 + a_4T + a_6 \equiv T^2(T + \tilde{a}_2)$, which has a double and a simple root: the type is I_m^* for some $m \geq 1$, and the model is minimal, with $v(\Delta) = \nu + 6$. By Proposition 10.46(iv), $f = 2$, and Ogg's formula (Theorem 10.47) gives $m + 5 = v(\Delta) + 1 - f = \nu + 5$, so $m = \nu$.

For the converse, let E have type I_n^*, $n \geq 1$, at odd p. As in the converse part of the proof of Proposition 10.39, there is a minimal model $y^2 = x^3 + a_2x^2 + a_4x + a_6$ with $\varpi \mid a_2$, $\varpi^2 \mid a_4$, $\varpi^3 \mid a_6$ whose cubic $\tilde{P}$ has a double and a simple root. The model $y^2 = x^3 + (a_2/\varpi)x^2 + (a_4/\varpi^2)x + a_6/\varpi^3$ of the twist by ϖ is integral and reduces to $y^2 = \tilde{P}(x)$, which has a node. So the twist has multiplicative reduction, and $v(j) = v(j(E^{(\varpi)})) < 0$.

(ii) We do not prove this. It is a special case of Lorenzini's description of the minimal regular model of a quadratic twist of a curve with multiplicative reduction [12] (we follow the citation in [18] and have not checked the numbering there); $f_2 = 2a(\chi)$ is the conductor exponent of the twist of the Steinberg representation by χ (Chapter 7), and $a(\chi) = 2$ for $\mathbb{Q}_2(\sqrt{-1})$, $\mathbb{Q}_2(\sqrt{3})$ and 3 for $\mathbb{Q}_2(\sqrt{\pm 2})$, $\mathbb{Q}_2(\sqrt{\pm 10})$. □

On Cremona's tables below conductor 10,000, all 17,761 additive pairs with $v_p(j) < 0$ have a symbol I_m^* (3,732 at $p = 2$, 7,805 at $p = 3$, the rest at $p \geq 5$); at odd p always $m = \nu$; at $p = 2$, $m - \nu = 4$ with $f_2 = 4$ in 2,742 cases and $m - \nu = 8$ with $f_2 = 6$ in 990 cases. Examples at 2: `208a1` (I_5^*, $\nu = 1$, $f_2 = 4$), `304b1` (I_7^*, $\nu = 3$, $f_2 = 4$), `832c1` (I_9^*, $\nu = 1$, $f_2 = 6$), `336d1` (I_{12}^*, $\nu = 8$, $f_2 = 4$).

Example 10.67 (Tate parameters of the paper's curves)**.** The Tate parameters $q_E = q^\nu u$ at the potentially multiplicative primes of the running examples, computed from the series of Proposition 10.58 (and, at odd q, also with PARI's `ellinit(E, O(q^k)).tate`, with the same result):

curve, prime	type	a_q	ν	$\bar{u} \bmod q$	remark
`11a1`, 11	I_5	1	5	$10 \equiv -1$	$(\mathbb{F}_{11}^\times)^5 = \{\pm 1\}$: $\bar{u}$ is a 5th power
`11a3`, 11	I_1	1	1	8	
`20a1`, 5	I_2	-1	2	1	nonsplit
`77c1`, 7	I_3	-1	3	5	same q_E as `539c1` at 7 (I_3^*)
`77c1` and `539c1`, 11	I_2	1	2	3	
`140b1`, 7	I_5	-1	5	1	same q_E as `980i1` at 7 (I_5^*)
`140b1`, 5	I_1	-1	1	2	nonsplit
`980i1`, 5	I_1	1	1	2	split: the twist by -7, $\chi_{-7}(5) = -1$
`726a1`, 2	I_6	-1	6	1	
`726a1`, 3	I_3	-1	3	2	
`121d1`, 11	I_1^*	0	1	8	same q_E as `11a3`
`121d2`, 11	I_5^*	0	5	10	same q_E as `11a1`

For instance $q_E = 10 \cdot 11^5 + 2 \cdot 11^6 + 6 \cdot 11^7 + \cdots$ for `11a1`. The values of t_ℓ: $t_5(\texttt{11a1}, 11) = 1$; $t_3(\texttt{539c1}, 7) = t_3(\texttt{77c1}, 7) = v_3(3) = 1$; $t_5(\texttt{980i1}, 7) = 1$; $t_3(\texttt{726a1}, 2) = v_3(6) = 1$. At $q = 3$, `726a1` has $v_3(\nu) = 1$, which the appendix's `rule` counts as $t_3(E, 3)$; the revised Definition def:terms defines $t_\ell(E, q)$ for every bad q, including $q = \ell$.

Warning 10.68 (PARI's Tate parameter at 2). In PARI/GP 2.17.2, the value `ellinit(E, O(2^k)).tate[3]` does not return the Tate parameter when $v_2(j) = -1$: in all 1,262 such pairs among the curves of conductor below 3,000 it returned q_E^5 (for instance $2^5 + 2^6 + \cdots$ for `208a1`, whose Tate parameter is $2 + 2^2 + 2^4 + \cdots$), while for every other pair with $v_p(j) < 0$ in that range ($\nu \geq 2$ at 2, and all odd p) it agreed with the series of Proposition 10.58. Before using PARI's value, check that $j(\texttt{q}) = j_E$, or compute q_E from the series. The paper's computations use $v_q(j)$ directly and are not affected.

10.8 Quadratic twists and reduction

The paper twists constantly: by χ_{q^*} at a prime of type I_0^* or I_n^* (Theorems thm:qual (b), (c) and thm:quant, Lemma lem:euler), by a product χ_S of such characters (Theorems thm:twist and thm:quant), and in the definition of the twist-symmetrised weight $w_\ell(E, q)$ of its Section 1.4. This section proves the paper's Lemma lem:twist (i) and (ii), which say what twisting does to Kodaira types, component groups and conductor exponents; part (iii) of that lemma (Atkin and Li) is about newforms and is proved in Chapter 13. Recall (Chapter 9) that for $d \in \mathbb{Q}^\times$ the twist $E^{(d)}$ of a model $y^2 = x^3 + a_2x^2 + a_4x + a_6$ is $y^2 = x^3 + da_2x^2 + d^2a_4x + d^3a_6$, with invariants d^2c_4, d^3c_6, $d^6\Delta$ and the same j, and that $\rho_{E^{(d)},\ell} \simeq \rho_{E,\ell} \otimes \chi$, where χ is the quadratic character of $\mathbb{Q}(\sqrt{d})$; following decision D3 it is written χ_D, indexed by the discriminant D of $\mathbb{Q}(\sqrt{d})$ (for $d = 11$, $D = 44$ and $\chi = \chi_{-4}\chi_{-11}$). For an odd prime q, the paper's $E^{(q)}$ is the twist by $q^* = (-1)^{(q-1)/2}q$, that is, by χ_{q^*} and the field $\mathbb{Q}(\sqrt{q^*})$, which is unramified outside q (Chapter 6).

Lemma 10.69 (unramified twists; the paper's Lemma lem:twist (i))**.** *Let $E/\mathbb{Q}$, p a prime and $d \in \mathbb{Q}^\times$ such that $\chi = \chi_D$, the quadratic character of $\mathbb{Q}(\sqrt{d})$ (D its discriminant), is unramified at p (for instance χ_{q^*} with $q \neq p$ an odd prime; at $p = 2$ this uses $q^* \equiv 1 \pmod 4$). Then E and $E^{(d)}$ have at p the same reduction type, the same Kodaira symbol, the same geometric component group Φ_p^{geom}, the same $v_p(\Delta_{\min})$, the same conductor exponent f_p, the same $v_p(j)$ and the same Tate parameter. The Tamagawa number c_p can change.*

Proof. E and $E^{(d)}$ become isomorphic over $\mathbb{Q}_p(\sqrt{d})$, which is $\mathbb{Q}_p$ or the unramified quadratic extension of $\mathbb{Q}_p$. By Proposition 10.31, the reduction type, the Kodaira symbol, Φ_p^{geom} and $v_p(\Delta_{\min})$ are unchanged by this extension, so they agree for E and $E^{(d)}$. The conductor exponents agree because $V_\ell E^{(d)} \simeq V_\ell E \otimes \chi$ and χ is trivial on I_p, so the two representations of I_p coincide. The j-invariant, hence $v_p(j)$ and the Tate parameter, does not change under twisting. An example where c_p changes: `11a1` has split multiplicative reduction of type I_5 at 11, with $c_{11} = 5$; its twist by 13 (and 13 is not a square modulo 11, so χ_{13} is unramified at 11 with $\chi_{13}(\mathrm{Frob}_{11}) = -1$) is `1859a2`, of type I_5 nonsplit at 11, with $c_{11} = 1$; so is its twist by -3, `99d2`. □

Lemma 10.70 (ramified twists at odd primes; the paper's Lemma lem:twist (ii))**.** *Let q be an odd prime and $E^{(q)} = E^{(q^*)}$.*

(a) If E has type I_n at q with $n \geq 1$, then $E^{(q)}$ has type I_n^* at q, with the same n, and conversely.

(b) If E has type I_0 at q, then $E^{(q)}$ has type I_0^*, and conversely.

(c) If $q \geq 5$, the twist exchanges $\mathrm{II} \leftrightarrow \mathrm{IV}^*$, $\mathrm{III} \leftrightarrow \mathrm{III}^*$, $\mathrm{IV} \leftrightarrow \mathrm{II}^*$, and $v_q(\Delta_{\min})$ changes by $+6$ modulo 12.

(d) The conductor exponent $f_q(E^{(q)})$ is 0 if $E^{(q)}$ has good reduction, 1 if it has type I_n with $n \geq 1$, and 2 if it has additive reduction and is one of the curves in (a), (b), (c).

At every prime $p \neq q$ the two curves have the same type, Φ_p^{geom} and f_p (Lemma 10.69).

Proof. Throughout, q is odd and we use the twisted model of a model $y^2 = x^3 + a_2x^2 + a_4x + a_6$ (complete the square first; for q odd this does not change minimality at q). Its invariants are $(q^*)^2c_4$, $(q^*)^3c_6$, $(q^*)^6\Delta$: the twist by q^*, not by q, though only q-adic valuations are used below, and those are the same for q and q^*.

(a) Here $v_q(j) < 0$. By Theorem 10.59 the reduction type of a curve with $v_q(j) < 0$ is determined by the square class of $\gamma = -c_4/c_6$: unramified for multiplicative, ramified for additive reduction. The twist multiplies γ by $(q^*)^2/(q^*)^3 = 1/q^*$, which moves an unramified class to a ramified one and back. So E is multiplicative if and only if $E^{(q)}$ is additive. By Theorem 10.66(i) the additive one has type I_ν^*, and the multiplicative one type I_ν, with the same $\nu = -v_q(j)$, as j is unchanged. (The identification of the index of I_ν^* with ν uses Ogg's formula, through Theorem 10.66.)

(b) If E has good reduction at q, Proposition 10.39 shows that $E^{(q)}$ has type I_0^* (with the sign $\epsilon = q^*/q$ in the cubic). Conversely, if E has type I_0^*, Proposition 10.39 gives a model $y^2 = x^3 + a_2x^2 + a_4x + a_6$ with $q \mid a_2$, $q^2 \mid a_4$, $q^3 \mid a_6$ and $\tilde{P}$ separable. The twist by q^* is $y^2 = x^3 + q^*a_2x^2 + (q^*)^2a_4x + (q^*)^3a_6$, and the change $(q, 0, 0, 0)$ turns it into $y^2 = x^3 + \epsilon a_{2,1}x^2 + a_{4,2}x + \epsilon a_{6,3}$ with $\epsilon = q^*/q$ and $a_{i,j} = a_i/q^j$. This model is integral, and its cubic has the roots $\epsilon\theta$ for the roots θ of P, which are distinct modulo q; so its discriminant is a unit and $E^{(q)}$ has good reduction.

(c) Let $q \geq 5$ and let E be potentially good and additive, with minimal $v = v_q(\Delta_{\min}) \in \{2, 3, 4, 6, 8, 9, 10\}$ (Corollary 10.38). The twisted model has $v_q(\Delta) = v + 6$ and $v_q(c_4)$ raised by 2. If $v + 6 < 12$ it is minimal (Proposition 10.5(i)). If $v + 6 \geq 12$, then, as $v_q(j) \geq 0$ is unchanged, $3v_q(c_4') \geq v_q(\Delta') \geq 12$ for the twisted model, so $v_q(c_4') \geq 4$ and one step of Proposition 10.5(ii) gives a minimal model with $v_q(\Delta) = v - 6 < 12$. So the new minimal valuation is $v + 6$ or $v - 6$, that is, $v + 6$ mod 12: $2 \leftrightarrow 8$, $3 \leftrightarrow 9$, $4 \leftrightarrow 10$ (and $0 \leftrightarrow 6$, which is (b)). By Corollary 10.38(iii) this is $\mathrm{II} \leftrightarrow \mathrm{IV}^*$, $\mathrm{III} \leftrightarrow \mathrm{III}^*$, $\mathrm{IV} \leftrightarrow \mathrm{II}^*$.

(d) The values 0 and 1 are Proposition 10.46(i), (ii). In the additive cases of (a), (b), (c) we have $V_\ell E^{(q)} = V_\ell E \otimes \chi_{q^*}$, where χ_{q^*} is tamely ramified at q (it is trivial on the pro-q group P_q, as q is odd), and $V_\ell E$ is tame at q: it is unramified (case (b)), multiplicative (case (a), Proposition 10.46(ii)), or additive with $q \geq 5$ (case (c), Proposition 10.46(iv)). So P_q acts trivially on $V_\ell E^{(q)}$, $\mathrm{sw} = 0$, and $f_q(E^{(q)}) = 2$ by Proposition 10.46(iii). □

All 90,970 pairs (curve numbered 1 in its isogeny class, odd bad prime q) of Cremona's tables below conductor 10,000 behave as stated: the same n in all 78,351 cases of (a), the shift of $v_q(\Delta_{\min})$ by 6 modulo 12 in all 5,264 potentially good additive cases with $q \geq 5$ other than I_0^*, and the same type, conductor exponent and $v_p(\Delta_{\min})$ at every other bad prime.

Warning 10.71. The paper's proof of Lemma lem:twist (ii) in its original version wrote the invariants of the twist as q^2c_4, q^3c_6, $q^6\Delta$, which are those of the twist by q; the twist by q^* has $(q^*)^2c_4$, $(q^*)^3c_6$, $(q^*)^6\Delta$. For $q \equiv 3 \pmod 4$ the two are different curves, differing by the twist by -1, which is ramified at 2: the twist of `121a1` by -11 is `121c2`, of conductor 121, while its twist by 11 is `1936k2`, of conductor $2^4 \cdot 11^2$ (type I_4^* at 2). Likewise the phrase "the conductor exponents are the standard ones for q odd (tame)" is right for the exchanges of (a), (b), (c), and wrong for general additive reduction at $q = 3$, which can be wild (Remark 10.79).

Proposition 10.72 (the exchanges at $q = 3$)**.** *Let $E/\mathbb{Q}$ have additive, potentially good reduction at 3 of type* II, III, IV, IV*, III* *or* II*, *and let $E^{(3)} = E^{(-3)}$. Then $f_3(E^{(3)}) = f_3(E)$, and the type of $E^{(3)}$ at 3 is* IV*, III*, II*, II, III, IV *respectively.*

Proof. The conductor: $V_\ell E^{(3)} = V_\ell E \otimes \chi_{-3}$ with χ_{-3} tamely ramified at 3, so P_3 acts on both through the same representation and the Swan conductors agree; $E^{(3)}$ is additive: it is not multiplicative, as $v_3(j) \geq 0$, and it does not have good reduction, since otherwise $E = (E^{(3)})^{(3)}$ would have type I_0^* by Lemma 10.70(b). So $V^{I_3} = 0$ for both and $f_3(E^{(3)}) = f_3(E) =: f$. The discriminants: by the formula of Chapter 9 for the twist of a general model, the twisted model has discriminant $2^{12}(-3)^6\Delta$, so $v_3(\Delta_{\min}(E^{(3)})) \equiv v_3(\Delta_{\min}(E)) + 6 \pmod{12}$. The components: both curves are potentially good (j is unchanged and $v_3(j) \geq 0$), so neither has type I_n or, by the converse in Theorem 10.66(i), I_n^* with $n \geq 1$; so both have $1 \leq m \leq 9$, and Ogg's formula (Theorem 10.47) gives $v_3(\Delta_{\min}) = m + f - 1 \in [f, f+8]$ for both. Two integers in an interval of length 8 that are congruent modulo 12 after adding 6 to one of them differ by exactly 6; so $m(E^{(3)}) = m(E) \pm 6$. The types with $m \in \{1,2,3\}$ and additive potentially good reduction are II, III, IV, and those with $m \in \{7,8,9\}$ are IV*, III*, II*. Since $m(E) \in \{1,2,3,7,8,9\}$ and $1 \leq m(E^{(3)}) = m(E) \pm 6 \leq 9$, $m(E^{(3)}) = m(E) + 6$ if $m(E) \leq 3$ and $m(E) - 6$ if $m(E) \geq 7$. This gives the exchanges. □

This answers, given Ogg's formula, the question left open in the revised paper (Open question 6) whether the exchanges of Lemma lem:twist (ii) hold at $q = 3$. All 5,122 such pairs below conductor 10,000 (curves numbered 1) behave as stated. Examples: `54b1` (type II, $f_3 = 3$) and its twist `54a1` (type IV*, $f_3 = 3$); `27a1` (IV*) and `27a3` (II); `36a1` (III, $f_3 = 2$) and `36a3` (III*).

In the paper 10.73. Section 1.4 of the paper lists the exchanges "the first and the last for every odd q, the three others for $q \geq 5$"; Proposition 10.72 shows that the three others also hold at $q = 3$. Lemma lem:euler ("$V = V' \otimes \chi_{q^*}$ with V' Steinberg at q" for I_n^*, and unramified for I_0^*) uses (a) and (b). Theorems thm:qual (b), (c) and thm:quant twist a type II, II*, I_0^* or I_n^* prime to a type IV*, IV, good or multiplicative one; Theorem thm:quant needs this at every odd q in S, which (a) and (b) provide. The proof of Theorem thm:main uses the multiplicative twist at

an I_n^* prime; at $q=2$, where Lemma 10.70 does not apply, the local statement Theorem 10.59 (a ramified quadratic twist of a Tate curve) is what is needed.

Proposition 10.74 (the twist-symmetrised weight)**.** *For a bad prime q and an odd prime ℓ, let $w_\ell(E,q)$ be the paper's weight: $v_\ell(n)$ if the type at q is I_n, or I_n^* with q odd; 1 if $\ell=3$ and the type is* IV *or* IV**, or* II *or* II* *with q odd; 0 otherwise. Then*

$$w_\ell(E,q)=\max\bigl(v_\ell(\#\Phi_q^{\mathrm{geom}}(E)),\, v_\ell(\#\Phi_q^{\mathrm{geom}}(E^{(q)}))\bigr) \text{ for odd } q, \qquad w_\ell(E,2)=v_\ell(\#\Phi_2^{\mathrm{geom}}(E)).$$

Proof. For odd q use the exchanges of Lemma 10.70 and Proposition 10.72 and the orders of Table 10.1. $\mathrm{I}_n \leftrightarrow \mathrm{I}_n^*$: $\max(v_\ell(n),v_\ell(4))=v_\ell(n)$, as ℓ is odd. $\mathrm{I}_0 \leftrightarrow \mathrm{I}_0^*$: $\max(0,v_\ell(4))=0$. II $\leftrightarrow$ IV* and IV $\leftrightarrow$ II*: $\max(v_\ell(1),v_\ell(3))=[\ell=3]$. III $\leftrightarrow$ III*: $v_\ell(2)=0$. At $q=2$: I_n gives $v_\ell(n)$, I_n^* and I_0^* give $v_\ell(4)=0$, IV and IV* give $v_\ell(3)=[\ell=3]$, the others 0, which is the definition at $q=2$. □

The paper states the max formula for $q\geq 5$ only; by Proposition 10.72 it holds at $q=3$ as well.

Example 10.75 (twist pairs)**.** Curve-level twists used in the paper, each identified by computing the twisted model, its minimal model and its label (all checked with PARI against Cremona's tables):

E	type at q	d	$E^{(d)}$	type at q	other bad primes
`11a1`	I_5 at 11	-11	`121d2`	I_5^* ($v=11$)	none
`11a3`	I_1 at 11	-11	`121d1`	I_1^* ($v=7$)	none
`121a1`	II ($v=2$) at 11	-11	`121c2`	IV* ($v=8$)	none
`121c1`	IV ($v=4$) at 11	-11	`121a2`	II* ($v=10$)	none
`539c1`	I_3^* at 7	-7	`77c1`	I_3	I_2 at 11 on both
`539d1`	I_0^* at 7	-7	`11a3`	I_0	I_1 at 11 on both
`980i1`	I_5^* at 7	-7	`140b1`	I_5	IV* at 2, I_1 at 5 on both
`20a1`	IV* at 2	-4	`80b1`	I_0^*, $f_2=4$	I_2 at 5 on both

In the row of `980i1`, the type IV* at 2 is preserved because $-7\equiv 1 \pmod 8$, so χ_{-7} is trivial on $G_{\mathbb{Q}_2}$. The last row is a wild twist at 2, to which no rule of Lemma 10.70 applies. The first row also shows that $e_\ell(E,q)$ is not twist invariant (the revised paper corrects the last sentence of Definition def:terms accordingly; see Chapter 21): `11a1` has $e_3(E,11)=0$, while `121d2` has $e_3(E,11)=v_3(11^2-1)=1$ (Chapter 19); the Tate parameters at 11 coincide (Example 10.67).

10.9 The wild primes 2 and 3

At $p\geq 5$ every additive reduction is tame and the Kodaira symbol determines $v(\Delta_{\min})$, f and the image of inertia (Corollary 10.38, Theorem 10.52). At $p=2$ and $p=3$ none of this holds. The paper meets this in Remark rem:wild ("At $q=2,3$ the Kodaira type does not determine P_q"), in Section sec:wild (the minima at 2 and the weights w_3 at 3), and in Definition def:terms (the index of a symbol I_n^* at 2 is not the valuation of the Tate parameter). This section collects what changes and gives the dictionary between the image of inertia at 2 and the factorisation of the 3-division polynomial over $\mathbb{Q}_2$, which is how the image is computed in practice.

Remark 10.76 (what changes at 2 and 3). Let $p\in\{2,3\}$.

(i) The type does not determine $v(\Delta_{\min})$ or f, although Ogg's formula $f = v(\Delta_{\min}) + 1 - m$ still links them (Theorem 10.47).

(ii) f can exceed 2: up to 8 at 2 and 5 at 3 (Theorem 10.48).

(iii) G_{ss} can be non-abelian, and of order divisible by p (Theorem 10.52(ii), (iii)).

(iv) The type does not determine G_{ss}, with two exceptions: IV, IV* at 2 force $G_{\mathrm{ss}} \simeq \mathbb{Z}/3$ (Corollary 10.54), and III, III*, I$_0^*$ at 3 force $G_{\mathrm{ss}} \simeq \mathbb{Z}/4$, $\mathbb{Z}/4$, $\mathbb{Z}/2$ (Remark 10.79).

(v) At 2 a symbol I$_m^*$ may be potentially good, and when it is potentially multiplicative $m \neq -v_2(j)$ (Theorem 10.66(ii), Remark 10.81).

Hence the Euler factor of the adjoint is not a function of the type at 2 and 3, as the paper says in Remark rem:wild; what determines it is the image of inertia (Chapter 19; the paper's Proposition prop:shapes).

Proposition 10.77 (inertia at 2 and the 3-division polynomial)**.** *Let $E/\mathbb{Q}_2$ have additive, potentially good reduction. Let $x_1, \dots, x_4$ be the roots of the 3-division polynomial ψ_3 (the x-coordinates of the four lines of $E[3]$, Chapter 9) and e_i the ramification index of $\mathbb{Q}_2(x_i)/\mathbb{Q}_2$. Then G_{ss} is one of $\mathbb{Z}/2$, $\mathbb{Z}/3$, $\mathbb{Z}/4$, $\mathbb{Z}/6$, Q_8, $\mathrm{SL}_2(\mathbb{F}_3)$, and the multiset $(e_1, \dots, e_4)$ is*

- $(1,1,1,1)$ *if $G_{\mathrm{ss}} \simeq \mathbb{Z}/2$; then E is the twist of a curve with good reduction at 2 by a ramified quadratic character χ, and $f_2 = 2a(\chi) \in \{4, 6\}$;*
- $(1,3,3,3)$ *if G_{ss} is cyclic of order 3 or 6; order 3 occurs exactly for the types* IV, IV**, with $f_2 = 2$, and order 6 is the twist of the order-3 case by a ramified quadratic character, with $f_2 \geq 3$;*
- $(2,2,2,2)$ *if $G_{\mathrm{ss}} \simeq \mathbb{Z}/4$;*
- $(4,4,4,4)$ *if $G_{\mathrm{ss}} \simeq Q_8$ or $\mathrm{SL}_2(\mathbb{F}_3)$.*

Proof. The list of groups is Theorem 10.52(iii) ($G_{\mathrm{ss}} \neq 1$ as the reduction is additive). By Theorem 10.51, G_{ss} is the image of I_2 in $\mathrm{Aut}(E[3])$, a subgroup of $\mathrm{SL}_2(\mathbb{F}_3)$, and the x_i correspond to the four lines of $E[3]$. For a root x_i, $e_i = [\mathbb{Q}_2^{\mathrm{ur}}(x_i) : \mathbb{Q}_2^{\mathrm{ur}}]$ (the residue degree is absorbed by $\mathbb{Q}_2^{\mathrm{ur}}$), which is the size of the orbit of x_i, that is of its line, under I_2. So (e_i) is the multiset of orbit sizes of G_{ss} on the four lines, each orbit counted once for each of its lines. The element -1 acts trivially on lines, and $\mathrm{SL}_2(\mathbb{F}_3)/\{\pm 1\} \simeq A_4$ acts on the four lines as A_4 on four points. The unique element of order 2 of $\mathrm{SL}_2(\mathbb{F}_3)$ is -1; elements of order 3 and 6 map to 3-cycles; elements of order 4 map to double transpositions; Q_8 maps onto the Klein group, which is transitive; and $\mathrm{SL}_2(\mathbb{F}_3)$ maps onto A_4. This gives the four patterns. If $G_{\mathrm{ss}} = \{\pm 1\}$, the character $\psi\colon I_2 \to \{\pm 1\}$ through which inertia acts is invariant under conjugation by $G_{\mathbb{Q}_2}$ (a conjugate of an element of I_2 acts by the conjugate matrix, and -1 is central), and it extends to a quadratic character χ of $G_{\mathbb{Q}_2}$. Explicitly: choose a Frobenius lift $\phi \in G_{\mathbb{Q}_2}$. Its image generates $G_{\mathbb{Q}_2}/I_2 \simeq \hat{\mathbb{Z}}$ topologically, and since $\hat{\mathbb{Z}}$ is free as a profinite group on one generator, $a \mapsto \phi^a$ is a section, so $G_{\mathbb{Q}_2} = I_2 \rtimes \overline{\langle\phi\rangle}$ with $\overline{\langle\phi\rangle} \simeq \hat{\mathbb{Z}}$ and every element written uniquely as $i\phi^a$ ($i \in I_2$, $a \in \hat{\mathbb{Z}}$). Put $\chi(i\phi^a) = \psi(i)$. Then $\chi(i\phi^a \cdot j\phi^b) = \chi(i(\phi^a j \phi^{-a})\phi^{a+b}) = \psi(i)\psi(\phi^a j \phi^{-a}) = \psi(i)\psi(j)$ by the invariance of ψ, so χ is a continuous character extending ψ (this is the construction of the paper's Proposition prop:shapes(ii)); then $\rho \otimes \chi$ is unramified, so $E \otimes \chi$ has good reduction (criterion of Néron, Ogg and Shafarevich), and $f_2(E) = a(V' \otimes \chi) = 2a(\chi)$ for V' unramified (Chapter 7), with

$a(\chi) \in \{2,3\}$. The same argument applied to the character $g \mapsto g^3 \in \{\pm 1\}$ of a cyclic G_{ss} of order 6 shows that E is the twist of a curve with $G_{\mathrm{ss}} \simeq \mathbb{Z}/3$; order 6 is wild since $2 \mid 6$, so $f_2 \geq 3$. Order 3 is tame, so $f_2 = 2$, and by Theorem 10.53 and Corollary 10.54 this happens exactly for IV, IV*. □

Table 10.4 records what occurs on Cremona's tables below conductor 10,000: all 20,979 pairs (curve, $p = 2$) with additive potentially good reduction at 2, by type, f_2 and pattern. The patterns were computed by factoring ψ_3 over $\mathbb{Q}$ and reading the ramification indices of the primes above 2 in each factor with PARI.

type at 2	f_2: pattern (number of pairs)
IV	2: $(1,3,3,3)$ 2,094
IV*	2: $(1,3,3,3)$ 2,656
II	4: $(1,3,3,3)$ 469, $(4,4,4,4)$ 478; 6: $(1,1,1,1)$ 136, $(4,4,4,4)$ 395; 7: $(4,4,4,4)$ 236
II*	3: $(4,4,4,4)$ 1,171; 4: $(1,1,1,1)$ 291; 6: $(1,3,3,3)$ 108, $(4,4,4,4)$ 96
III	3: $(4,4,4,4)$ 1,062; 5: $(4,4,4,4)$ 644; 7: $(4,4,4,4)$ 210; 8: $(2,2,2,2)$ 98, $(4,4,4,4)$ 96
III*	3: $(4,4,4,4)$ 1,582; 5: $(4,4,4,4)$ 314; 7: $(4,4,4,4)$ 210; 8: $(2,2,2,2)$ 98, $(4,4,4,4)$ 96
I_0^*	4: $(1,3,3,3)$ 596, $(4,4,4,4)$ 720; 5: $(4,4,4,4)$ 792; 6: $(1,3,3,3)$ 196, $(4,4,4,4)$ 200
I_m^* (potentially good)	3: $(4,4,4,4)$ 1,575; 4: $(1,1,1,1)$ 466, $(4,4,4,4)$ 1,305; 5: $(4,4,4,4)$ 528; 6: $(1,1,1,1)$ 250, $(1,3,3,3)$ 146, $(4,4,4,4)$ 1,429; 7: $(4,4,4,4)$ 236

Table 10.4: Additive potentially good reduction at 2 below conductor 10,000: conductor exponent and ramification pattern of the roots of ψ_3 over $\mathbb{Q}_2$. The pattern $(1,3,3,3)$ with $f_2 = 2$ is $G_{\mathrm{ss}} \simeq \mathbb{Z}/3$, with $f_2 \geq 4$ it is $\mathbb{Z}/6$.

In particular $f_2 = 2$ occurs only for IV and IV*, in accordance with Theorem 10.53.

In the paper 10.78. The Euler factor $P_2(X)$ (Chapter 19; the paper's Proposition prop:shapes) is determined by G_{ss} and the action of Frobenius on $(\mathrm{ad}^0 V)^{I_2}$: cyclic of order 3 or 6 gives $P_2(X) = 1 + 2X$ and $e_3(E,2) = 1$ (the order-6 case reduces to the order-3 case, as $\mathrm{ad}^0 V$ is unchanged by a quadratic twist); $\mathbb{Z}/2$ gives the cubic $(1-2X)(1-(a^2-4)X+4X^2)$ of the twist E' with good reduction, $a = a_2(E')$, and $e_3(E,2) = v_3(9-a^2)$; $\mathbb{Z}/4$ gives $1 \pm 2X$; Q_8 and $\mathrm{SL}_2(\mathbb{F}_3)$ act absolutely irreducibly, so $(\mathrm{ad}^0 V)^{I_2} = 0$ and $P_2 = 1$. So a type II or II* at 2 can have $e_3(E,2) \in \{0,1,2\}$: `336f1` has type II at 2, $f_2 = 4$, pattern $(1,3,3,3)$, $P_2 = 1+2X$, $e_3(E,2) = 1$ and $\deg \varphi_E = 24$; `1232l1`, the twist of `77a1` by χ_{-4} with $a_2(\texttt{77a1}) = 0$, has type II* at 2, $f_2 = 4$, pattern $(1,1,1,1)$, $P_2 = (1-2X)(1+2X)^2 = 1+2X-4X^2-8X^3$, $e_3(E,2) = 2$ and $\deg \varphi_E = 288 = 2^5 \cdot 3^2$. The original version of Remark rem:wild said that "a type II at 2 has $P_2 = 1$ and $e_3 = 0$"; the revised version says "usually", with these two curves as counterexamples. A potentially good I_m^* at 2 has the pattern $(4,4,4,4)$, hence $P_2 = 1$, in most cases (Table 10.4).

Remark 10.79 (the prime 3). Let $E/\mathbb{Q}_3$ have additive reduction.

(i) Types III, III*, I_0^* are tame ($f_3 = 2$), with G_{ss} cyclic of order 4, 4, 2.

(ii) Types II, II*, IV, IV* are wild: $f_3 \in \{3,4,5\}$ and $3 \mid e$.

(iii) Type I_n^* ($n \geq 1$) is potentially multiplicative, with $n = -v_3(j)$ and $f_3 = 2$.

Justification. (iii) is Theorem 10.66(i) and Proposition 10.46(iv). The tame/wild dichotomy in (i), (ii) is Theorem 10.53. In (ii), wild means that the pro-3 group P_3 acts nontrivially, so its image, a nontrivial 3-group, lies in G_{ss}. In (i), G_{ss} is a quotient of the procyclic tame inertia of order prime to 3, so it is cyclic of order 2 or 4 (Theorem 10.51(iii)). It has order 2 if and only if E is a ramified quadratic twist of a curve with good reduction (the argument of Proposition 10.77), that is, up to an unramified twist, the twist by χ_{-3} of such a curve, which has type I_0^* by Lemma 10.70(b) and Lemma 10.69; conversely type I_0^* is such a twist. So III, III* have order 4. □

Below conductor 10,000, at 3: III, III* (1,553 pairs each), I_0^* (2,283) and I_n^* (7,805) always have $f_3 = 2$; II and IV* both have $f_3 = 3, 4, 5$ in 879, 280, 107 pairs, and IV and II* both in 628, 177, 71 pairs. The equal distributions are Proposition 10.72: the twist by χ_{-3} is unramified outside 3, preserves the conductor, and exchanges these types.

In the paper 10.80. This is why Lemma lem:IV excludes $q = 3$ (at 3 the types IV, IV* are wild, and the conclusion "tamely ramified, Serre conductor exponent 1" fails), and why the weights w_3 at $q = \ell = 3$ in Conjecture conj:euler are not explained by Lemma lem:euler. The Remark after the Example of Section 3 of the paper mentions $q = 3$ in cases (a) and (b) of Theorem thm:qual; there the obstruction is $q = \ell$, and the ramification at 3 of the types II, IV is also wild by (ii).

Remark 10.81 (symbols I_m^* at 2). At 2 a symbol I_m^* is potentially good in 5,935 of the 9,667 pairs below conductor 10,000 (61%; among curves numbered 1 in their class, which below conductor 10,000 are the optimal curves except in class `990h` (whose optimal curve is `990h3`; it is multiplicative at 2, as $2 \parallel 990$, and so does not enter this count), 2,340 of 4,037, that is 58%, the figure quoted in the revised Remark rem:wild; the six curves with complex multiplication among them are all potentially good, so the optimal non-CM curves give 2,334 of 4,031, again 58%). When it is potentially multiplicative, $t_\ell(E, 2) = v_\ell(\nu)$ with $\nu = -v_2(j) = m - 4$ or $m - 8$ (Theorem 10.66(ii)), so $v_\ell(m)$ is the wrong invariant: `304b1` has type I_7^* and $\nu = 3$, so $t_3(E, 2) = 1$ although $v_3(7) = 0$; `832c1` has type I_9^* and $\nu = 1$, so $t_3(E, 2) = 0$ although $v_3(9) = 2$.

In the paper 10.82. Definition def:terms: "a symbol I_n^* at $q = 2$ can have potentially good reduction, and then $t_\ell = 0$"; Section sec:wild: "the index n of the symbol is not the valuation of the Tate parameter at 2, which is why n was not the right invariant". In the proof of Theorem thm:main a potentially multiplicative prime 2 is handled by Theorem 10.59 (a ramified quadratic twist of a Tate curve with $\nu = -v_2(j)$), not by Lemma lem:twist, which is for odd q (the original proof of Theorem thm:main applied Lemma lem:twist there; see Chapter 21).

10.10 Cremona's tables and PARI

The paper's computations take the curves from Cremona's tables and recompute every local invariant with PARI (Section sec:data and the appendix). This short section says what the tables contain and what PARI computes, and collects the local data of the running examples in one table. File formats and the practical use of PARI are in Chapter 22.

Remark 10.83 (what the tables contain). Cremona's tables [4, 5] list, for every elliptic curve over $\mathbb{Q}$ up to their conductor bound, the global minimal model in the normalisation of Theorem 10.6, the conductor, the isogeny class, rank and torsion. The file `allbsd` adds the product $\prod_p c_p$ of the Tamagawa numbers, the real period and the other quantities of the conjecture of Birch and Swinnerton-Dyer; other files give the isogeny matrices, the modular degrees (`degphi`) and the mod

ℓ images. Section sec:data of the paper describes the `ecdata` repository as containing every curve "with its Kodaira symbols, Tamagawa numbers, modular degree, isogeny class, torsion" and mod ℓ images, and adds that "the Kodaira types and discriminant exponents at the bad primes were recomputed with PARI/GP 2.15 from the a-invariants" (its appendix calls `ellglobalred`). The files used in this book (`allcurves`, `allbsd`, `allisog`, `alllabels`, `degphi`, `galrep`, conductor below 10,000) are a subset of that repository and, unlike the paper's description of the full repository, do not list the Kodaira symbols or the Tamagawa numbers prime by prime: here these are recomputed with Tate's algorithm (`ellglobalred`), as in the paper's own check. File formats are in Chapter 22. The minimal discriminant and j follow from the model.

Example 10.84 (a PARI session)**.** With PARI/GP 2.17.2, for `726a1`, `20a1` and `11a1`:

```
E = ellinit([1,1,0,-35,-51]);          \\ 726a1
R = ellglobalred(E)
  \\ [726, [1, 0, 0, 0], 4, [2, 1; 3, 1; 11, 2], [[1, 10, 0, 2], [1, 7, 0, 1], [2, 3,
     0, 2]]]
R[4]                                    \\ [2, 1; 3, 1; 11, 2]
elllocalred(ellinit([0,1,0,4,4]), 2)  \\ 20a1: [2, -4, [1, 0, 0, 0], 3]
elllocalred(ellinit([1,0,1,170,-3237]), 7)   \\ 539c1: [2, -7, [1, 0, 0, 0], 4]
E11 = ellinit([0,-1,1,-10,-20]);      \\ 11a1
ellinit(E11, O(11^8)).tate[3]
  \\ 10*11^5 + 2*11^6 + 6*11^7 + 6*11^8 + 5*11^9 + ... + O(11^13)
valuation(E11.j, 11)                    \\ -5
```

The local data of `726a1` are $[f,k,*,c]=[1,10,*,2]$, $[1,7,*,1]$, $[2,3,*,2]$ at $2,3,11$: types I_6, I_3, III (codes $10=6+4$, $7=3+4$, 3), Tamagawa numbers $2,1,2$ with product 4, and conductor $2\cdot 3\cdot 11^2$. For `20a1` at 2 the code -4 is IV*, $f_2=2$, $c_2=3$. For `539c1` at 7 the code $-7=-(3+4)$ is I_3^*. For `11a1` the Tate parameter has $\nu=5$ and $\bar{u}=10$ (Example 10.67).

Example 10.85 (local data of the running examples)**.** For each curve: conductor, minimal discriminant, and at each bad prime q the type, f_q, c_q, $\#\Phi_q^{\mathrm{geom}}$ and $v_q(j)$ (all computed with PARI; the hand computations of Example 10.42 agree).

curve	N	$\Delta_{\min}$	q: type, f_q, c_q, $\#\Phi_q^{\mathrm{geom}}$, $v_q(j)$
`11a1`	11	-11^5	11: I_5, 1, 5, 5, -5
`20a1`	$2^2\cdot 5$	$-2^8\cdot 5^2$	2: IV*, 2, 3, 3, 4; 5: I_2, 1, 2, 2, -2
`99a1`	$3^2\cdot 11$	$3^3\cdot 11$	3: III, 2, 2, 2, 9; 11: I_1, 1, 1, 1, -1
`121a1`	11^2	-11^2	11: II, 2, 1, 1, 1
`121c1`	11^2	-11^4	11: IV, 2, 1, 3, 2
`121c2`	11^2	-11^8	11: IV*, 2, 1, 3, 1
`121d1`	11^2	-11^7	11: I_1^*, 2, 2, 4, -1
`539c1`	$7^2\cdot 11$	$-7^9\cdot 11^2$	7: I_3^*, 2, 4, 4, -3; 11: I_2, 1, 2, 2, -2
`539d1`	$7^2\cdot 11$	$-7^6\cdot 11$	7: I_0^*, 2, 2, 4, 0; 11: I_1, 1, 1, 1, -1
`726a1`	$2\cdot 3\cdot 11^2$	$2^6\cdot 3^3\cdot 11^3$	2: I_6, 1, 2, 6, -6; 3: I_3, 1, 1, 3, -3; 11: III, 2, 2, 2, 0
`980i1`	$2^2\cdot 5\cdot 7^2$	$-2^8\cdot 5\cdot 7^{11}$	2: IV*, 2, 1, 3, 19; 5: I_1, 1, 1, 1, -1; 7: I_5^*, 2, 2, 4, -5

The curves `121c1`, `121c2` at 11 and `980i1` at 2 have $c_q=1<3=\#\Phi_q^{\mathrm{geom}}$; `726a1` at 2 and 3 has $c_q<\#\Phi_q^{\mathrm{geom}}$ because the reduction is nonsplit.

10.11 Exercises

The exercises go from routine to harder. Several ask the reader to check a claim of the paper; PARI is useful for the numerical ones (Chapter 22).

Exercise 10.86. Show that an integral model with $v(\Delta) < 12$ is minimal, and that for $p \geq 5$ an integral model with $v(c_4) \geq 4$ and $v(\Delta) \geq 12$ is not (Proposition 10.5). Decide whether the model $y^2 = x^3 - 27c_4x - 54c_6$ of `11a1` is minimal at 2 and at 3.

Exercise 10.87. For `11a1` at 11 and `20a1` at 5, decide whether the reduction is split, once from the polynomial $T^2 + a_1T - a_2$ after moving the node to $(0,0)$, and once from the Legendre symbol $\left(\frac{-c_6}{p}\right)$ (Proposition 10.12). Compare with $a_{11} = 1$ and $a_5 = -1$.

Exercise 10.88. Write down explicitly the isomorphisms $\tilde{E}_{\mathrm{ns}}(k) \simeq k^+$ for $y^2 = x^3$ and $\tilde{E}_{\mathrm{ns}}(k) \simeq k^\times$ for $y^2 = x^3 + x^2$, and check directly that three collinear points are sent to elements with sum 0, respectively product 1 (Proposition 10.10).

Exercise 10.89. Let $E =$ `11a1`, which has good reduction at 3 and 7. Using Corollary 10.17, show that the prime-to-3 part of $E(\mathbb{Q})_{\mathrm{tors}}$ injects into $\tilde{E}(\mathbb{F}_3)$ and the prime-to-7 part into $\tilde{E}(\mathbb{F}_7)$. Count these groups (orders 5 and 10), deduce that $E(\mathbb{Q})_{\mathrm{tors}}$ has order dividing 5, and conclude that $E(\mathbb{Q})_{\mathrm{tors}} \simeq \mathbb{Z}/5$ by exhibiting a point of order 5 (for instance $(5,5)$ on $y^2+y = x^3-x^2-10x-20$).

Exercise 10.90. Run Tate's algorithm by hand for `121a1` and `121c2` at 11 and for `539d1` and `539c1` at 7, and check the types used in the Example of Section 3 of the paper (Example 10.42).

Exercise 10.91. Check the paper's claim (Lemma lem:twist (ii)) that the twist of `121a1` by -11 has type IV* at 11. Then show that its twist by $+11$ has additive reduction at 2 (it is `1936k2`, of type I_4^* at 2 with $v_2(j) = 0$), so that the twist by q and the twist by q^* really are different curves (Warning 10.71).

Exercise 10.92. Check the paper's sentence "the cubic $T^3 + a_2T^2 + a_4T + a_6$ of Tate's algorithm has distinct roots modulo q" for the twist of `11a3` by -7 (it is `539d1`): compute a model $y^2 = x^3 + a_2x^2 + a_4x + a_6$ of `11a3`, the cubic of the twist with the sign $\epsilon = -1$, and its roots modulo 7 (Proposition 10.39).

Exercise 10.93. Prove $w_\ell(E,q) = \max\big(v_\ell(\#\Phi_q^{\mathrm{geom}}(E)), v_\ell(\#\Phi_q^{\mathrm{geom}}(E^{(q)}))\big)$ for $q \geq 5$ from Table 10.1 and Lemma 10.70 (Proposition 10.74). Which curves of Example 10.85 have $c_q \neq \#\Phi_q^{\mathrm{geom}}$, and why?

Exercise 10.94. Verify Ogg's formula $f_q = v_q(\Delta_{\min}) + 1 - m_q$ at every bad prime of the curves of Example 10.85 (Theorem 10.47).

Exercise 10.95. Using Theorem 10.52, compute the semistability defect at each additive prime $q \geq 5$ of the running examples. Show directly, following the proof of Theorem 10.52, that `726a1` acquires good reduction at 11 over $\mathbb{Q}_{11}^{\mathrm{ur}}(11^{1/4})$ and not over $\mathbb{Q}_{11}^{\mathrm{ur}}(\sqrt{11})$.

Exercise 10.96. Write out the proof of Corollary 10.54 for `20a1`: identify $E[3]^{I_2}$ with $\Phi_2^{\mathrm{geom}}[3]$, and check that the 3-division polynomial of `20a1` has one root in $\mathbb{Q}_2$ and three roots generating a totally ramified cubic extension. Then check the paper's claim (Remark rem:wild, Proposition prop:forms) that a type IV or IV* at 2 has $P_2(X) = 1 + 2X$: deduce it from Corollary 10.54(ii), (iv) as in the paper link after it, and compare with PARI's `lfunsympow` for `20a1` and `52a1`.

Exercise 10.97. Show that $a_6(\mathsf{q}) = -(5s_3 + 7s_5)/12$ has integer coefficients, and compute $j(E_{\mathsf{q}})$ up to the term $196884\,\mathsf{q}$ (Theorem 10.57). Invert the series to obtain $\mathsf{q}(J) = J + 744J^2 + 750420J^3 + \cdots$ (Proposition 10.58).

Exercise 10.98. Check the paper's claim (Example of Section 3) that $t_3(\texttt{539c1}, 7) = 1$: compute $v_7(j)$, the Tate parameter of `539c1` at 7 to a few 7-adic digits and its unit part, and show that they coincide with those of `77c1` (Example 10.67).

Exercise 10.99. For `11a1` at 11, show that the unit part of q_E is $\equiv -1 \pmod{11}$ and that it is a 5th power in $\mathbb{F}_{11}^{\times}$. This is the situation of the extra term of Lemma lem:tamfactor at $\ell = 5$ (although $\ell = 5$ is an Eisenstein prime for `11a1`, so the lemma applies and Theorem thm:main does not).

Exercise 10.100. Let E/K have multiplicative reduction of type I_n, with k finite, and let K'/K be the unramified quadratic extension, with $\mathrm{Gal}(K'/K) = \langle\sigma\rangle$. Using Theorem 10.59, $E(K) \simeq K^{\times}/q_E^{\mathbb{Z}}$ in the split case, and $E(K) \simeq \{w \in K'^{\times}/q_E^{\mathbb{Z}} : \sigma(w) = w^{-1}\}$ in the nonsplit case (the Frobenius action twisted by $w \mapsto w^{-1}$). Identify $E_0(K)$ with the classes represented by units, and compute $E(K)/E_0(K)$ in both cases for general n: it is $\mathbb{Z}/n$ (split), and $\mathbb{Z}/2$ or 0 according as n is even or odd (nonsplit). This reproves Proposition 10.65(ii). Check $c_2 = 2$ and $c_3 = 1$ for `726a1`, which is nonsplit of type I_6 at 2 and I_3 at 3.

Exercise 10.101. Check the paper's statement (Definition def:terms, Section sec:wild) that the index of a symbol I_m^* at 2 is not the valuation of the Tate parameter, on `304b1` and `832c1`: compute m, $v_2(j)$, f_2 and $t_3(E, 2)$ (Theorem 10.66, Remark 10.81). Then compute PARI's `ellinit(E, O(2^20)).tate[3]` for `832c1` and explain the result (Warning 10.68).

Exercise 10.102. Check the claim of the original version of the paper (Remark rem:wild) that "a type II at 2 has $P_2 = 1$" against `336f1` and the type II* curve `1232l1`, using PARI's `lfunsympow` and the dictionary of Proposition 10.77. Explain the answer through the inertia images ($\mathbb{Z}/6$, respectively $\mathbb{Z}/2$), and verify that $v_3(\deg\varphi_E) = e_3(E, 2) = 2$ for `1232l1` ($\deg\varphi_E = 288$).

10.12 Notes and references

Sources followed. Sections 10.1 to 10.3 follow Silverman's first book [23, Chapter VII] (minimal models in VII.1, reduction in VII.2 and VII.5, the formal group in Chapter IV and VII.2, torsion in VII.3 and VII.7, potential good reduction in VII.5); the proofs of Proposition 10.10 and Proposition 10.15 are the standard ones written out. Néron models, Kodaira symbols and Tate's algorithm follow [22, Chapter IV] (Sections IV.5 to IV.9, with the table of Kodaira types and Tamagawa numbers in IV.9 and Ogg's formula in IV.11), with [2] for the general theory and [4, Section 3.2] for the explicit form of the algorithm. The Tate curve follows [22, Chapter V] (V.3 for the uniformisation, V.5 for curves with $v(j) < 0$). The original sources are Néron [13], Kodaira [7], Tate [24], Ogg [14], Saito [19] and Serre and Tate [21]; Serre's paper [20, Section 5.6] discusses the image of inertia at 2 and 3.

Theorem numbers. We cite theorem numbers only where they have been checked: Dokchitser and Dokchitser [6, Theorem 3.1] and its source Kramer and Tunnell [8, Proposition 8.20] (the number was checked in the arXiv version of [6] when this book was planned). Elsewhere we cite sections. In particular Kraus's paper [10], which gives the semistability defect at all primes and the full 2- and 3-adic classification, is cited as a whole: the numbering of its theorems is quoted inconsistently in later papers (see Billerey [1] for the 2-adic statements and [17] for a recent restatement). Kraus's conditions on c_4, c_6 (Proposition 10.5(iii)) are in a different paper [9]. Papadopoulos [15] tabulates the Kodaira types at 2 and 3 in terms of valuations and

congruences. The relation between Kodaira symbols and wild ramification for twists of curves with multiplicative reduction at 2 is in Lorenzini [12]; see also [18].

The types IV, IV* at 2. Corollary 10.54 has two independent proofs: the one given here, through the component group (Proposition 10.33), which is the proof of Lemma lem:IV in the revised paper; and one through the conductor, from Theorem 10.53: $f_2 = 2$ forces $\mathrm{sw} = 0$, so the image of inertia is a cyclic group of odd order dividing 24, hence of order 3. The original version of the paper left this case as an exception in Proposition prop:forms.

The exchanges at 3. Proposition 10.72 is proved here from Ogg's formula; we have not found it stated in the literature in this form, though it can presumably be read off the tables of [15].

Computations. All numbers in the chapter were computed with PARI/GP 2.17.2 [16] through `cypari2`, on Cremona's tables [5] for conductor below 10,000 (64,687 curves, 211,455 pairs of a curve and a bad prime). The population checks quoted in the text are: Kraus's conditions (Proposition 10.5); $a_p = \left(\frac{-c_6}{p}\right)$ (Proposition 10.12); the formulas for Step 2 of Tate's algorithm; the table for $p \geq 5$; the values of c by type; the twist exchanges (90,970 pairs); the types at additive potentially multiplicative primes; the conductor exponents at 2 and 3; and the inertia patterns at 2 (20,979 pairs). The Tate parameters were computed from the series of Proposition 10.58; the discrepancy of PARI's `.tate` at 2 is recorded in Warning 10.68.

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